California Polytechnic State University
Campus Facility Maintenance and Renovation Standards

Scope
This policy applies to the purchasing, renovation and management of site and buildings on the main campus for California Polytechnic University, San Luis Obispo (Cal Poly). It includes guidelines for purchasing materials related to these activities, disposing of waste generated from these activities, and managing indoor air quality during the work itself. This plan specifically addresses:

- Base building elements permanently or semi-permanently attached to the building
- Furniture and furnishings as well as the components and parts needed to maintain them
- Construction related waste management
- Indoor Air Quality Policy and Procedures for Maintenance and Renovation

Goals

Operational element	Goal	Measurement
Purchasing Policy for Maintenance and Renovation	• Annual review, update and confirmation of Cal Poly Facility Standards to confirm compliance with, California State University Sustainability Policy Proposal (RJEP/CPBG 05-14-01), LEED O+M, CalGreen Requirements, and other relevant campus standards and preferences • Demonstrated compliance by review of purchases that meet the stated environmentally preferable standards • Purchase at least 50%, by cost, of the total maintenance and renovation materials that meet or exceed the environmentally preferable standards documented in the Campus Facility Standards	Annual reporting
Purchasing Furniture and Furnishings	• Purchase at least 75%, by cost, of total furniture and furnishings that meet one or more of the environmentally preferable criteria (recycled, wood, bio-based, reused, etc.) documented in the Campus Facility Standards	Annual reporting
Waste Management for Maintenance and Renovation	• Divert at least 75% of the total construction and demolition material • Reduce the solid waste disposal rate by 50 percent (PRC § 42921) by 2016, by 80 percent by 2020, and move to zero waste. (14-New) • The CSU will encourage the reduction of hazardous waste to the extent possible while supporting the academic program. (14-New)	Annual Reporting by Category as required by Cal Recycle

Performance Measurement and Schedule for Reassessment
All activity will be recorded annually by the responsible party and reported to the appropriate agency (usually the Chancellor's Office). The criteria met by the products or equipment used will also be recorded.

Standard Operating Procedures and Implementation Strategies
For all maintenance and renovation projects (regardless of size) the University shall purchase/utilize a minimum of 50% of environmentally preferable products. Environmental preferable criteria include:

- Recycled Content
- Forest Stewardship Council (FSC) Certified Wood
- Sustainable Agricultural Network's Sustainable Agriculture Standard Bio-based Materials
- Re-used or Salvaged Materials
- Extended Producer Responsibility Program Products
- GreenScreen v1.2 Benchmark
- Cradle to Cradle Certification
- Product Manufacturer Supply Chain Optimization
- Low Emission (VOC) Standards
- Regionally Produced Products (100 miles)

Quality Assurance/Quality Control Processes
Prior to project start, the responsible Project Manager will confirm receipt and understanding of Campus Facility Standards by the contractor or vendor. The Project Manager will ensure compliance and documentation is completed through the submittal, periodic pay review and oversight processes in place on the campus.

At the conclusion of each fiscal year the Associate Directors and Directors will review reporting information to confirm standards compliance and evaluation of policy relevance and accuracy against LEED requirements, California State University Sustainability Policy (RJEP/CPBG 05-14-01), and other state legislation and requirements. If any implementation goals are not being met, the responsible party will investigate the situation and will work with the individuals performing the maintenance activities to resolve the issue. The responsible party will evaluate whether updates are necessary to the plan or the maintenance process in order to achieve the implementation goals.

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Division 00 – Procurement and Contracting Supplementary Requirements

Not Used

Division 01 - General Requirements

01 00 00 - General Requirements

Campus Standards are noted when a sole source is required to match an existing system on campus (keying, energy management controls, etc.). The following words shall be included in the specifications: *No Substitutions Allowed*, or *Substitutions: Not Allowed*. The sole source may include a manufacturer, or a manufacturer and a specific model.

Campus Preferences are noted when certain items have met with success on campus. In the specifications are listed first under Products. The campus will attempt to list the required or desired features that make these items a preference.

Products that are ***Not Recommended*** are noted when certain items have failed to meet the campus needs, require high or frequent maintenance, have a short life, or are not compatible with campus systems, etc. The campus will attempt to list the reasons these items are not recommended.

01 35 46 Indoor Air Quality Procedures

For all maintenance and renovation projects (regardless of size) the following indoor air quality standards shall be adhered to, to support occupant, contractor and vendor health:

- Follow the recommended control measures of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) IAQ Guidelines for Occupied Buildings under Construction, 2nd edition (2007), ANSI/SMACNA 008–2008, Chapter 3
  - Protect stored on-site and installed absorptive materials from moisture damage.
  - Do not operate permanently-installed air handling equipment during construction unless filtration media with a minimum efficiency reporting value (MERV) of 8, as determined by ASHRAE 52.2–2007, with errata (or equivalent filtration media class of F5 or higher, as defined by CEN Standard EN 779–2002, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance), are installed at each return air grille and return or transfer duct inlet opening such that there is no bypass around the filtration media.
- Build into the schedule a procedure to, before occupancy, replace all filtration media with the final design filtration media.
- Evaluate whether a flush-out or air quality testing is needed after construction ends and all interior finishes are installed but prior to occupancy.

01 80 00 – Performance Requirements

Third Party Verified Certification - LEED Certification

New construction shall be designed and implemented to achieve LEED for New Construction Gold Certification. Design and construction teams are to provide all necessary services to complete the full certification process. All costs to achieve certification shall be the responsibility of the project team.

Performative Modeling

With each new construction or major renovation design submittal, the design / construction team shall complete the following iterative modeling at a minimum:

- Predictive Energy Modeling: Site EUI benchmarked against other projects of similar type and size.
- Predictive Daylight Modeling: The University prioritizes the use of daylighting in all regularly occupied spaces (unless indicated otherwise programmatically). While avoiding glare/contrast is necessary, spaces should meet the most current IESNA standards for daylight autonomy (300 – 3000 lux) at the work plane. Design Teams should demonstrate spatial daylight autonomy through the use of annual computer simulations.
- Predictive Water Use: For both indoor and outdoor water, the project should be benchmarked against other projects of similar size and type using a theoretical baseline.

01 74 19 – Construction Waste Management

A minimum of 75% of the total project waste should be diverted from landfill, in order of preference 1) weight, 2) volume, whichever is most feasible to measure.

- Provide contract documents, including a waste management plan, to show evidence of recycling, and reuse of recovered materials.
- Inform the University and architect where and if Construction and Demolition Waste Management requirements could detrimentally impact Construction and Demolition schedule.
- Provide separate itemization of cost related to Construction Waste Management.
- Effect optimum management of solid wastes via a materials management hierarchy.
- The materials management hierarchy shall be: reduce, reuse, and recycle.
- Prevent environmental pollution and damage.

For each facility renovation project, the Project Manager will coordinate with the contractor or personnel to discuss the scope of the renovation.

- The scope of the renovation must be determined and the materials to be used and discarded during the renovation must be identified. Packaging will be a consideration in the materials that will be discarded.
- The approximate volume of each type of waste will be broken out. Separate categories may include cardboard, wood products and cabinetry, drywall, tile, etc.
- From this material flow, the five largest waste categories will be determined.
    - The Project Manager will coordinate proper waste disposal and landfill diversion for these waste categories. This will involve contacting the appropriate vendors, scheduling haul dates, and ensuring properly sized storage areas for the construction waste.
    - If necessary, a separate secured storage area will be secured for hazardous waste, such as paint.
    - Once the waste disposal has been coordinated, the renovation manager will write waste disposal instructions for each waste category and will distribute to the appropriate vendors.
- For regular maintenance activities, the facility manager will ensure that the proper materials are recycled or composted.
- For all work, including renovation and maintenance, document solid waste disposal and diversion. Include the quantity by weight of waste generated; waste diverted through sale, reuse, or recycling; and waste disposed by landfill or incineration. Identify landfills, recycling centers, waste processors, and other organizations that process or receive the solid waste.

Division 02 - Existing Conditions

02 00 00 - Existing Conditions

Existing underground utilities in areas of work shall be located by potholing. Potholing is required to establish critical elevations, especially with gravity lines such as storm or sanitary sewer. Potholing shall include points of connection to existing utilities, utilities near footings or which are being crossed and utilities being relocated. It will be necessary to identify existing utilities that must remain active during construction and provide temporary routing of utilities if necessary to accommodate construction activities without limiting service to existing facilities.

02 33 00 Geotechnical Investigations

As determined by project size and scope, the contractor is responsibility to conduct their own subsurface geotechnical investigation and prepare a soils engineering report for use in the final design of a project.

02 40 00 - Demolition and Structure Moving

The contractor is responsible for the removal and disposal of all demolished materials from the site at approved disposal or recycling facilities.

Even if no hazardous materials or contaminated soils are identified on the project site, the contractor shall still identify appropriate disposal facilities in case any of these materials are discovered during construction. Demolished materials such as asphalt paving, aggregate base, and concrete should be recycled and reused if possible.

Contractor shall coordinate with the University to identify salvageable items and to identify the party responsible for salvage prior to commencing demolition operations.

Dust control shall be employed during demolition operations and BMPs shall be employed to minimize tracking of dirt and debris off of the project site.

Division 03 - Concrete

03 00 00 - Concrete

Fly ash may be substituted for Portland cement. The following recommendations apply to concrete mix designs available in the area and consider the quality of locally available aggregate. 15% fly ash substitution may be made without impacting strength for concrete with up to f'c = 5 ksi. At this level of substitution, it improves the workability of the concrete and is less expensive than the Portland cement it replaces. Up to 25% substitution for f'c less than or equal to 5 ksi may be used although concrete quality begins to degrade because it is harder to work and strength is impacted. 35% is may be used in foundations or retaining walls where there is little need to work the concrete but strength is limited to f'c = 4 ksi and is determined based on 56 days rather than the traditional 28 days.

Cast-in-place concrete will utilize cementitious and aggregate materials produced locally.

03 20 00 - Concrete Reinforcing

Recycled content for reinforcing steel shall be a minim of 80%.

Division 04 – Masonry

Not Used

Division 05 - Metals

05 41 00 - Structural Metal Stud Framing

Provide backing in area where equipment will be installed on walls. Confirm locations with Project Manager

Provide backing in additional areas where equipment may be installed on walls in the future. Provide minimum 20 gage steel studs at 16 inches on center.

Division 06 - Wood, Plastics, Composites

06 10 00 - Rough Carpentry

Provide backing in areas where equipment will be installed on walls. Confirm locations with Project Manager.

Provide backing in additional areas where equipment may be installed on walls in the future. Provide minimum 20 gage steel studs at 16 inches on center.

06 41 00 - Architectural Wood Casework

Locks for Drawers and Doors: Provide two keys per room of differently keyed drawers and doors where non-state keys are used.

06 61 16 - Solid Surfacing Fabrication

Lavatory Countertops:

- Solid polymer components
- LEED™ NC v4: Credit EQ 2 – Low emitting materials; and Credit MR 3 – Building Product disclosure and optimization – sourcing of raw materials
- Fire response characteristics: Class A per UL 723 (ASTM E84); Flame Spread Index of 25 or less, and Smoke Development Index of 450 or less
- Warranty: 10 years, minimum
- Provide surface and performance characteristics as Corian® surfaces from the DuPont company

Division 07 - Thermal and Moisture Protection

07 63 00 - Sheet Metal Roofing Specialties

Roof penetrations of service utilities:
Preferred Manufacturer: LSC Corp.

Pipe Chase Housing:

Removable lid with gasket. Pipe seals for pipes from ¼-inch to 1½ -inches outside diameter. Housing with gasket to curb. 3-inch skirt protects leading edge of roof. Constructed of welded power coated aluminum and stainless steel bolted connections. 2-inch flange out onto roof. Tower extension allows for space to mount disconnects and control boxes. Some models are equipped with hose bibb and knock-outs for GFCI.

07 84 13 - Penetration Firestopping

Wall and floor penetrations of cabling for telecommunications, fire alarm and security.

Manufacturer: Specified Technologies Inc. Model Series: EZ Path Fire Rated Pathways. Website: http://sti.fmpdata.net/ftp/Estimation_Installation/ZIS1029_Grid_Install_Sheet.pdf

- Wall – Model: EZDP433GK-C with four colored pathways to be organized as follows: ORANGE = Fire Alarm, BLUE = Communications-Horizontal Cabling, YELLOW = Security and Door Access, and WHITE = Communications-Backbone Cabling (between telecom rooms).
- Floor – Model: Series 44 Modular Grid System in either one, two or four modules with pathways in banks of four with the option to utilize blank firestop filler panels in multi-slot grids. Four colored pathways to be organized as follows: ORANGE = Fire Alarm, YELLOW = Security and Door Access, and WHITE = Communications-Backbone Cabling (between telecom rooms).

Firestopping at fire rated openings for conduit and cable trays.

Through Penetration Firestopping of Fire Rated Assemblies: ASTM E814 with 0.10 inch water gage (24.9 Pa) minimum positive pressure differential to achieve fire F-Ratings and temperature T-Ratings not less than 1-hour.
- Wall Penetrations: Fire F-Ratings not less than 1-hour.
- Floor Penetrations: Fire F-Ratings and temperature T-Ratings not less than 1-hour.
- Through Penetration Firestopping of Non-Fire Rated Floor Assemblies: Materials to resist free passage of flame and products of combustion.
- Noncombustible Penetrating Items: Noncombustible materials for penetrating items connecting maximum of three stories.
- Penetrating Items: Materials approved by authorities having jurisdiction for penetrating items connecting maximum of two stories.

Firestopping Products:
Preferred Manufacturers and Models:
- 3M Fire Protection Products
- Tremco Inc.

Cable Tray:
- 3M, Model W-L-4004 - 24" x 4" steel or aluminum cable tray, 32% fill. Also referred as WL4004.
- Tremco, Model TREMstop PS, fire containment pillow system.
- Prohibited: Intumescent putty.

Metal Pipe:
- 3M, Model C-AJ-1001 or CAJ-1009
- Tremco, Model C-AJ-1179 or C-AJ-1187

Non-Metallic Pipe:
- 3M, Model C-AJ-2005
- 3M Fire Barrier FS-195+ Wrap/Strip – 2”x24”, Intumescent elastomeric strip with foil on one side (UPC # 00051115071157), and Plastic Pipe Device PPD6 (UPC # 00051115082535) for 6” pipe, or Restricting Collar RC-1-2” (UPC #00054007083245) for 2” pipe
- Tremco, Model C-AJ-2082 or FA-2024

Cabling:
- 3M, Models

Single, Fire Barrier Pass-Through Devices:

- 2-1/2” Square (UPC # 00051115165962) and Mounting Brackets 2-1/2” SQ Single Mount (Pair) (UPC # 00051115187506)
- Fire Barrier Pass-Through Device 4” Square (UPC # 00051115165979) and Mounting Brackets 4” SQ Single Mounting (Pair) (UPC # 00051115187520)
- Fire Barrier Pass-Through Device 4” Round (UPC #00051115165986) and Mounting Brackets 4” Round Single Mount (Pair) (UPC # 00051115187544)

Multiple, Square Pass-Through:

- Fire Barrier Pass-Through Device 4” Square and Mounting Brackets 4” SQ Triple Mount (Pair) UPC # 00051115187537).
- Tremco

Division 08 - Openings

08 06 70 - Hardware Schedule

Key Schedule

Use this area to list all keys provided on project referencing appropriate Sections. The Campus prefers to have one listing for all keys. The purpose of Key Schedule is to allow the Architect to coordinate requirements and specifications for all keys provided on the project (doors, cabinets, drawers, panels, accessories, toilet room fixtures, equipment, etc.), have one listing for all keys, organize key submittal to the University, and provide manufacturer model numbers for future key replacement.

Key Schedule for {project name} {building number}						
Purpose / Location	**Room Number**	**Keys Required**	**Total Keys**	**Model Number**	**Manufacturer**	**Section Reference**
Cabinet locks						06 40 00 – Architectural Woodwork
Construction - Master		20	20			08 71 00 – Door Hardware
Construction - Control		2	2			08 71 00 – Door Hardware
Control		4	4			08 71 00 – Door Hardware
Master		6 per each master set				08 71 00 – Door Hardware
Grand Master		4	4			08 71 00 – Door Hardware
Extra Blank		1 per each lock				08 71 00 – Door Hardware
Building Entrance		2/each lock or cylinder				08 71 00 – Door Hardware
Interior Doors		___ /each lock or cylinder				08 71 00 – Door Hardware

Key Schedule for {project name} {building number}						
Purpose / Location	**Room Number**	**Keys Required**	**Total Keys**	**Model Number**	**Manufacturer**	**Section Reference**
Toilet Accessories		6/ each building restroom/ accessory keyed differently				10 28 00 Toilet, Bath, and Laundry Accessories
Equipment						Division 11 – Equipment sections
Furnishings						Division 12 – Furnishings sections
Elevators		3 keys, 1 installed in elevator pit.				14 20 00 – Elevators
Fire Control Cabinets						21 00 00 – Fire Suppression
Fire Control Padlocks		2/ each Padlock (IC cores)				21 00 00 – Fire Suppression
Plumbing						22 00 00 – Plumbing
Plumbing Padlocks		2/ each Padlock (IC cores)				22 00 00 – Plumbing
HVAC Equipment Panels		2/ each Panel				23 00 00 – HVAC
HVAC Yard, Doors and Equipment Padlocks		2/ each Padlock (IC cores)				23 00 00 – HVAC
Electrical Panels						26 00 00 – Electrical
Emergency Generator Panels and Equipment		2/ each Padlock (IC cores)				26 00 00 – Electrical
Lighting Control Panels		2/ each Panel				26 00 00 – Electrical
Lighting Switches		2/ each switch				26 00 00 – Electrical
Electrical Yard, Equipment, & Switches Padlocks		2/ each Padlock (IC cores)				26 00 00 – Electrical

Key Schedule for {project name} {building number}						
Purpose / Location	**Room Number**	**Keys Required**	**Total Keys**	**Model Number**	**Manufacturer**	**Section Reference**
Communications Equipment						27 00 00 - Communications
Emergency Telephones (Blue Light Phones)		2/ each				27 00 00 - Communications
Communications Equipment Padlocks		2/ each Padlock (IC cores)				27 00 00 - Communications
Electronic Safety and Security Panels						28 00 00 – Electronic Safety and Security
Electronic Safety and Security Equipment Padlocks		2/ each Padlock (IC cores)				28 00 00 – Electronic Safety and Security
Irrigation Equipment						32 80 00 – Irrigation
Utilities						33 00 00 - Utilities

Closeout Submittal: Key Schedule – Provide hard copies and one electronic copy of completed Key Schedule along with keys separated and labeled.

08 11 13 - Hollow Metal Doors and Frames

Steel Doors:

- Exterior: Full flush doors (ANSI A250.8-1998 (SDI-100)), 1-3/4 inches thick, galvannealed steel (ASTM A653/A653M-07), 14 gage closer reinforcement top and bottom with snap-in top caps.
- Interior: Full flush doors (ANSI A250.8-1998 (SDI-100)), 1-3/4 inches thick, 14 gage closer reinforcement top and bottom.
- Hardware preparation and reinforcement: Prepare for commercial hardware (ANSI A250.6-1997) and mortised locks (ANSI A115.1), no face holes unless coordinated with hardware schedule. Factory reinforce for surface applied hardware – 14 gage for closer. Locations per ANSI/DHI A115.
- Preferred Manufacturer and Model: Steelcraft L16 Series

Steel Door Frames:

- Exterior: 16 gage welded door frames; A60 hot dipped galvannealed steel.
- Flush Frames: F-Series Frames, 16 gage. http://www.steelcraft.com/downloads/pdfs/steelcraft_spec_l1.pdf
- Interior: 16 gage knock-down door frames.
    - Drywall Frames: DW-Series (using Baseboards) and K-Series (Not using Baseboards), 16 gage.
        - http://www.steelcraft.com/pc_frames_drywall.asp

  - http://www.steelcraft.com/downloads/pdfs/steelcraft_spec_dw1.pdf
- Reinforcement: 14 gage for closer, 4-7/8 inches ASA strike, and 4-1/2 inches hinges.
- Preferred Manufacturer: Steelcraft.

08 14 00 - Wood Doors
36 inches minimum, solid core, birch veneer.

Closeout Submittals: Samples for University Paint Shop: Submit documentation of painting system used on “pre-finished” doors, including two manufacturer’s sample finishes for each color and system used on the project.

08 41 00 - Entrances and Storefronts
No underfloor closures

08 71 00 - Door Hardware

Keying System:
University Facility Services Campus Key Coordinator designs keying system.

Contractor:
General: Submit proposed keying system to Architect for review and written approval by Trustees, and copy of Best Access (Stanley Security Solutions) Purchase Order Number to Architect for Trustees Records.

Closeout Submittal: Submit three hard copies and one electronic copy of final keying schedule.

Products: Key locks and cores at factory and maintain permanent records of information.

Execution: After submittal and approval of keying system by Trustees, submit manufacturer’s Purchase Order Number for Trustees to order cores and keys. Manufacturer shall deliver permanent keys, control keys and cylinder cores to Trustees Representative.

Coordinate installation of permanent cores by University Lock Shop with Trustees Representative.

Keys:
Material: Nickel-silver. Stamp keys “Do Not Duplicate.”

Quantity: Cylinder Keys (3 per core); and Extra Blank Keys (1 per core or lock).

Keying:
Telecommunication Spaces (Rooms): Use 112-series key lock. (Coordinate with Division 27 – Communications.)

Master Padlocks: Use where less security is required. Coordinate with Trustees Representative during design process for specific Campus Project requirements, and Divisions 21, 22, 23, 26 (Electrical Panels), 27, and 32 (Bollards). Key alike to Campus Specifications #2359, and provide 2 keys per panel, etc.

State Padlocks: Use where higher security is required, such as Electrical Yards and Switches, HVAC Yards and Equipment, etc. Coordinate with Trustees Representative during design process for specific Campus Project requirements, and Divisions 21, 22, 23 (HVAC Air Handling Units), 26 (Electrical

Switchgear, Transformers), 27, and 32. Requires interchangeable cores complying with Campus Cylinders and Keying Standards. Interchangeable core.

Hinges:
Campus Preferred Manufacturers:

- Hager Companies: Exterior -- Stainless Steel, Ball Bearing Part #BB1191; Interior – Plated with 626 Finish, #BB1279BB
- McKinney Products Company, Division of Assa Abloy
- Stanley.

Pivots and Pivot Hinges:
Campus Preferred Manufacturers:

- Ives; Division of Allegion
- Rixson-Firemark, Inc.; Div. of Assa Abloy
- National Manufacturing 101 Retro Kit

Continuous Pin and Barrel Hinges:
Campus Preferred Manufacturers:

- Pemko
- Markar Products, Inc.
- McKinney Products Company; Div. of Assa Abloy

Mechanical Locks and Latches:
Campus Preferred Manufacturers:

- Best Access Systems – 45H Series
- Corbin Russwin – ML2000 Series
- Schlage Lock Company; an Ingersoll-Rand Company – L94 Series
- Assa Abloy 8200 Series mortise locks

Electric Mortise Locksets:
Campus Preferred Manufacturer:

- Best Access Systems – 45H Series.

Surface Bolt:
Campus Preferred Manufacturer:

- Door Controls International
- Don-Jo Manufacturing
- Hager Companies
- Ives: H.B. Ives

- Rockwood Manufacturing Company
- Triangle Brass Manufacturing Company, Inc.

Flush Bolts:
Campus Preferred Manufacturers:

- Door Controls International
- Don-Jo Manufacturing
- Hager Companies
- Ives: H.B. Ives
- Rockwood Manufacturing Company
- Triangle Brass Manufacturing Company, Inc.

Exit Devices:
Prefer rim device over vertical rods. Preferred Manufacturer:

- Von Duprin, Division of Allegion – Series 98 or 99.
- Corbin Russwin – Series ED5000.
- Not Recommended: Adams Rite Manufacturing Co. – Products are low quality and don't hold up.

Cylinders:
Small format interchangeable core type, constructed from brass or bronze, stainless steel, or nickel silver.

Number of Pins: Seven

Mortise Type: Threaded cylinders with rings and appropriate cam to suit lock function and type.

Function: D-Storage; R-Classroom; N-Passage, A-Office, TA-Dormitory.

Required Manufacturer:

- Best Access Systems.
- Medeco Keymark

Door Cylinders:
Best Model 1E Seriesor equivalent (Arrow, general).

Construction Keying:
Construction Master Keys:

Provide temporary construction master keys (20 construction master keys and 2 control keys). See Section 08 06 05 – Key Schedule.

Construction Cylinders:

- Provide construction cores.
- Permanent Keying: University installs permanent cores.

Electric Strikes:
Campus Preferred Manufacturers:

- Hes Innovations (1006, 9600)
- Von Duprin

Operating Trim (push/pulls):
Campus Preferred Manufacturers:

- Don-Jo Manufacturing
- Forms & Surfaces
- Triangle Brass Manufacturing Company, Inc.

Accessories for Pairs of Doors:
Campus Preferred Manufacturers:

- Coordinators:
  - Don-Jo Manufacturing
  - Hager Companies
  - Ives: H.B. Ives
  - Door Controls Internationals
- Removable Mullions:
  - Von Duprin, Division of Allegion
  - Corbin Russwin Architectural Hardware, Division of Assa Abloy.
  - Note: “Key Removable” when needed.
- Astragals:
  - National Guard Products, Inc.
  - Pemko Manufacturing Company, Inc.
  - Reese Enterprises, Inc.

Closers:
Campus Preferred Manufacturers:

- Surface-Mounted Closers:
  - LCN, Division of Ingersol-Rand, 4040 Series, mounting: inside of door, parallel arm preferred.
  - Corbin Russwin, DC6200 Series.
- Electromechanical Closers:
  - LCN, Division of Allegion, 4041 Series
  - Norton, Division of Assa Abloy
- Concealed Floor Closers: Not preferred because replacement is expensive and difficult.
- Closer Holder Release Devices:
  - Not preferred because of wear and tear on top hinge.

  - Corbin Russwin Architectural Hardware, Division of Assa Abloy.
  - Rixson-Firemark, Inc., Division of Assa Abloy.
- Overhead Stops: Not allowed because pull hinges out causing door to sag, and damages door jambs.

Protective Trim Units:
Campus Preferred Manufacturers:

Metal Protective Trim Units:

- Don-Jo Manufacturing
- IPC Door and Wall Protection Systems, Inc.
- Triangle Brass Manufacturing Company, Inc.

Stops and Holders:
Preferred Manufacturers: Metal Protective Trim Units:

- Don-Jo Manufacturing
- Ives: H.B. Ives
- Norton Door Controls, Division of Assa Abloy.
- Rixson-Firemark, Inc., Division of Assa Abloy.
- Triangle Brass Manufacturing Company, Inc.

Door Gasketing:
Campus Preferred Manufacturers:

- Door Gasketing:
  - National Guard Products, Inc.
  - Pemko Manufacturing Company, Inc.
- Door Bottoms:
  - National Guard Products, Inc.
  - Pemko Manufacturing Company, Inc.

Thresholds:
Campus Preferred Manufacturers:

- Thresholds: Meet ADA Access requirements.
  - National Guard Products, Inc.
  - Pemko Manufacturing Company, Inc.
  - Rixson-Firemark, Inc., Division of Assa Abloy.

<u>Division 09 - Finishes</u>
09 29 00 - Gypsum Board
Standard: 5/8-inch, Type X.

Preference: 5/8-inch, Fire-Rated Mold-Resistant Board (XP Gypsum Board panels) complying with ASTM C 1396, Type X

09 30 13 - Ceramic Tiling
Standard: Grout: Sanded.

Execution: Seal grout, as soon as possible, after grouting is completed. Purpose: To prevent permanent stains from being imbedded in grout.

09 60 00 – Floorings
Maintenance Instructions: Provide Manufacturer's detailed floor finishing and maintenance instructions for each type of flooring – concrete, ceramic tile, carpeting, linoleum, vinyl composition tile, hardwood, terrazzo, etc.

09 68 00 - Carpeting
Standard: Carpet System must meet or exceed the current VOC limits of the Carpet and Rug Institute's

Green Label Indoor Air Quality Test Program.
Maintenance Instructions: Provide Manufacturer's detailed floor finishing and maintenance instructions.

09 91 00 - Painting
General: Close-out Submittals: Submit two painted samples, 6 inches by 6 inches illustrating each color and system used on the project.

Extra Materials: University has first right of refusal for left-over paint. 2 Fresh gallons of each color for Paint Shop.

General Notes to Design Team
- Recycled Paint for Interior Surfaces: The use of recycled paint for interior surfaces is not supported by the Standards for technical and quality control reasons.
- The Standard is based on the Green Seal's Standard GS-11 requirements, and the products listed below meet these requirements. Recycled paint, as offered by Kelly-Moore, is made from Kelly-Moore paint returned by contractors. Kelly-Moore sorts by sheen and mixes the paint to provide 14 standard colors. However, none are an off-white normally used in offices, and the darker colors are not acceptable. MSDS are not provided, and the products are not guaranteed to be Low VOC product. Website: http://www.greenseal.org/certification/environmental.cfm

Painting and Coating Systems must not exceed the current VOC and chemical component limits of the Green Seal's Standard GS-11 requirements.

09 91 13 - Exterior Painting
Wood:
Provide the following paint system over exterior wood surfaces:

Option 1:
1st Coat: Multi-Purpose Int/Ext Acrylic Primer B51W450 Series

2nd Coat: Super-Paint Exterior Acrylic Satin Enamel A89W1151 Series
3rd Coat: Super-Paint Exterior Acrylic Satin Enamel A89W1151 Series

Option 2:
1st Coat: DE Ultra-Grip Premium Int/Ext Primer UGPR 00-1
2nd Coat: DE Evershield Exterior Eg-Shel Paint EVSH 30-2
3rd Coat: DE Evershield Exterior Eg-Shel Paint EVSH 30-2

Ferrous Metal:
Provide the following finish system over exterior ferrous metal. Primer is not require on shop-primed items:

Option 1:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series
3rd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series

Option 2:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474
3rd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474

Zinc-Coated Metal:
Provide the following finish system over exterior zinc-c coated (galvanized) metal surfaces:

Option 1:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series
3rd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series

Option 2:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474
3rd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474

Aluminum:
Provide the following finish system over exterior aluminum surfaces:

Option 1:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series
3rd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series

Option 2:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474
3rd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474

09 91 23 - Interior Painting

Gypsum Board – flat finish on ceilings:
Provide the following system over interior gypsum board ceiling surfaces:
1st Coat: Premium Interior Acrylic Wall & Wood Primer B28W8111
2nd Coat: Pro Mar 400 0 VOC Interior Acrylic Flat B30W4651 Series
3rd Coat: Pro Mar 400 0 VOC Interior Acrylic Flat B30W4651 Series

Gypsum Board – satin finish:
Provide the following system over interior gypsum board:

Option 1:
1st Coat: Premium Interior Acrylic Wall & Wood Primer B28W8111
2nd Coat: Super-Paint Interior Acrylic Satin Enamel A87W1151 Series
3rd Coat: Super-Paint Interior Acrylic Satin Enamel A87W1151 Series

Option 2:
1st Coat: DE Ultra-Grip Premium Int/Ext Primer UGPR 00-1
2nd Coat: DE Everest Interior Eg-Shel Paint EVER 30
3rd Coat: DE Everest Interior Eg-Shel Paint EVER 30

Gypsum Board – semi-gloss finish:
Provide the following system over interior gypsum board:

Option 1:
1st Coat: Premium Interior Acrylic Wall & Wood Primer B28W8111
2nd Coat: Super-Paint Interior Acrylic Semi-Gloss Enamel A88W1151 Series
3rd Coat: Super-Paint Interior Acrylic Semi-Gloss Enamel A88W1151 Series

Option 2:
1st Coat: DE Ultra-Grip Premium Int/Ext Primer UGPR 001
2nd Coat: DE Everest Interior Semi-Gloss Paint EVER 50
3rd Coat: DE Everest Interior Semi-Gloss Paint EVER 50

Wood – paint-grade:
Provide the following paint finish system over new interior wood surfaces:
Option 1:
1st Coat: Premium Interior Acrylic Wall & Wood Primer B28W8111
2nd Coat: Pro Classic Interior Acrylic Semi-Gloss Enamel B31W1151 Series
3rd Coat: Pro Classic Interior Acrylic Semi-Gloss Enamel B31W1151 Series

Option 2:
1st Coat: DE Ultra-Grip Premium Int/Ext Primer UGPR 001
2nd Coat: Everest Interior Semi-Gloss Paint EVER 50
3rd Coat: Everest Interior Semi-Gloss Paint EVER 50

Ferrous Metal:
Provide the following paint finish system over ferrous metal:

Option 1:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series
3rd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series

Option 2:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474
3rd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474

Zinc-Coated Metal:
Provide the following finish system over interior zinc-coated (galvanized) metal:

Option 1:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series
3rd Coat: Pro Industrial HP Acrylic Semi-Gloss Coating B66W651 Series

Option 2:
1st Coat: Pro-Cryl Universal Acrylic Metal Primer B66W310 Series
2nd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474
3rd Coat: PPG Pitt-Tech Satin Industrial Enamel 90-474

09 93 13 - Exterior Staining and Finishing

Wood – Semi-transparent:
Provide the following semi-transparent stain finish system over exterior wood:

1st Coat: WoodScapes Waterbased Polyurethane Semi-Transparent Stain A15T5

2nd Coat: WoodScapes Waterbased Polyurethane Semi-Transparent Stain A15T5

Wood – Solid Color Stain:
Provide the following solid color stain finish system over exterior wood:

1st Coat: WoodScapes Exterior Acrylic Solid Color Stain A15W51 Series

2nd Coat: WoodScapes Exterior Acrylic Solid Color Stain A15W51 Series

09 93 23 - Interior Staining and Finishing

Wood – stain-grade:
Provide the following stain system over new interior wood surfaces:

1st Coat: Wood Classics Interior Oil Stain A49T804

2nd Coat: Zar Ultra Max Waterborne Oil Modified Sanding Sealer

3rd Coat: Zar Ultra Max Int. Waterborne Oil Modified Polyurethane Varnish - Satin

4th Coat: Zar Ultra Max Int. Waterborne Oil Modified Polyurethane Varnish - Satin

09 97 23 - Concrete and Masonry Coatings

Exterior Concrete Masonry Units:
Provide the following finish systems over exterior concrete masonry units:

Option 1:

1st Coat: Heavy Duty Int/Ext Acrylic Block Filler B42W46

2nd Coat: Super-Paint Exterior Acrylic Satin Enamel A89W1151 Series

3rd Coat: Super-Paint Exterior Acrylic Satin Enamel A89W1151 Series

Option 2:

1st Coat: Heavy Duty Int/Ext Acrylic Block Filler B42W46

2nd Coat: DE Evershield Exterior Eg-Shel Paint EVSH 30-2

3rd Coat: DE Evershield Exterior Eg-Shel Paint EVSH 30-2

Interior Concrete Masonry Units:
Provide the following finish system over interior concrete masonry block units:

Option 1:

1st Coat: PrepRite Int/Ext Acrylic Block Filler B25W25

2nd Coat: Super-Paint Interior Acrylic Satin Enamel A87W1151 Series

3rd Coat: Super-Paint Interior Acrylic Satin Enamel A87W1151 Series

Option 2:

1st Coat: PrepRite Int/Ext Acrylic Block Filler B25W25

2nd Coat: DE Everest Interior Eg-Shel Paint EVER 30

3rd Coat: DE Everest Interior Eg-Shel Paint EVER 30

09 97 26 - Cementitious Coatings

Exterior Concrete, Stucco and Masonry (Other than Concrete Masonry Units):
Provide the following finish systems over exterior concrete, stucco and brick masonry surfaces:

Option 1:

1st Coat: Loxon Int/Ext Acrylic Masonry Primer A24W8300

2nd Coat: Super-Paint Exterior Acrylic Satin Enamel A89W1151 Series

3rd Coat: Super-Paint Exterior Acrylic Satin Enamel A89W1151 Series

Option 2:

1st Coat: Loxon Int/Ext Acrylic Masonry Primer A24W8300

2nd Coat: DE Evershield Exterior Eg-Shel Paint EVSH 30-2

3rd Coat: DE Evershield Exterior Eg-Shel Paint EVSH 30-2

Interior Concrete and Masonry (Other than Concrete Masonry Units):
Provide the following paint system over interior concrete and brick masonry surfaces:

Option 1:

1st Coat: Loxon Int/Ext Acrylic Masonry Primer A24W8300

2nd Coat: Super-Paint Interior Acrylic Satin Enamel A87W1151 Series

3rd Coat: Super-Paint Interior Acrylic Satin Enamel A87W1151 Series

Option 2:

1st Coat: Loxon Int/Ext Acrylic Masonry Primer A24W8300

2nd Coat: DE Everest Interior Eg-Shel Paint EVER 30

3rd Coat: DE Everest Interior Eg-Shel Paint EVER 30

Division 10 - Specialties

10 11 16 - Markerboards

Standards: Porcelain steel magnetic white markerboards; 28 gage porcelain enameled steel surface; 50-year manufacturer's warranty.

10 17 00 - Telephone Specialties

Campus Standard: Emergency "Talk a Phone" Telephone stations shall be placed as needed with one touch direct "ring down" communications to the UPD central for emergency assistance. Stations should be placed in parking areas and exterior common areas.

10 21 13 - Toilet Compartments:

Campus Standards: Doors, panels and pilasters:
1-inch thick High Density Polyethylene (HDPE) resins.

Hardware:
Continuous mounting brackets, pilaster shoe, slide latch, "U" shaped pull handle.

10 28 13 - Toilet Accessories:

Closeout Submittal - Keys:

- Provide manufacturer name and model numbers for keys and coordinate closeout submittal with Section 08 06 05 – Key Schedule.
- Standards: Keys: Provide 2 keys per dispenser. Keyed to match CAT-74 key.

Toilet Tissue Dispenser:
Surface-mounted type; roll-in-reserve dispenser with hinged front secured with tumbler lockset; two standard-core tissue rolls, up to 5-1/4 diameter (1800 sheets); extra roll automatically drops in place with bottom roll is depleted. Example: Bobrick Contura B-4288.

Installation in Handicapped Accessible Stalls: Mount sufficient distance from hand rails to allow access to dispenser for refilling.

Sanitary Napkin Disposal:
Surface-mounted type; stainless-steel; seamless exposed walls with self-closing top cover with continuous hinge; locking bottom panel; removable reusable receptacle. Example: Bobrick Contura B-270.

Installation in Handicapped Accessible Stalls: Mount sufficient distance from hand rails to allow access to dispenser for refilling.

Seat-Cover Dispenser:
Surface-mounted type; stainless-steel unit with concealed opening at bottom for filling; 250-seat-cover capacity, minimum. Example: Bobrick Contura B-4221. Installation: 18 inches, minimum above top of flush valve.

If placed too close to the flush valve, dispenser, which is loaded from the bottom, cannot be refilled.

Installation in Handicapped Accessible Stalls: Mount sufficient distance from hand rails to allow access to dispenser for refilling.

Combination Seat-Cover Disposal / Toilet Tissue Dispenser / Sanitary Napkin Disposal, or Combination Seat-Cover Disposal / Toilet Tissue Dispenser:
Not allowed. The process to fill them is much more laborious and requires that a custodian have two keys at once to open the unit; he/she must pull out filled dispenser from one side to fill the opposite side and often results in toilet tissue falling in the toilet. Often the spindles may become obsolete and difficult to replace. Over time, the doors to the waste disposal become damaged and don't close or open properly. They become unsightly and we end up mounting the surface mounted dispensers next to the combination units. Units are difficult to repair and parts are difficult to come by.

Soap Dispenser:
Surface-mounted type: AeroFoam Soap Dispenser. Example: DEB SBS soap dispenser.

Installation: Mount sufficient distance from mirror edges to allow access to dispenser for refilling.

Key cores are typically located on the side of the dispenser.

Sanitary Napkin Vendor:
All-welded construction; seamless door with returned edges and secured by tumbler lockset; identification reading "Napkins" and "Tampons," brand-name advertising not allowed; 30 napkins and 20 tampons capacity, minimum; double-coin operation, 50 cents. Example: Bobrick B-3500x2.

Paper Towel Dispenser:
Surface-mounted type: roll pull towel dispenser; translucent plastic, 12¾ inches wide by 10¼ inches deep by 16½ inches high. Example: Kimberly-Clark IN-SIGHT, Model #KIM09990.

Campus prefers hand dryers over paper towel dispensers. Only specify paper towel dispensers if required for a lab or other special need.

Electric Hand Dryers:
Touch free surface mounted. Example: Dyson Airblade AB02

10 43 21 - Accessibility Evacuation Chair
Campus Standard: Manufacturer: Garaventa Accessibility, www.evacutrac.com,
Model: Garaventa Accessibility - Model Evacu-Trac CD7. No known equal.

Not Recommended: Manufacturer and Model: EZ Glide™ Evacuation Stair Chair
Reasons: Capacity and speed control -- Person's weight capacity is limited to 200 pounds with one person assisting, verses 300 pounds with Garaventa. No breaks to stop chair on stairs, verses speed control of 1.13 feet per second with Garaventa.

Division 11 – Equipment
Not Used

Division 12 - Furnishings
12 30 00 – Casework
Locks for Drawers and Doors:
Standards: Provide 2 keys per room of differently keyed drawers and doors where non-State keys are used.

Closeout Submittal: Keys: Provide manufacturer name and model numbers for keys and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

12 50 00 – Furniture
Locks for Drawers and Doors:
Standards: Provide 2 keys per room of differently keyed drawers and doors where non-State keys are used.

Closeout Submittal: Keys: Provide manufacturer name and model numbers for keys and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

For all furniture purchases (regardless of size) the University prioritizes the purchase/selection of furnishings that meet one more of the following environmentally preferable product standards:
- Recycled Content
- Forest Stewardship Council (FSC) Certified Wood
- Sustainable Agricultural Network's Sustainable Agriculture Standard Bio-based Materials
- Re-used or Salvaged Materials
- Extended Producer Responsibility Program Products
- GreenScreen v1.2 Benchmark

- Cradle to Cradle Certification
- Product Manufacturer Supply Chain Optimization
- Low Emission (VOC) Standards
- Regionally Produced Products (100 miles)

<u>Division 13 – Special Construction</u>
Not Used

<u>Division 14 - Conveying Equipment</u>
14 24 00 - Hydraulic Elevators
Signal Equipment:
Use non-proprietary and universally maintainable controls and software. Equipment and software necessary for long-term maintenance shall be available for purchase by any licensed elevator maintenance organization. (Examples include Car control station and in-car position indicators, Hall push-button stations, Lanterns and gongs, Hall position indicators, etc.)

Special Features:
Access for persons with disabilities, Fire emergency service, Medical emergency service, Earthquake requirements, and Emergency exit service.

Closeout Submittals:
Maintenance Tools: Provide complete list of specialty tools required for general maintenance, including programmable and diagnostic tools.

Quality Assurance:
Installer Qualifications: Elevator manufacturer installers, or firm with written approval from manufacturer and with five years, minimum of successful experience installing elevators similar to those required for the Project.

Elevator Maintenance Period:
- Maintain complete elevator installation for 12 months, minimum beginning at start of warranty period per Procurement and Contracting Requirements.
- Provide monthly systematic examination, adjustment and lubrication of elevator equipment. Repair or replace worn electrical and mechanical parts using parts produced by elevator equipment manufacturer. Exception: Greater frequency required if problems occur with elevator operation.
- Perform work during non-peak traffic periods.
- Provide 24-hour emergency call-back service with 2-hour response if the elevator is empty, and 1-hour response time for trapped occupants.
- Parts: Locally maintain an adequate stock of parts for replacement and emergency purposes.
- Maintenance Personnel: Qualified and approved by elevator manufacturer, and with 5 years, minimum experience.
- Maintenance Logs: Prepare logs of each visit and submit to University within 72 hours after visit.
- Extended Elevator Maintenance Proposal: Submit proposal to extend maintenance by three years.. Include stipulated sum with premiums due annually.

Manufacturer/Products:
Elevator: Capable of producing premium quality commercial hydraulic elevators and accessories to provide a complete installation.

Experience: Five years, minimum.

Option: Manufacturer shall either manufacture the major components, or shall maintain a service facility within California with complete service parts for a period of 20 years, minimum after completion of project.

Fabrication:
- Doors: Power operated hollow metal doors with track, rollers and frame; two-point suspension, nonmetallic sheaves; 3-inch diameter, minimum for car doors and 2-1/2-inch diameter for hoistway doors. Finish: Stainless steel.
- Hoistway Entrances: Formed metal with struts, hanger headers, fascia plates, toe guards and Underwriters' Laboratory labels. Finish: Match doors.
- Entrance Protection: Provide infrared door detectors, complying with ADA and applicable codes.
- Car Finishes Front Returns: Stainless steel with inset buttons, swing return panels.
- Pads: Provide wall attachment buttons with protective pads.

Operating Fixtures and Signals:
- Car Control Station: Provide service cabinet with built-in, hands-free communications via touch-button for automatic dialing from elevator car, and automatic answering for calling from outside. Provide door hold-open button.
- Hall Call Buttons: Provide illuminated mechanical hall buttons.
- In-Car Position Indicator: Provide above elevator hoistway entrance or above control panel for cars serving more than two stops.
- Lanterns: Provide wall mounted lanterns at center opening doors, above each hoistway entrance and jamb mounted lanterns at side opening doors with audible signal, one for up-travel, two for down-travel.

Miscellaneous Items:
- Battery Operated Emergency Lighting: Provide in car.
- Two-Speed Fan: Provide in car.
- Other Items: Provide as required by applicable codes.

Special Features:
- Access for Persons with Disabilities: Comply with CCR Title 24, and ADAAG.
- Fire Emergency Service:
  - Provide control circuits to comply with CBC.
  - Comply with CCR Title 24.
  - Provide three-position "on-off-bypass" key switch at ground floor.
  - Provide instructions for operation in ground floor cabinet.
  - Connect elevators to smoke detectors in lobbies.
- Medical Emergency Operation: Comply with CCR Title 24.
- Earthquake Requirements: Comply with applicable codes.
- Emergency Exit Service: Provide system to lower elevators to next lower floor level, or to Ground Floor, and doors to open without assistance in power outage.

Finishes:
Exposed-to-View Surfaces in Car, Machine Room and Hoistway Entrances:
- Stainless Steel: Number 4 finish (satin directional polish).

- Baked Enamel: Clean, degrease zinc-coated metal surface; one coat zinc oxide primer sprayed and baked; two coats semi-gloss enamel sprayed and baked.

14 28 16 - Elevator Controls

Standard: Open protocol, non-proprietary, specified with trouble shooting and operational software, and serviceable by all licensed elevator contractors.

Manufacturer: Motion Control Engineering. http://www.mceinc.com/.

Elevator Keys: Standard: Provide 3 keys, installing 1 in elevator pit.

Closeout Submittal: Keys: Provide manufacturer name and model numbers for keys and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

Division 21 - Fire Suppression

21 00 00 - Fire Suppression

Note: This document is a listing of University requirements and is not a complete specification. Consultant shall incorporate these requirements into a full technical specification.

General

Generally, the University requires that the consultant prepare a specification for a Contractor to design and construct (design-build) the fire protection systems to meet the specifications contained herein, including the various design and performance criteria delineated and to be responsible for the actual performance of the system according to these criteria. Verify with Project Manager that project will be design-build.

Quality Assurance

- Contractor to have C-16 license.
- Work shall be done in accordance with the N.F.P.A., the California Administrative Code, the Campus Standards including those of the Deputy State Fire Marshal, the California Building and Fire codes, and the requirements of the California State Fire Marshal. Plans shall be stamped by a California State licensed Fire Protection Engineer or mechanical engineer.

Submittals

- Provide standpipe and wet pipe fire sprinkler systems installation shop drawings, component submittal sheets and hydraulic calculations for approval of University representative and State Fire Marshal. University's Representative shall submit to Designated Campus Fire Marshal who is State Fire Marshal Representative. Do not begin installation until approval has been received.
- Shop drawings shall indicate standpipe and sprinkler assembly, wiring diagrams, including zone control and detection, flow, tamper switches and supervisory devices, inspector's test valve, fire department connection, post indicating valve, etc.
- Shop drawings shall indicate fire protection system connection to site water system including check assemblies.
- Contractor shall notify fire alarm subcontractor if waterflow or tamper switch or P.I.V. devices are relocated from locations shown on bid documents.

Design Criteria

- Contractor may with permission of the Designated State Fire Marshal, in lieu of the hydraulic calculations specified, use the prescriptive sizing method outlined in NFPA No. 13.
- The sprinkler system shall provide complete automatic wet-pipe sprinkler protection throughout the building including the attic space.
- The sprinkler system shall be zoned. Generally a zone should cover no more than a single floor.
- The University's Representative shall work with the Deputy State Fire Marshal to determine the Project hydraulic requirements (including occupancy classification), allowing for future expansion of system, system pressure variations, and the desired margin of safety.
- The following minimum pressure values should be used for sprinkler system design regardless of available site pressure values.
  - Static - 50 psi
  - Residual - 40 psi
  - Flow pressure - 35 psi at 1000 gpm
- Sprinklers shall be provided in combustible construction both below and above ceilings. Quick response heads shall be provided where required by the State Fire Marshal. On-off heads shall be provided in extra hazard areas.
- Backflow prevention - Conform to AWWA recommendations for protection of potable water system. Currently this consists of a single check valve assembly.
- Gongs/Bells - Use fire alarm system horns for annunciation of flow switches and monitoring of tamper switches. Do not install water gongs or bells.
- Fire protection water connections to a building shall be made with a Post Indicating Valve, check valve and fire department connection that are unique to that building. For very small clusters of buildings, an exception to this policy may be made by the Fire Marshal after receiving a written request by the Project Manager and after evaluation of the site plan.
- Access to main shutoff valves shall be through no more than one door which is located on the exterior of the building.

21 05 19 - Meters and Gages of Fire-Suppression Systems
Refer to CSU Building Metering Guide:
http://www.calstate.edu/cpdc/ae/gsf/documents/CSU_Metering_Guide.pdf

21 09 00 - Instrumentation and Control for Fire Suppression
Standards: Conduit: ¾-inch, minimum

21 12 00 - Fire-Suppression Standpipes
Buried piping and fittings: Cement-mortar lined, ductile iron pipe conforming to AWWA C-151, Class 200, 21/45 with Tyton or as approved, mechanical joints and fittings conforming to NFPA-24. Install pipe joints anchored with tie rods and restraints. Provide concrete thrust blocks at elbows and offsets conforming to NFPA-13 and NFPA-24. Other types of UL/FM reviewed piping including PVC C900 Class 200 may be used as approved by University's Representative and the State Fire Marshal. Thrust block installation shall be inspected by State Fire Marshal.

Above ground piping and fittings: Standard weight, Schedule 40, welded and seamless black steel, ASTM A-135 or A-53 pipe conforming to NFPA-13. Fittings for pipe sizes 2-1/2 inch and smaller shall be malleable iron, black, 150 lb. standard or cast iron, 175 lb. WP, screwed ends. Fittings for pipe sizes 3 inch and larger may be as above or weld fittings, ANSI 16.9, Schedule 40, flanged or welded fittings, cast iron, 175 lb. WP. Clamp type UL/FM approved fittings may be used on concealed piping where listed by the State Fire Marshal. Piping installed outdoors shall be galvanized.

Alternate piping system to be Victaulic, Grinnel, or equal grooved piping systems.
CPVC piping may be used in locations approved by NFPA 13 and the State Fire Marshal.

21 13 00 - Fire Suppression Sprinkler Systems
Preference: Dry-Pipe Sprinkler Systems:
Use where water could damage equipment in room.

Standards: Alarm Devices:
Water-Motor Operated Alarm, Electrically Operated Alarm, Water-Flow Indicator, Valve Supervisory Switch, and Indicator-Post Supervisory Switch.

Sprinkler and Valves
- Flanged 175 psi minimum water pressure rating.
- Gate (2-1/2 inch and larger): Iron body, bronze fitted 175 lb. WWP, BB, OS & Y, solid wedge flanged. Standpipe valves required shall be size 2-1/2 inch and shall have hose ends with caps and chains as required by the Deputy State Fire Marshal, bronze finish outdoors.
- Check (2-1/2 inch and larger): Iron body, bronze fitted 175 lb. WWP, bronze disc, horizontal swing, BB, flanged.
- Gate (2 inch and smaller): Bronze 175 lb. WWP, block pattern, OS & Y, solid wedge, screwed ends.
- Sprinkler Heads - Heads subject to vandalism shall be protected with wire guards. Heads installed in hazardous storage areas shall be on-off type and shall be submitted to Deputy State Fire Marshall for approval.
- Escutcheons: Install for exposed to view pipe penetrations through finished partitions, walls and floors.
- Pressure Gauges: Dial type with 0-300 lb. range, with brass case, glass face and attachment.

Backflow Prevention Assembly (where required):
BEECO-Hersey, Watts, Febco or equal, reduced pressure type assembly, UL/FM approved and City of San Luis Obispo Public Works Engineering Standards approved and listed with OS & Y gate vales and tamper switches.
Execution
- Piping runs shall be checked beforehand and with other trades to insure clearance. Provide maximum head room and run piping to maintain proper clearance for maintenance and to clear openings in exposed areas. Piping shall be run in strict coordination with mechanical ducts and equipment, structural, and architectural conditions.
- Cutting structural members for passing sprinkler pipe or supports will not be permitted except with approval from the University.
- Piping fittings shall not be located over electrical machinery or equipment, unless adequate insulating protection (16-gauge galvanized sheet metal pans with hemmed edges) is provided against drip caused by leaks.
- In general, shut-off valves shall be oriented so that the valve stem is horizontal. If valves controlling sprinkler systems are inaccessible or over an elevation of 7'-0"or greater above the finished floor or grade, install permanent ladders, clamped threads on risers, chain operators or similar devices to provide access to authorized personnel.

- Gate valves shall be installed with the position indicator located so it can easily be read, and the crank is within reach and easily operable.
- Supports - Provide seismic restraints for piping and equipment as required by NFPA and C.C.R. Title 24, whichever is stricter.
- Inspector's Test Connection - Provide test connections approximately 6 feet above the floor for the sprinkler system or portion of each sprinkler system equipped with an alarm device. Locate at the hydraulic most remote part of each system. Provide test connection piping to a location where the discharge will be readily visible and where water may be discharged without property damage. Provide discharge orifice of same size as smallest sprinkler orifice.
- Main Drain - Provide separate drain piping to discharge at set point outside building or to sight cones attached to drain of adequate size to readily receive the full flow from drain under maximum pressure. Main drain shall be piped to a sanitary sewer drain large enough to accept the full flow from the drain.
- Fire Department Connections - Provide connections approximately 3 feet above finish grade, of the University approved two-way type, with National Standard female 4" hose threads with plug, chain, and identifying fire department connection escutcheon plate. Connection shall be free standing (or wall mounted - coordinate with Fire Marshal).
- Flushing and disinfection - Disinfect potable portions of the Fire Protection System as required by the State Plumbing and Health Codes and the Deputy State Fire Marshal. Newly installed piping is to be kept isolated from the system until bacteriologically acceptable. If isolation is provided by a closed gate valve, pressure testing for leakage in the new facilities shall be conducted before bacteriological acceptance. The University shall designate method and sequence of connecting mains to minimize contamination danger.
- Tests - Tests shall be conducted at such times as directed by and in the presence of the University's Representative and the State Fire Marshal and shall be provided as required by NFPA-13, 24, and the State Fire Marshal. Piping shall be hydrostatically tested as required by NFPA-13 and NFPA-24but not less than 150 psi for two (2) hours.
- Operating parts, including valves, water flow detectors, and valve supervisory switches shall be tested for proper operation.
- Contractor to submit system certification forms per NFPA-13

Division 22 - Plumbing

22 05 00 - Common Work Results for Plumbing

Following is a brief description of the features which should be incorporated in the specifications for various plumbing materials and equipment. Where a specific product is stated with model number, that product should be used as the basis of design and listed as the first named in the specifications. Manufacturers listed without model number are intended to give the designer input on manufacturers whose products have been found to be acceptable.

General Quality Level:

Generally, plumbing material and equipment selected should be institutional grade. Long life and simple low cost maintenance are critical attributes for nearly all CAL POLY projects. Plumbing fixtures in public areas should also be selected to endure rough use and daily janitorial cleaning.

General-Duty Valves Requirements:

- Isolation Valves: Provide sufficient number of valves for ease of service, and to reduce inconvenience to users due to outages and draining of systems for minor repairs.
- Ball valve: Provide with stainless steel ball and stem from ½”-1.5” and from 2” up either bronze gate valve or epoxy coated resilient wedge iron body gate valve.

- Relief valves: Daylight to a conspicuous location. Plumb to sewer.

Identification for Plumbing Piping and Equipment Requirements:

- Identification Charts: Location should be reviewed and approved by Project Manager in consultation with Facility Services.
- Piping Service: Identification Tags: Provide with abbreviated legend on 1st line and pipe size on 2nd line. Locate to be visible from exposed points of observation. Where 2 or more pipes run parallel, place printed legend and other markers in same relative location.
- Valves Service Identification Tags: Provide with abbreviated legend on 1st line and valve service chart number on 2nd line. Identification Charts: Provide two (2) satin finished extruded aluminum frames with rigid clear plastic glazing; 8-1/2 x 11 inches, minimum for each chart. In addition, provide electronic copy of each chart.

Keys for Cabinets and Padlocks:

- Cabinets and Equipment: Provide 2 keys per panel. Coordinate with Section 08 06 05 – Key Schedule.
- Padlocks: Coordinate with Section 08 06 05 – Key Schedule.
- Closeout Submittal: Provide panel keys separated and labeled. Provide location, room number, quantity, manufacturer name and model numbers of keys, and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

Testing

- All new piping shall be tested prior to tie in to existing systems.
- Do not test against existing valves when connecting into an existing system. Provide a slip blind at the valve flange or other suitable isolation.
- Test gauges shall have 3” minimum dial, with oil fill and gauge cock. Gauge calibration shall be verified to the satisfaction of the University’s Inspector prior to commencing testing.
- Domestic and Industrial Hot & Cold Water Piping
- 150 PSIG test pressure for a period of 4 hours using a test medium of water.
- 200 PSIG test pressure for a period of 4 hours using a test medium of water for portions of the system which serve both the fire suppression system and the domestic water .
- Drain, Waste & Vent Piping Including Lab Waste: 10 feet of head for a period of 1 hour using a test medium of Water.
- Natural Gas Piping: 50 PSIG test pressure for a period of 4 hours using a test medium of air.
- Compressed Air Piping: 150 PSIG test pressure for a period of 4 hours using a test medium of air.
- Pure Water Piping Systems: 100 PSIG test pressure for a period of 4 hours using a test medium of purified water.22 05 19 Meters and Gages for Plumbing Piping
    - Water meters for domestic water should be nutating disk type sufficiently accurate to record the lowest expected flow (typically the flush of a single low flush toilet). Provide compound metering if necessary to accommodate a large range of flows.
    - Provide a permanently piped full size meter bypass on all domestic water services 3” and larger. Valving should be arranged so that the meter can be removed without a disruption of water service.
    - All water meters should record in cubic feet.
    - Meters for irrigation may be either nutating disk or turbine type depending on expected flow.

22 06 00 - Schedule for Plumbing

Boiler / Chiller Room Plumbing

Provide floor drains / sinks for relief valve drains and to allow for drain down of closed loop systems. Coordinate locations with equipment layout so that piped equipment drains will not be a trip hazard in service aisles. Select floor drains/sinks to contain spill over and splashing due to equipment drain discharge.

Provide make up water through a RP backflow preventer for each closed loop system.

Plaster / Sand Use.

Rooms where plaster or other sediment material is used shall be equipped with a dedicated above grade sand trap on the drain outlet of each sink. Typical applications include art studios using plaster, anthropology labs using plaster for bone castings, etc.

Dark Rooms

Provide RP back flow preventers for hot and cold water to dark room sinks.

Fume Hoods

For each plumbing fume hood plumbing service, provide a manual isolation valve and union at the piping drop above each fume hood so that the hood can be easily removed with minimal disruption of building piping.

Commercial Kitchens

Per City of San Luis Obispo Waste Water Discharge Ordinance, a large outdoor grease trap is typically required. Position grease trap to allow for easy access for pumping and minimal nuisance odors at occupied locations while pumping.

22 06 10 - Schedule for Plumbing Piping and Pumps

Piping: Standard: U.S.-manufactured.

22 10 00 - Plumbing Piping

All underground plumbing utility piping shall have a minimum of 12” sand encasement.
All underground plumbing utility piping shall have a minimum depth of 24” unless otherwise approved by the administrative authority after all existing utilities have been potholed by the contractor.

Design Documentation

Programming; For wet laboratories, studios, dark rooms, commercial kitchens, laundries, and other special use rooms requiring plumbing utilities to accommodate special functions, the Design Professional shall gather and document all information required to confirm Plumbing requirements as part of room programming. This information shall be included with the Design Development Submittal. Obtain and document the following information for each building room where special use plumbing services will be required.

- Room name
- Person requesting the plumbing services, Interviewer, Date
- Room use
- A list of required plumbing services. Include qualitative and quantitative data if available.

- Functions / processes which the required plumbing services are intended to accommodate. Provide sufficient information to ascertain the need for special plumbing provisions such as back flow prevention, pressure regulation, drain sediment traps, etc.
- A list of all equipment / apparatus which will require plumbing services. Include equipment manufacturer's data (connection size, use rate, etc.) if available.
- The above programming information may be omitted for rest rooms and residential kitchens where all required design data is covered by applicable codes.

Calculations:
The Design Professional shall include final plumbing calculations with the 50% Working Drawing submittal. These calculations shall include the following:

- The room programming documents covered above.
- All assumptions clearly stated. In particular, document assumptions that had to be made in order to proceed with the design in the case of insufficient data being provided by the users. The intent here is to flag assumptions which the University should verify as correct.
- Applicable code requirements
- Tabulated design criteria for each plumbing service by system including:
    - System
    - Location of Service (Room number)
    - Required flow quantity for each location including: water use fixture units, drainage fixture units, GPM flow rates for specialized equipment, natural gas load by both BTU/HR and CFH, CFH for air, vacuum & specialized gasses.
- Calculated data as required to select each piece of plumbing equipment including:
    - Hot water heater demand and storage capacity.
    - Sump pump sizing data.
    - Lift station sizing data.
    - Circulation pump sizing data.
- Pipe sizing data including: combined flow for each pipe run, developed length for each pipe run, selected pipe size based on stated criteria (sizing chart, pressure drop, fluid velocity, etc. This data may be either in tabulated form or rough pipe schematic.
- Rainwater leader calculations

Plumbing Working Drawings
General: Plumbing Working Drawings shall be sufficiently complete to assure high quality, fully functional plumbing systems, no contractual ambiguity, and also shall serve as a long term record of the building's plumbing systems. The following guidelines apply to all Plumbing

- No work shall be called out in a manner which is not contractually enforceable. Plumbing quality level shall be fully identified to assure fair competitive bidding as well as a quality installation.
- Design / build format for the plumbing section of the work as is sometimes used in residential plumbing construction shall not be used on Cal Poly projects. An exception to this guideline shall be the case of the entire building project being design / build format when approved by the University's Representative.
- All points of connection to existing piping systems shall be identified.
- All piping shall be sized.
- The routing of all piping shall be indicated. The indicated routing shall be verified to be feasible within the furring spaces indicated in the architectural drawings.
- All equipment and fixtures shall be located. Maintenance access space as recommended by the manufacturer shall be indicated on the drawings.

- All piping penetrations of the building shell shall be indicated.
- Concrete core drills of existing structures shall be identified.

Provide a Plumbing Legend covering all symbols and abbreviations used in the plumbing drawings in order to have a fully enforceable contract.

Provide a Plumbing Schedule covering factory assembled plumbing equipment and fixtures. The intent should be the easy determination of the plumbing equipment included in the project to encourage competition among comparable product suppliers. Note that when a product manufacturer's name and model are called out on the schedule to serve as the basis of design, the specifications must be also be coordinated so that the same manufacturer is the first specified. See Section 01630; Product Options and Substitutions of Cal Poly's standard Division 1 for further information.

Provide Plumbing Floor Plans to scale for each level where plumbing services are to be provided. Provide enlarged partial plans for congested areas such as public rest rooms where the normal scale plans will not adequately depict the work.

Sections to scale should be provided when the vertical arrangement of equipment and piping relative to other building components is critical for proper function or adequate maintenance clearance, as well as to assure the feasibility of pipe routing through tight spaces.

Plumbing Riser Diagrams: Plumbing riser diagrams shall be provided for multi-story buildings whenever the relative configuration of piping components cannot be depicted in the plan view alone without ambiguity. As a minimum, riser diagrams shall be provided in the Plumbing Working Drawings for the following systems: all drain waste & vent systems including lab waste, all hot water systems with circulation, all pure water systems, and any other system which includes forced circulation by pumping.

Plumbing Details: Plumbing details shall be provided as necessary to contractually assure a given level of quality. As a minimum, details shall be provided to cover all pipe penetrations of roofs, equipment supports, piping supports on roofs, backflow preventers, equipment installations where the relative position of plumbing appurtenances such as valves, unions, thermometer, drains, etc. are important to function, ease of service, and future replacement, pipe crossings of building expansion joints, pressure reducing assemblies, backflow prevention assemblies, roof drainage assemblies, floor drain assemblies, all pipe penetrations of exterior walls above and below grade.

Piping Diagrams (Schematics) shall be provided whenever the relative arrangements of plumbing components and equipment cannot be clearly depicted in the plans. As a minimum, piping diagrams shall be provided for in the Plumbing Working Drawings for water heating systems with circulation, pure water systems, and lift stations.

22 11 00 - Facility Water Distribution

All back flow preventers will be tested by a certified tester prior to the system by put in service.

Cal Poly owns and maintains the underground water distribution system throughout the campus.

Provide a domestic water pressure reducing valve and meter at each new building.

Provide a separate water meter and RP backflow preventer for irrigation at each new building.

Fire water should not be metered or reduced in pressure. Back flow prevention is not required for fire water.

DCDA assemblies are required by Cal Poly and the County of San Luis Obispo.

For new buildings to be connected to the campus water system, the anticipated additional water demand should be identified early during preliminary planning. This water demand should be submitted to the Principal Engineer for CAL POLY Facilities Planning and Capital Projects.

Improvements to the campus system may be required to accommodate the additional demand. The Principal Engineer shall identify a suitable point of connection to the campus system and what system improvements may be necessary to accommodate the new building.

Design Conditions:
Water System Pressure: The water pressure assumed for system pipe sizing and design shall be the lower of the following:

- 60 PSI
- The actual system pressure.

Since the water pressure in the campus water mains varies throughout the campus depending on elevation within four water pressure zones, the Plumbing Designer should request the water pressure from the University's Representative for each project. The University's Representative will provide the known pressure at a given point (usually a fire hydrant). The Plumbing Designer will be responsible for determining the pressure at the point of use and should take into account the change in elevation between the known reference point and the point of use.

22 11 16 - Domestic Water Piping
Acceptable Piping Materials

- Above Grade: Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder for joining 2" and smaller pipe. 15% silver brazing conforming to AWS classification BCuP-5 for joining 2-1/2" and larger piping.
- Below Grade: Type K hard drawn or soft copper tubing, with wrought copper sweat fittings, all connections brazed with 15% silver brazing conforming to AWS classification BCuP-5 , wrap pipe with 1 layer of 10 mil tape. Soft copper tubing with radius bends shall be used to minimize below grade fittings. (Note: Pipe layouts which place water piping under the building slab should be avoided whenever other alternatives are feasible.)
- Below grade, beyond building footing, 3" and smaller piping, alternate material: Schedule 80 PVC pipe with socket fittings and solvent cement joints or HDPE.
- Below grade, beyond building footing, 4" and larger: Cement-mortar lined Class 150 ductile iron pipe with bell and spigot joints and cast iron fittings. Specify restraining method: thrust blocks or restrained joints. Restrained joints should be called out for pipe configurations where thrust blocking will be difficult.
- Below grade, beyond building footing, 4" and larger, alternate material: Class 200 AWWA C-900 PVC with bell and spigot joints and cast iron fittings. Specify restraining method: thrust blocks or restrained joints. Restrained joints should be called out for pipe configurations where thrust blocking will be difficult.

Water Pressure Reducing Valves
Provide a pressure reducing valve (PRV) which will limit the supply pressure to 80 PSI at the domestic water entry point to each building. This PRV shall be provided on water services with pressure below 80 PSI as well as above in order to provide additional protection in the event of a loss of pressure control in the campus water mains.

For low system pressures where a PRV may cause an undesirable pressure loss an exception to this requirement may be granted upon obtaining approval from the University's Representative.
Provide compound PRV's where required to accommodate a large range of flows.
Fire water should not be reduced in pressure. Separate fire water from domestic water ahead of the PRV.

Valving:
Water piping systems shall be provided with manual isolation valves at the following locations:
- At branch connections to underground system mains.
- At all building points of entry.
- On both sides of piped in devices which may need to be removed for servicing including water meters, back flow preventers, pressure reducing valves, strainers, pumps, etc. The devices shall be removable without draining down the building system.
- Exception: On small branch lines (1 inch and less) where the quantity of building water is small, the valve on the downstream side of the device may be omitted.
- At each floor where branch piping connects to a riser.

Domestic / Industrial Hot Water
For water conservation, all systems shall be designed to provide near immediate hot water at fixtures by using either a hot water circulation pump or electric heat tape. An exception to this is small systems with a short distance (50 feet or less) between the water heater and the fixture. Verify with the University Representative whether heat tape or circulation will be used on a project by project basis.

For systems with a hot water circulation, verify that the pump is not over selected and that the pipe velocities due to circulation will be reasonably low. (Copper fitting failure due to high velocity scouring has been a problem). Grunfos Alpha Series Pumps are preferred.

Hot water mixing valves where required shall be Powers Intellistation. Intellistation Jr., or equal meeting the criteria below:
- Configurable on location. Does not require factory pre-programming or special software and laptop
- Controls water temperature to +/- 2°F during periods of low/zero demand
- 3.5° full-color, user-selectable touch screen display
- User programmable high-temperature sanitization mode
- In case of power or cold water failure, valve flows full cold for safety with manual override feature to set mixed outlet temperature
- Settings can be adjusted/monitored at the controller or remotely through BAS (Building Automation System)
- Displays pressure, temperature and flow/BTU data
- Pass code protected for security
- User programmable high temperature alarm

Domestic hot water heaters shall be gas fired, to minimize campus electrical demand as well as energy use cost.

Exceptions:

- Domestic hot water in buildings served by the campus central heating water system should be provided from a storage tank equipped with a double wall heat exchanger. (An exception to this may be allowed in the case of buildings with very low anticipated domestic hot water usage).
- Lavatories with small usage which are remotely located from a central system may be provided with hot water from a small electric storage type water heater with a maximum electrical demand of 1.5 kW. This exception shall not apply to shower usage. Instantaneous electric water heaters are not permitted due to high electrical demand.

Provide separate heating sources for domestic hot water and space heating water except for buildings served by the campus central heating water system.

Water heaters shall be installed in a sheet metal drain pan (smitty pan) with drain outlet piped to a safe location on the building exterior or floor drain.

Exception; Water heaters installed on a concrete floor slab where leakage would not damage the floor or building. Provide a floor area drain in the vicinity of the water heater.

All hot water piping shall be insulated.

Water systems shall be designed to prevent water hammer. Provide properly sized shock arrestors adjacent to all quickly closing valves including toilet flush valves, washing machines, dish washers, and solenoid valves. Provide a ball valve below each shock arrestor to allow for removal without system drain down. Provide access doors for all shock arrestors in concealed locations. Shock arrestor locations and sizes should be positively identified within the contract documents.

Disinfection

All potable water systems shall be disinfected and analyzed for bacteriological content. (Note: On Cal Poly projects, fire protection systems should also be disinfected, since they are installed without approved backflow protection) No fire system shall be installed without back flow protection.

Bacteriological analysis shall be completed by a third party laboratory approved by the Cal Poly Office of Environmental Health & Safety (EH&S). The laboratory shall be submitted via the University's Representative for approval by EH&S a minimum of 72-hours prior to conducting disinfection. Analysis results shall be submitted via the University Representative to EH&S to certify compliance with the specifications. Disinfection procedure shall be repeated should EH&S indicate that compliance with the applicable regulations has not been achieved. Plumbing shop representative will need to be given a copy of the results before the system can be put into service.

Provide an industrial water system for non-potable uses at all wet laboratories, faucets with serrated tip for hose connections, dark rooms, and all other locations using toxic or hazardous materials and requiring water service for uses other than drinking. Backflow protection shall be by a reduced pressure principal type back flow preventer.

22 13 00 - Facility Sanitary Sewerage

Cal Poly owns and maintains the sanitary sewer system throughout the campus.

For new buildings to be connected to the sanitary sewer system, the anticipated additional sewer load should be identified early during preliminary planning. This load should be submitted to the Principal Engineer for Cal Poly Facilities Planning and Capital Projects. Improvements to the campus system may be required to accommodate the additional load. The Principal Engineer shall identify a suitable point of connection to the campus system and what system improvements may be necessary to accommodate the new building.

Systems discharging to the campus sewer system shall be in compliance with the requirements of the City of San Luis Obispo Waste Water Treatment Plant. Mechanical systems discharging unusual wastes may require special provisions or may not be allowed. Triggers include; high or low pH, oil, grease, chemical contamination, biological contamination, and rain water to the sewer. Mechanical systems typically impacted include: commercial kitchen waste, elevator sump pump discharge, lab process waste, and water purification system waste. The Design Professional shall be responsible for confirming impacts to the project due to the City Waste Water Discharge Ordinance. Alternatives for compliance shall be discussed with the University's Representative. Negotiations involving specific project issues shall be channeled through the Cal Polly Office of Environmental Health & Safety (Director).

For wet laboratories, the method which will be used for dealing with contaminated liquid wastes should be determined early in the design process in consultation with the University's Representative and Cal Poly Environmental Health and Safety (EH&S) . Contaminated liquid waste from laboratories are not allowed to be poured directly into drainage systems discharging to the campus sewers. EH&S should be consulted for procedures for dealing with liquid contaminated wastes. Contaminated wastes are typically containerized and held in the laboratory for disposal by EH&S as hazardous waste. Lab waste piping systems are provided in wet laboratories only for disposal of non-hazardous liquids and as a precaution in case of an accidental spill.

Soil and Waste Piping within a building and outside within five feet (5') of the foundation, except where indicated otherwise, shall be No-Hub cast iron pipe and fittings, asphalt coated, free from defects, and shall conform to the requirements of CISPI Standard 301 ASTM A-888 or ASTM A-74 and manufactured by AB & I, Charlotte or Tyler. Fittings shall be up with Stainless Steel, Heavy-Duty No-Hub Couplings and shall be in compliance with ASTM C-1540 and ASTM C-564 Standards, except all above ground vent piping joints may be made up with Standard-Duty No-Hub Coupling in compliance with CISPI –310, and ASTM C-564 Standards. For fats, oils and grease waste, Blucher Pipe systems are preferred.

Complete soil/waste drainage piping will be provided to each domestic plumbing fixture and will discharge by gravity. Sanitary drainage ventilation piping will be provided to each domestic plumbing fixture or trap and will terminate at various locations on the roof.

Horizontal sanitary, waste and vent piping will be designed for a uniform grade of 2% (¼" per foot).

HVAC condensate drainage piping (CD) will be provided to each separate HVAC unit, such piping will drain to an indirect waste connection to the sanitary soil/waste system via either tailpiece connection at the nearest lavatory or sink or indirectly to a floor sink.

Floor drains will be provided in laundry rooms and all toilet rooms. Drains will not be located under the partitions and will be provided with clearance for service work.

Underground cast iron pipe will be wrapped with 8 mil polypropylene pipe wrap to protect against low pH soil on campus.

All floor drains will be served by a trap primer. The trap primers will be flush-o-meter valve flushing tube or lavatory p-trap type. (No mechanical type trap primers will be provided).

Wall clean-outs at the end of runs for main lines should be provided, where possible above rim height of highest fixture on that line of all floors.

Shower drains will be fitted with removable hair screens within floor drain for ease of maintenance.

Sink drains will be cleanable without disassembly of the associated trap with cleanout above rim height of sink on all floors.

Routing drainage piping overhead of electronic equipment, telecomm equipment and electrical equipment will be avoided.

Cleanouts will be the same nominal size as the pipe they serve; where they occur in piping eight inches and larger, six inches size will be used. All cleanouts will be accessible.

Cleanouts will be provided on all floors at the following locations:

- Horizontal offsets.
- End of soil, waste, water or storm drains more than five feet in length.
- Maximum of 50 foot intervals of horizontal runs within the building.
- Base of vertical sanitary stacks and storm drain leader lines.
- Each change of direction if the total aggregate change exceeds 90 degrees.
- Main sewer connection outside of building. A 2-way cleanout with dual access plugs (threaded bronze, thermoplastic, or PVC) will be provided at this location. Cleanout will be installed in round cast iron valve box marked “Sewer”.
- Above rim height of all sinks and lavatories on all floors.
- Above rim height of all urinals.
- Up stream of all doubles santees and combinations.

Acceptable Piping Materials

Above Grade Sanitary Waste and Vent Piping: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, neoprene coupler with stainless steel clamp and shield.

Below Grade Sanitary Waste and Vent Piping: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, extra heavy neoprene coupler with stainless steel clamp and shield (Husky, Clamp All, or equal)

Above Grade Vent Piping (Alternate materials)

- Schedule 40 galvanized steel pipe with recessed screwed cast iron drainage fittings.
- Type DWV copper with DWV solder fittings, 50/50 equivalent lead free solder.

Above / Below Grade Waste and Vent Piping (Alternate material for student housing buildings up to 3 stories only) Schedule 40 PVC DWV pipe and DWV fittings with solvent cement joints.

Waste vents shall be located a minimum distance of 30 feet horizontally from HVAC air intakes and occupied roof areas. Additionally, waste vents shall be extended with a support system to a level seven feet above the roof when the roof is occupied (such as in the case of roof mounted greenhouses). Waste vents shall not be located in confined roof wells where HVAC air intakes are located.

Provide accessible cleanouts at all levels of the building the same as required by code for first floor and lower levels.

Avoid sewage lift stations where possible. Gravity flow of sewage is preferred and is usually achievable due to the significant elevation change of most building sites on the campus. If a lift station is required, minimize the required capacity by segregating out upper level waste which can be gravity flow. Lift stations should be duplex pump type, powered by emergency power, with alarm contacts provided as an input to the building management system. Lift stations which handle rest rooms sewage shall be grinder type.

Floor drains shall be equipped with trap primers. Trap primer shall be accessible and equipped with a shut off valve to allow for servicing without system drain down.

22 14 00 - Facility Storm Drainage
Acceptable Piping Materials
Above Grade: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, neoprene coupler with stainless steel clamp and shield.

Below Grade Sanitary Waste: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, extra heavy neoprene coupler with stainless steel clamp and shield (Husky, Clamp All, or equal).

Insulate building storm water piping inside buildings to prevent building damage due to condensation on the pipe exterior.

22 15 00 - General Service Compressed-Air Systems
Acceptable Piping Materials;
Laboratory & Instrument Air Applications: Hard drawn ACR copper tubing with wrought copper sweat fittings. Tubing pre-washed, degreased and capped at both ends by manufacture. Fittings pre-washed, degreased and wrapped by the manufacturer. All joints brazed with 15% silver brazing conforming to AWS classification BCuP-5. Brazing shall be done with nitrogen purge to be prevent oxidation.

Alternate material for Shop Air Applications Only: Schedule 40 galvanized steel pipe with treaded 150 pound galvanized malleable iron fittings.

Alternate material for Shop Air Applications Only: Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder.

Building (house) compressed air shall not be shared with the building pneumatic compressed air system for HVAC controls. Building (house) compressed air and HVAC pneumatic control air shall be separate stand-alone systems complete with their own dedicated compressors.

For laboratory and instrument air, provide oil filter with automatic drain, and refrigerated air drier. For shop air applications, confirm requirements with users.

Valving: Compressed air systems shall be provided with manual isolation valves at the following locations:

- At each floor where branch piping connects to a riser.
- At the compressor.
- At branch connections to mains.
- Upstream of any equipment connected to compressed air.

Shop air systems typically require quick couple disconnect fittings for connection of flexible hoses. Confirm requirements with users.

Equipment:
Large house systems: Liquid ring type. (Nash or equal). Provide water saving circulation option with cooling water heat exchanger.

Requirements:
Drain Lines: Daylight to a conspicuous location. Plumb to sewer.
Hammer Arrestors: Install in a maintenance-accessible location for.

22 30 00 - Plumbing Equipment
Domestic Water System
Cold water systems will be sized using CPC flush valve curves, and hot water systems using flush tank curves. Equipment branches and main pipe size will be based on flow requirements without diversity. All plumbing valves will be located behind a lockable access panels when they are in a concealed location.

Hose bibs will be provided on the roof of each building. The hose bibs will be spaced 50 ft apart, maximum.

Chrome plated stops with gasket seats will be provided for sinks, lavatories, and wash basins when exposed to public view. A hose bibb with vacuum breaker will be provided under the lavatories in each toilet room. Exposed branch water supply piping in toilet rooms and custodial rooms will be chromium plated.

Water hammer arrestors will be provided in the wall, as required, behind a lockable access panel. Trap primers from lavatory tailpieces or water closet flush-o-meter valves for floor drains will also be provided. Isolation valves and unions at equipment connections will be provided.

The water piping will be designed to provide a minimum residual pressure at the most remote water closet of at least 30 psig. Pressure regulators will be installed to comply with California Plumbing Code.

- The maximum water pipe velocity will be 5 ft/sec. for hot water and 8 ft./sec for cold water.
- The minimum supply pipe size will be ½” for one plumbing fixture with a maximum flow of 1.25 gpm.
- Fixture pipe sizes will be as follows: 1-1/2” for a flush valve water closet, ¾” for a shower, ½” for sink and ¾” for hose bibs.

Domestic water piping within the buildings will be “Viega” pro-press or equal copper tubing and shall conform to ASTM B 75 or ASTM B88, Fittings shall be copper in conformance with ASTM B16.18, ASTM B16.22 or ASTM B16.26. O-rings for copper press fittings shall be EPDM.22 32 00 Domestic Water Filtration Equipment

Building Pure Water Systems
Building (house) pure water systems shall be designed to guarantee a water quality level of at least Lab Grade 3 per National Committee for Clinical Laboratory Standards (NCCLS) at all times. Individual laboratories requiring a guarantee of purer water shall install additional purification equipment in the laboratory as part of the building equipment (reference; Letter of Understanding dated 3-19-90, J. West; Physical Plant to K. McCaffrey; Natural Sciences).

The makeup water shall include pre-treatment consisting of a prefilter, a water softener, an activated carbon absorption filter, and a five (5) micron filter followed by treatment which shall consist of a reverse osmosis (RO) unit, mixed bed deionizer exchange service tanks, an ultra-violet light, and a 0.2 micron filter. Make up water capacity shall be determined in conjunction with sizing of the system storage tank.

The entire piping system shall be continually circulated. Dead legs shall be limited to drops to individual lab outlets. (Confirm maximum dead leg distance for each application) The entire circulating volume shall be exposed to ultra-violet light on each circulation. Provide dual stainless steel circulation pumps to allow for continued circulation in the event of pump failure.

The system shall include a storage tank sized to allow make up to occur over a 12 hour period while accommodating maximum anticipated use with 20 % excess capacity. Determination of maximum system use shall include a diversity factors which shall be determined with the University's Representative for each system.

The system shall include a side stream polishing loop which will continually circulate a portion of the water circulating through the system back through the mixed bed deionizers prior to return to the storage tank.

Provide a back pressure valve on the return piping to the storage tank set to assure adequate pressure at the highest most laboratory faucet.

Provide pressure gauges on both sides of all filters and pumps.

Building (house) pure water systems shall be monitored by the Building Management System to provide alarms to the Physical Plant watch stander at the campus Central Heat Plant. Alarms shall be generated in the case of Water quality (conductivity) out of range, low tank level, and no flow.

Acceptable Piping Materials:
Socket fused polypropylene prepared specifically for deionized service. Pipe shall be sterilized and capped immediately after production. Fittings, valves, and unions shall be sterilized and individually wrapped immediately after production. Continuous trough support system.

Schedule 80 PVC with solvent weld fittings prepared specifically for deionized service. Pipe shall be sterilized and capped immediately after production. Fittings, valves, and unions shall be sterilized and individually wrapped immediately after production. Continuous trough support system. Stainless steel could be used in some applications.

Confirm piping material with the University's representative on a case by case basis.

Valves shall be diaphragm type.

Valving:
Pure water piping systems shall be provided with manual isolation valves at the following locations:
At each floor where branch piping connects to a riser.

On both sides of piping elements which may need to be removed for servicing including filters, pumps, ultraviolet lights, mixed bed deionizers, and RO Units.

At branch connections to mains.
Provide provisions for system drain down to a floor drain including drain down of storage tanks.

22 35 00 - Domestic Water Heat Exchangers
Tube Bundles: Standard: Cupronickel Alloy. (Note: Standard grade copper tube bundles corrode within 3-5 years.)

Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

Equipment: Water Heaters: 10 Year Tank Warranty; State, A.O. Smith, or equal

22 40 00 - Plumbing Fixtures
Lavatory

- Student Housing: Heavy duty commercial grade single handle faucet; deck mounted with 4" centerset, chrome plated finish, vandal resistant handle, brass body, washerless design, rotating ball control mechanism with replaceable nonmetallic seats and stainless steel socket, vandal resistant 0.5 GPM flow restrictors spray outlet, left rod operated metal drain.
    - Preferred Manufacturer: American Standard
        - Lavatory – Wall Hung: American Standard Lucerne Model Numbers:0356.028.020, 20” by 18”, high back.
        - Lavatory – Countertop: American Standard Rondalyn Countertop Sink Model Numbers: 0490.156.020 round, vitreous china, self-rimming front overflow design.

    - Faucets: Preferred Manufacturer and Model: Moen CA83005 Non-mixing Electronic Faucet (0.325 GPM Vandal Resistant Flow). http://www.moen.com/search?search_scope=0&search_terms=CA8305
- Public: Heavy duty commercial grade single handle faucet; deck mounted with 4" centerset, chrome plated finish, vandal resistant handle, brass body, washerless design, rotating ball control mechanism with replaceable nonmetallic seats and stainless steel socket, vandal resistant 0.5 GPM flow restrictor spray outlet, and grid drain.
    - Campus Lavatory Standard (as appropriate):
        - 0.375 gpm Tamper Resistant Laminar Flow Control for Existing Lavatory faucet: Neoperl 40.7059.733
        - Moen 8413 F03 Comm 4” cc .35 GPM Lav Faucet w/o
        - Moen 8551 Sensored Lay Fct
    - Campus Basin Standard: Lavatory Basin with 4” Centerset Holes – American Standard Lucerne 0355.012.020
- Kitchen, Student Housing: Heavy duty commercial grade, single handle faucet, deck mounted with 8" centers, 8" swing spout, chrome plated finish, vandal resistant handle, brass body, rotating stainless steel ball control mechanism with replaceable non-metallic seats and stainless steel socket, vandal resistant 2 GPM flow restrictor aerator.

Shower Valves

- Student Housing: Institutional grade, pressure balancing, adjustable safety limit temperature stop control.
- Public: Institutional grade, pressure balancing, adjustable safety limit temperature stop control.
- Campus Standard: 1.5 GPM Low-Flow Pressure Balancing Surface Mounted Shower System – Hydapipe 1-901

Fixture Traps:
Chrome plated, 17 gauge cast bronze.

Faucets, Supplies, and Trim:
Mechanical Trap Primers:
Standard: Reduce or eliminate use.

Hose Bibbs:
3/4” hose connection, chrome plated, with vacuum breaker, loose key.

Water Meters:
Domestic Water: Nutating disk, magnetic drive, bronze case

Irrigation Water: Turbine type, magnetic drive, bronze case

Pressure Reducing Valve: Bronze with Teflon disk and diaphragm Valves, General Purpose Ball Valves, 2-1/2" and smaller. Provide threaded valve with downstream union.

22 42 13 - Commercial Water Closets, Urinals, and Bidets
Toilets

- Student Apartments:
  - Toilet shall be 1.28 GPF vitreous china round front bowl, with direct flow priming jet and reverse trap. Hydraulic performance exceeding ANSI A112.19.2 requirements.
  - Seat: All plastic, closed front with cover, stainless steel hinge.
  - Campus Preference:
    - 1.28 HET Floor Mount Floor Discharge – American Standard 2234.001.020 1.1 - 1.6
    - 1.28 HET Floor Mount Floor Discharge (ADA) – American Standard 3461.001.020 1.1 - 1.6
- Public Restrooms (including dormitories):
  - Toilets shall be 1.28 GPF, wall hung with support carrier, vitreous china, elongated, flush valve type exceeding ANSI A112.19.2 requirements.
  - Campus Preference:
    - American Standard 2257.101.020 with Afwall 1.1 – 1.6 (1.28)
    - 1.28 GPF HET manual Toilet (No Piston Valve) – Zurn Z6000 AV Model
    - 1.28 GPF HET Sensor Toilet (No Piston) – ZER6000 CPM Medel
  - Flush valve: 1.28 GPF
  - Flush Valves - Toilets (with infrared sensors):
    - Zurn ZER6000AV-ONE-CPM high efficiency flushometer valve
    - Campus no longer installs Flush Valves - Toilets without infrared sensors.
  - Seat: All plastic, open front elongated without cover; stainless steel hinge.

- Alternate toilet for single stall public rest rooms where light use is expected: Toilet shall be 1.28 GPF with hydro-pneumatic pressure assisted flush, vitreous china, floor mounted, elongated bowl, exceeding ANSI A112.19.2 requirements. Seat: All plastic, open front elongated without cover; stainless steel hinge.
- Toilet carrier manufactures Zurn, Josam, Mifab and J.R. Smith
- Not Recommended:
  - Toilet: Zurn Z5615 Series, Zurn Z5655 Series, and Zurn Z5665 Series 1.28 GPF Elongated vitreous china toilet. The toilet bowls are too shallow for proper water scouring when flushed.
  - Toilet Seat: Enlongated Open Front Seat – Bemis 1955SSCT

Urinals

Campus Standard, as appropriate for use/scope:

- Washbrook 0.125 GPF FloWise Washout Top Spud Urinal, or
- *Z5758* "The Retrofit Pint" 0.125 gpf Ultra Low Consumption Urinal System (21" – 24" Footprint)
- Zurn-ZER 6003AV-CPM Model Sensor Operated Battery Powered Flush Valve for ¾" Urinals

Trap Adapters:

Standard: Use when waste line has a tubular trap. Purpose is to allow for easier maintenance.
Preferred Manufacturer and Model: Arrowhead Brass Products, #6001A, 1-1/2" M.I.P. x 1-1/2" Tube Size – Brass Nut, Brass Ferrule with stop. Male in Female UPC-1003.2.
http://www.arrowheadbrass.com/products/

Flushometers:

Flush Valves: Standard: Zurn

22 45 00 - Emergency Plumbing Fixtures

Locations where hazardous materials are used or stored shall be provided with emergency eye washes in compliance with California Code of Regulations, Title 8 (CalOSHA) Section 5162, ANSI Z358.1 - 1990 and applicable ADA requirements.

Emergency eye washes and showers shall be supplied from the potable water system. Industrial water shall not be used to supply these fixtures.

Emergency eye wash and shower control valves shall remain open once activated until intentionally shut off.

Emergency eye washes and showers shall be located a distance no greater than 100 feet from the hazard and shall require no more than 10 seconds for an injured person to reach.

Floor drains shall not be provided at emergency showers in order to contain hazardous materials. The adjacent floor and walls should be designed to provide containment and also to be resistant to water damage. The plumbing design professional should coordinate this issue with the project Executive Architect.

22 47 00 - Drinking Fountains and Water Coolers

Water-Station Water Coolers:

- Hydration Station: Elkay Models LZSTL8WSSP and LZSTL8WSLPSurface Mount bottle filling station. Sanitary, touchless activation with auto 20-second shut-off

- WaterSentry Plus 3000-gallon capacity filtration system, certified to NSF/ANSI 42 & 53 (Lead, Class 1 Partiulate, Chlorine, Taste 7 Odor)
- Integrated silver ion anti-microbial protection
- Fill rate: 1.5 gpm; 1.1 gpm when connected to a remote chiller
- Laminar flow to reduce splash
- Real drain system
- Visual user interface display includes:
- Green Ticker counts bottles saved from waste
- LED visual filter monitor indicates when to replace filter
- Optional remote chiller can be installed within 15 feet of unit to deliver chilled drinking water.

22 62 00 - Vacuum Systems for Laboratory and Healthcare Facilities
Acceptable Piping Materials
Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder for joining 2" and smaller pipe. 15% silver brazing conforming to AWS classification BCuP-5 for joining 2-1/2" and larger piping.

Likely constituents of vacuum exhaust shall be determined and documented with the users to assist determination of vacuum exhaust location and degree of hazard due to discharged gasses.

Vacuum exhaust shall either be piped into the fume exhaust system or directly piped to a safe exterior location 7 feet minimum above the building roof. Vacuum exhaust discharge location should be carefully chosen so as not to create nuisance odors or hazards at operable windows, paths around the building, or other occupied exterior spaces. Vacuum exhausts which may create a nuisance or hazard should be piped to the same point as fume exhaust discharge for entrainment in the fume exhaust plume. House vacuum pump should be electrically interlocked with the fume exhaust fan in cases where the vacuum exhaust is directly connected to the fume exhaust system.

House vacuum pumps shall be protected by a drop out pot on the vacuum side of the unit.

Equipment:
Vacuum Pumps: Large house systems: Liquid ring type. (Nash or equal). Provide water saving circulation option with cooling water heat exchanger.

22 63 00 - Gas Systems for Laboratory and Healthcare Facilities
For Laboratory Buildings with a B Occupancy classification, verify that the quantity of stored gases classified as hazardous materials does not exceed the exempt quantities called out in the Building Code. H occupancy classification may otherwise be required.

Systems and storage for toxic gases shall be in compliance with the Uniform Fire Code including California Amendments in particular Section 8003, covering toxic compressed gases. Additionally, such systems shall be in compliance with the Toxic Gas Ordinance as written by the Western Fire Chiefs Association. Inform both the Executive Architect and University's Representative of requirements for gas storage cabinets, double containment piping, monitoring systems, and maximum quantity limitations in order to assure coordination among all disciplines.

Storage and piping concepts for laboratory gasses should be reviewed with Cal Poly Environmental Health and Safety via the University's Representative early in the design process. Issues include the possibility of asphyxiation (includes inert gasses) in confined spaces, as well as hazardous material concerns.

Laboratory gasses will not be used or stored in rooms having a HVAC system which returns air for recirculation.

Provide seismic restraint systems for all gas canisters and dewars.

Acceptable Piping Materials
Inert, Nontoxic Gases (nitrogen): Hard drawn ACR copper tubing with wrought copper sweat fittings. Tubing shall be pre-washed, degreased and capped at both ends by the manufacturer. Fittings shall be pre-washed, degreased and wrapped by the manufacturer. All joints shall be brazed with 15% silver brazing conforming to AWS classification BCuP-5. Brazing shall be done with nitrogen purge to prevent oxidation.

Other Gasses: Piping materials shall be selected on a project by project basis to meet specific program, compatibility, and regulatory requirements. Design for all toxic or hazardous gasses shall be reviewed and approved by EH&S.

Valving:
Laboratory gas systems shall be provided with manual isolation valves at the following locations:
At each floor where branch piping connects to a riser.
At the gas canister connections.
At branch connections to mains.
Upstream of any equipment connected to the laboratory gas system.

22 66 53 - Laboratory Chemical-Waste and Vent Piping
All wet laboratories, all fume hoods with cup sinks, and all other locations using chemicals which could damage standard DWV piping in the event of an accidental spill shall be provided with a lab waste and vent system. The lab waste and vent system shall be made of chemically resistant piping materials. The lab waste and vent system shall merge with the building sewer at a location outside of the building. Prior to combining the two systems, provide a lab waste sampling station (typically a manhole with a sump for sample collection).

Acceptable Piping Materials

Above and below grade Lab Waste and Vent Piping; (Alternate material); Chemical resistant coated cast iron pipe with matching mechanical joint DWV fittings. (Duriron or equal).
Above grade Lab Waste and Vent Piping; (Alternate material); Chemical resistant glass acid piping with glass DWV mechanical joint fittings. (Pyrex , Kymax, or equal).

Above and below grade Lab Waste and Vent Piping
Charlotte Pipe STM F 2618: Standard specification for CPVC Pipe and Fittings for Chemical Waste Drainage. Scope: this specification covers the performance requirements of CPVC pipe, fittings, and solvent cements used in chemical waste drainage systems. F493: Specification for solvent cements for CPVC pipe and fittings. Scope: This specification provides requirements for CPVC solvent cement to be used in joining CPVC pipe and socket fittings.

Lab waste vents shall be located a minimum distance of 50 feet horizontally from HVAC air intakes and occupied roof areas. Additionally, lab waste vents shall be extended with a support system to a level seven feet above the roof when the roof is occupied (such as in the case of roof mounted greenhouses). Lab waste vents shall not be located in confined roof wells where HVAC air intakes are located. Provide accessible cleanouts at all levels of the building the same as required by code for the first floor and lower levels.

Floor drains shall not be provided in laboratories using hazardous materials as precaution against such materials being discharged to the sewer system. The floor system should be designed to provide containment of such materials in the event of an accidental spill.

Division 23 - Heating, Ventilating, and Air Conditioning (HVAC)

23 00 00 - Heating, Ventilating, and Air Conditioning (HVAC)

Code Rules and Safety Orders:

Construction as called out on working drawings shall be in full accordance with the rules and regulations of all ten parts of the California Code of Regulations, Title 24 (The California Building Standards Code) per University administrative policy. Title 24 includes amended versions of the Uniform Building Code, Uniform Mechanical Code, Uniform Plumbing Code, National Electric Code and the Uniform Fire Code. Also included is the California Energy Code (Title 24, Part 6).

Additional government codes and regulations shall be complied with as applicable to the project.

The regional air quality management district with jurisdiction over the Cal Poly Campus is the San Luis Obispo County Air Pollution Control District ((805) 781-5912) with headquarters in San Luis Obispo, California. Mechanical systems regulated by the district include: fume exhaust systems, cooling towers, fuel burning equipment (depending on size & fuel type), paint spray booths, incinerators, and dust handling equipment. A “Permit to Construct” must be obtained from the District prior to beginning construction or modification on any system regulated by the District. The Design Professional shall be responsible for identifying project impacts requiring compliance with Air Pollution Control District regulations Alternatives for compliance shall be discussed with the Cal Poly Project Manager. The Design Professional shall provide Cal Poly with all information required to obtain a “Permit to Construct”. Permit applications, and related correspondence shall be channeled through the Cal Poly Environmental Health & Safety Office (Environmental Programs Manager).

Systems discharging to the campus sewer system shall be in compliance with the requirements of the City of San Luis Obispo Waste Water Treatment Plant (Industrial Waste Inspector (805) 781-7215). Mechanical systems discharging unusual wastes may require special provisions or may not be allowed. Triggers include high or low pH, oil, grease, chemical contamination, biological contamination, and rain water to the sewer. Mechanical systems typically impacted include commercial kitchen waste, elevator sump pump discharge, lab process waste, and water purification system waste. The Design Professional shall be responsible for confirming impacts to the project due to the City Waste Water Discharge Ordinance. Alternatives for compliance shall be discussed with the University’s Representative. Negotiations involving specific project issues shall be channeled through the CAL POLY Environmental Health & Safety Office (Director).

Basic Design Priorities:

Program Requirements / Design Guidelines / Government Codes: Mechanical systems shall be designed to meet the more stringent of: the user requirements as identified in the Program, the requirements these

Design Guidelines, or applicable government codes and regulations. The Design Professional shall be responsible for notifying the University's Representative of conflicting requirements. The University's Representative shall provide the final decision as to how conflicts will be resolved.

Ease of Maintenance / Operation: Mechanical systems shall be designed to be easily maintained and operated. Access and service space shall be key considerations of the design. Equipment requiring the movement of heavy items for service shall be located such that it is assessable by a hand truck from service vehicle parking locations. Equipment requiring frequent monitoring shall be located so as to be easily accessible from service vehicle parking locations.

Equipment Replaceability:
Mechanical equipment shall be located such that replacement is possible without building demolition. Buildings shall be provided with special provisions (oversized doors, removable louvers, access plates, etc.) as required to meet this requirement.

System Life:
Mechanical systems shall be designed to last the life of the building. Components shall typically be institutional grade. With approval of the University's Representative, an exception to this requirement may be granted for systems to support specialized applications with a short life expectancy.

Water Conservation:
Mechanical system shall be designed to minimize water use. CAL POLY has a long standing tradition of frugal water use. Design decisions which significantly impact water use shall be brought to the attention of the University's Representative. The University's Representative shall provide the final decision on such matters.

Utility Metering:
All mechanical utilities serving a building shall be metered. This shall include Water, Natural Gas, Campus Heating Hot Water, Campus Chilled Water and Chilled Water (when produced at a plant outside the building). See applicable Sections for metering specifics. A complex of residential buildings may be master metered with approval of the University's Representative.

Design Documentation:
Design documentation shall be provided for all mechanical systems to assure efficiency in the mechanical design process and to serve as a reference after construction.

Conceptual Design Report / Mechanical Systems: At the Schematic Design phase, submit a "Conceptual Design Report" for the project's mechanical systems, covering each anticipated mechanical system in the project. A key element of the report shall be a discussion of the alternative systems available to meet the identified needs. Design Development work for the mechanical systems shall not begin until the University's Representative has approved a conceptual alternative for each mechanical system. The

Conceptual Design Report shall include:
- Identification of the mechanical services required in each type of space included in the project.
- The base criteria to be used for capacity sizing each mechanical service to each space type within the project. Identify the source of the criteria (e.g., consultant recommendation, program requirement, building user interview, code requirement, campus standard, equipment manufacturer). State extent to which services will be oversized to accommodate future growth.

- A discussion of the pro and cons of the design alternatives available for each mechanical system.
- A recommended alternative for each mechanical system.
- Space allocation required for each recommended alternative.
- A description of access to be provided to each major piece of mechanical equipment.

Design Development Report / Mechanical Systems:
The Design Development submittal shall include a "Design Development Report" covering each mechanical system in the project. This report shall include:

- Room by room documentation of required mechanical services.
- A written description of each mechanical system covering system purpose, system type, utility inputs, and locations of all major equipment.
- The criteria to be used in sizing and laying out each mechanical system and the source of the criteria (similar to Conceptual Design Report).
- Rough schematics of the proposed systems indicating major equipment, points of connection to existing utilities, and the rooms served by each mechanical utility.
- A description of the required space to accommodate each major element of the proposed mechanical systems, including mechanical rooms, exterior mechanical yards, roof mounted mechanical equipment, mechanical shafts, and ceiling space provisions.
- A complete description of access for maintenance and replacement to each piece of mechanical equipment.
- Rough equipment sizing calculations.
- Catalog cut sheets on all major mechanical equipment.

Title 24 Building Envelope Plan Check Compliance Documents: The Design Professional shall submit Title 24, Part 6; Building Envelope Plan Check Compliance Documentation no later than the 50% Working Drawing Submittal. This documentation shall include applicable portions of forms; ENV-1, ENV-2, & ENV-3.

Title 24 Mechanical Plan Check Compliance Documents: Title 24 The Design Professional shall submit Title 24, Part 6; Mechanical Plan Check Compliance Documentation no later than the 50% Working Drawing Submittal. This documentation shall include applicable portions of forms; MECH-1, MECH-2, MECH-3 and MECH-4.

Final Mechanical Calculations & Equipment Documentation: Final mechanical calculations shall be submitted no later than the 50% Working Drawing Submittal. These calculations shall document the final basis of design for all mechanical equipment and systems. Title 24 plan check documentation may be used to supplement the calculations where applicable. Also included shall be manufacturer’s catalog cut sheet and sizing tables on all major mechanical equipment being used as the basis of design.

Working Mechanical Drawings
Maintenance access space as recommended by the manufacturer shall be indicated on the drawings for all mechanical equipment Designated access spaces shall include as a minimum: equipment access door swings, control panel access, tube pulls, coil removal space, fan shaft removal space, lubrication point access, and fire box access.

No abbreviation or symbol shall be used on the drawings unless included in the legend.
No work shall be called out in a manner which is not contractually enforceable.

Work shall not be called out in a design / build format on drawings intended for competitive bidding unless specifically approved by the University's Representative.

Mechanical Equipment shall have all important features specified. (Model number alone shall not be considered sufficient) Specifications for University Projects must be open to more than one bidder per Public Works Contract Law except in very specific situations. The specifications should be written in a manner which encourages competition among vendors of equivalent mechanical equipment yet precludes inferior products. To this end the quality level of all major features should be specified. See Part 1 of the Guidelines for further guidance on the method of calling out equivalent products for University contracts.

Natural and Mechanical Ventilation
2013 CMC Section 402.2 requires that naturally ventilated space must also be provided with mechanical ventilation unless the system meets any of the following exceptions:

- Exception #1: An engineered natural ventilation system where approved by the AHJ need not comply with Section 402.2.
- Exception #2: A mechanical ventilation system is not required where natural ventilation openings comply with the requirements of Section 402.2 and are permanently open or have controls that prevent the openings from being closed during occupancy.
- Exception #3: A mechanical ventilation system is not required where the zone is not served by heating or cooling equipment.

Energy Design Goals:
Energy Efficiency: Mechanical system shall be designed to minimize energy use. Cal Poly has a long standing tradition of frugal energy use. Design decisions which significantly impact energy use or cost shall be brought to the attention of the University's Representative. The University's Representative shall provide the final decision on such matters.

Depending on the project size, the ER is responsible for performing the energy model to demonstrate a project's energy efficiency to meet requirements of 2013 Title 24 Part 6 Energy Code, LEED Gold, and the CSU Energy Policy.

The following systems may be considered to help reach sustainability goals:

- Cogeneration
- Condensing boilers
- Solar thermal for domestic hot water
- Photovoltaic panels
- Phase Change Material
- Wall insulation
- Glazing performance
- Sunshades
- Variable air volume system for Make-up Air and Exhaust

Energy Modeling and Life Cycle Cost Analysis (LCCA)
Depending on the project size, the design team is responsible for the energy modeling and performing additional LCCAs associated with any of the sustainability options listed.

For LEED documentation, at least one year of data collection is required, consisting of:

- Total plant building and equipment electricity used, in kWh.

- Total plant building and equipment gas used, in therms for MBTU.
- Total heating energy output, in pounds or MBTU.

Mechanical Load Basis of Design

Outdoor Design Conditions

- Summer: 87°F db/65°F WB (ASHRAE 0.5%, San Luis Obispo, CA)
- Winter: 33°F (ASHRAE 0.2%, San Luis Obispo, CA)
- Elevation: 320 ft.
- California Climate Zone: 5

Indoor Design Conditions

- Lecture Halls: heat to 70°F, cool to 76°F
- Lab Space: heat to 70°F, cool to 76°F
- Other spaces: heat to 70°F, cool to 76°F
- Bedroom: Heating only, 70°F +/-2°F
- Living Room: Heating only, 70°F +/-2°F
- Study Room: Heating only, 70°F +/-2°F
- Laundry Room: Ventilation and Heating only
- Entry/Lobby (in dorm buildings): Ventilation and Heating only
- Admin Spaces: 75°F cooling, 70°F heating
- Elec. Rooms: Cooling only, 85°F
- Data/telecom rooms: Cooling only, 75°F
- Elevator machine rooms: Cooling only, 85°F

Indoor Humidity Control

- All areas, unless otherwise noted: no humidifiers required.

California Minimum Ventilation Criteria

- All areas: 15 cfm/person or 0.15 cfm/sq.ft. minimum, whichever is greater.
- Comply with 2013 California Building Energy Efficiency Standard.
- Comply with Chapter 4 of 2013 CMC.

Exhaust to Outdoors (Minimum Rates)

- Toilet rooms: 12 air changes per hour or 75 cfm/fixture, whichever is greater.
- Janitor closet: 100 cfm or 10 air changes per hour, whichever is greater.
- Shower rooms: 12 air changes per hour
- Main trash rooms: 12-15 air changes per hour
- Floor trash closets: 50 to 100 cfm per floor
- Kitchenette: 0.3 cfm per sf or hood exhaust rate, whichever is greater.

Building Envelope

- Glazing: glass/frame combination
- Typical vertical:
    - Description: double pane, low-e, and thermal break frame
    - U= 0.26 (center-of-glass summer daytime)
    - = 0.45 (with frame)
    - Solar heat gain factor = 0.27 (center-of-glass)

- Typical skylight:
  - Description: double pane, low-e, and thermal break frame
  - U= 0.26 (center-of-glass summer daytime)
  - U = 0.45 (with frame)
  - Solar heat gain factor = 0.27 (center-of-glass)
- Wall construction
  - Description: 4” metal framing at 16” O.C.
  - Insulation: R-13 batt insulation and 1” continuous rigid insulation at R-5 per inch.
  - Overall U-value = 0.105
- Roof construction:
  - Insulation: R-30
  - Overall U-value = 0.030

Internal Heat Gains

- Lighting (The D-B Team shall confirm and coordination the following values with the Lighting Designer):
  - Bedroom: 1 W/sq.ft.
  - Corridor: 0.6 W/sq.ft.
  - Restroom: 0.6 W/sq.ft.
  - Living Room: 1.1 W/sq.ft.
  - Study Room: 1.2 W/sq.ft.
  - Community Kitchen: 1.6 W/sq.ft.
- Receptacle power:
  - Bedroom: 2.5 W/sq.ft.
  - Living Room: 1 W/sq.ft.
  - Study Room: 1.5 W/sq.ft.
  - Light Recreation Space: 1 W/sq.ft.
- Occupant heat gain:
  - All areas: 250 Btuh sensible/250 Btuh latent
  - Light Recreation Space: 275 Btuh sensible/475 Btuh latent
- Electrical transformers: 2% loss
- Elevator machine rooms: refer to elevator drawings.
- Telecom/MDF/IDF rooms: refer to low-voltage drawings.

Ductwork Design Criteria (Maximum Allowable Values)

- Air velocities above these maximum values require acoustical treatment.
- Supply ducts:
  - Exposed in occupied spaces: 0.08” wg/100 ft and 1500 ft/min velocity
  - Above ceiling in occupied spaces: 0.08” wg/100 ft and 1800 ft/min velocity
  - In shafts adjacent occupied spaces: 0.20” wg/100 ft and 2000 ft/min velocity
  - In shafts adjacent unoccupied spaces: 0.30” wg/100 ft and 3000 ft/min velocity.
- Return/exhaust ducts:
  - Exposed in occupied spaces: 0.06” wg/100 ft and 1000 ft/min velocity
  - Above ceiling in occupied spaces: 0.06” wg/100 ft and 1500 ft/min velocity
  - In shafts: 0.10” wg/100 ft and 2000 ft/min velocity
  - In mechanical rooms: 0.20” wg/100 ft and 3000 ft/min velocity
- General building/toilet exhaust ducts: 0.10” wg/100 ft and 1500 ft/min velocity

Coils

- Maximum face velocity: 450 fpm
- Maximum fins per inch: 12
- Maximum air pressure drop – heating coil: 0.25" wc
- Maximum tube pressure drop: 10 ft. wg.
- Minimum tube pressure drop: 2 ft. wg.
- Minimum tube rows – heating coil in air handler: 1
- Minimum tube rows – reheat coil in VAV terminal unit: 2

Hydronic Piping Design Criteria

- Max water pressure drop: 4 ft.wg./100 ft.
- Max water velocity: 6 ft/sec
- Max allowable water velocity: 10 ft/sec in mechanical rooms
- Provide shut off valve for isolation of major areas, at each piece of equipment, and each air handling system.

Heating Hot Water Temperatures

- Plant heating water supply/return temperatures: 180°F/140°F
- Radiant heating panels supply/return water temperatures: 180°F/160°F
- Fin-Tube Baseboard Heaters supply/return water temperatures: 180°F/160°F

Redundancy Assumptions

- Boilers: provide N+1 boiler redundancy
- Heating water pumps: provide N+1 pump redundancy
- If Cogen is used, provide full back-up capacity with boilers.

Metering and Controls

- Provide BTU meter at each building to monitor heating hot water usage and DHW usage.
- Building Automation System (BAS) DDC control system for complete system optimization of major equipment. Campus Standard is Siemens Apogee.

Economizers:

Evaluate life cycle costs for using economizers on units smaller than 75,000 Btuh, especially on 24/7 loads.

24-Hour Cooling:

- Provide dedicated HVAC equipment/systems to spaces requiring 24-hour cooling (allowing unoccupied parts of the building to be shut down).
- Types of Spaces: Review program to determine spaces requiring 24-hour cooling (such as main data frame rooms, server rooms, telephone equipment rooms, elevator equipment rooms, labs and other areas containing special equipment).
- Equipment Function: Based upon critical functions of equipment, evaluate and analyze cooling options and redundancy -- DX Units verses Central Plant Chilled Water verses Hybrid CHW/DX units.

Equipment Access:

- Provide clear access to equipment for inspection and maintenance.
- Access Panels: Provide adequate, unobstructed access to equipment and ceiling access panels.

- Volume/Balancing Dampers: Locate access panels within 18 inches of service point.
- Pumps in Mechanical Rooms: Orient pumps with motor end towards center of room, and pump end towards wall (making sure suction strainer is accessible).
- Building Isolation Valves: Valves need to be raised, accessible and housed in a covered box, flush with ground level.

23 05 13 - Meters and Gages for HVAC Piping
Requirements: Refer to CSU Building Metering Guide. Website:
http://www.calstate.edu/cpdc/ae/gsf/documents/CSU_Metering_Guide.pdf

Intent: Metering to meet the performance-based LEED NC/EB Energy & Atmosphere Credit – Building-level energy metering standards for ongoing accountability and optimization of building energy and water consumption performance over time.

Refer to and coordinate with Section 25 00 00 – Integrated Automation.

Metering Equipment: Allow Campus the ability to identify problems and achieve improved system performance. Select meters for future connection to a remote-read automated metering network.

BTU meter shall be installed at each building and for each system (DHW and HHW). Controlotron, EMCO ST30 Ultrasonic, or approved equal.

Flow Meters:
- Function: Non-invasive flow measurement with a transmitter with 2-line backlit display with 4-button keypad. Connect to Campus Energy Management System (EMS) by Siemens.
- Execution*:* Locate on straight run of pipe with no flow interruptions such as valves or direction change. Installed location shall typically have straight uninterrupted pipe for 10 diameters upstream and 5 diameters downstream. Consult manufacturer's installation instructions for specific installation requirements and other piping configurations.
- Website: http://www.emcoflow.com/em_prods/sono_trak.aspx

Keys for Cabinets and Padlocks:
- Cabinets and Equipment: Provide 2 keys per panel. Coordinate with Section 08 06 05 – Key Schedule.
- Padlocks: Coordinate with Section 08 06 05 – Key Schedule.
- Closeout Submittal: Provide panel keys separated and labeled. Provide location, room number, quantity, manufacturer name and model numbers of keys, and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

23 05 53 - Identification for Plumbing and HVAC Equipment:
Volume/Balancing Dampers:
Plan Notations: Clearly identify on plans.
Field Identification: Provide visible, accessible identification. Example: Bright-colored streamers.

Property ID Labels for Equipment:
Submittals:
List of Equipment for Property ID Labels:
- Submit list of equipment for Property ID Labels, including location, function, equipment manufacturer's name, and model and serial numbers for verification by Engineer.

- After acceptance by Engineer: Submit hard and electronic copies for assignment of bar-coding identification numbers by Campus Facility Services.

Samples: Submit two Property ID Labels.

Closeout Submittals: Project Record Documents: Record actual locations of labeled items; include bar-code numbers, and provide as-built electronic copy of list of equipment with location, function, equipment manufacturer's name, and model and serial numbers.

Pre-Installation Meetings:
Convene minimum two weeks prior to commencing work of this section.

Labels:
Description: Property ID Labels.
- Material: Anodized Aluminum.
- Size: 2 x 0.875 inches.
- Attachment: Adhesive backed.
- Message:
  - Printed identification:
    - First Line: "[University Name] MEP"
    - Second Line: "Facility Services"
  - Bar code: Code numbers shall be provided by the University Facility Services via the Trustees Representative.

23 06 20.13 - Hydronic Pump Schedule
Flex Connectors / Expansion Joints:
Campus Standard: Stainless steel T321 bellows type expansion joint with guide rods/travel stops.

Campus Preferred Manufacturer: DME Incorporated model 551R fixed flange bellows type pump connector, type 321 stainless steel, ANSI 150 lb flanges with raised face, integral tie rod travel limits. Rated for operating pressure of 150 psi @ 800 degrees F. Website:
http://www.dmeexpansionjoints.com/pump-connectors.htm

Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

23 09 13 - Instrumentation and Control Devices for HVAC
Temperature Sensors /
Resistance Temperature Detectors (RTD's):
- Standard: Siemens 1000 Ohm Platinum RTD, Model Number PTM6.2P1K.
- Exception: Thermistors are required in some zone applications such as VAV boxes, and may only be used in those areas.
- Function: Use for air handlers, Hot Water (HW) and Chilled Water (CHW). Do not install in piping tee. Connect to Energy Management System (EMS) by Siemens.

Differential Pressure Sensors:
Resistance Temperature Detectors (RTD's):
- Siemens Sitrans 7MF4433 with 3-valve manifold and local display.

- Standard: Emerson Process Management, Rosemount Model #1151 Pressure Transmitter. Connect to Energy Management System (EMS) by Siemens. Specify pressure range for 150% of maximum expected differential pressure.
- Function: Measure differential pressure for monitoring and control of variable speed pumps.
- Website: http://www.emersonprocess.com/rosemount/Products/Pressure/m1151.html

Motor Controls:
Variable Frequency Drives (VFD's):
- Campus Standard: ABB ACH550. Enclosure: UL (NEMA) Type 12. Integral disconnect. No bypass. No known equal.
- Function: Connect to Campus Energy Management System (EMS) by Siemens as a Field Level Network (FLN) device to provide direct communication (not hardwired start/stop/status). Proof through drive and via current switches.

23 09 23 - Direct-Digital Control System for HVAC:
Room Numbering Standard:
Siemens shall confirm room numbering with Project Manager prior to initiating work. Room Numbering Standard are established that must be used on all documents developed for a project. The Architect shall submit the floor plans showing the proposed room numbers to the Project Manager for review and approval. It is critical that the room numbering scheme be established and finalized before Siemens begins creating the point database, as room numbers are integrated into the point names in the Energy Management System. Renumbering of rooms late in the design process can create significant rework by Siemens. .

Acceptable Control Contractor:
Campus Standard: Siemens Apogee (no substitutions).

System Description:
Connect to: Existing Energy Management System by Siemens.

Function: Control system 100 % DDC (Direct Digital Control). Modular Stand-along DDCP (Direct Digital Control Panel) capable of future BAS (Building Automation System) architecture with low/medium speed communication networks. BAS architecture to consist of communication network, user workstations, modular design DDCP with addressable and modifiable points from user workstations, or from master DDCP user interface panels; and be expandable for addition of hardware and software without removal of existing DDCP, sensors, actuators, and communication networks. Support and be capable of additional workstations, to systems maximum node capacity. System intelligence to allow user workstation(s) to program controls, perform analysis on field data, and generate maintenance and operational reports; and provide permanent program and data storage.

Communication: Connect control panels to Campus EMS network via Ethernet. Provide each panel with an adjacent 2-port Ethernet face plate. Coordinate with Campus IT for available ports and IP addressing.

Ethernet cable length (from Ethernet switch in IT closet) shall be 328 feet (100 meters), maximum.

Field Panels:
Campus Standard: Siemens Apogee PXC Modular, flemible Direct Digital Control (DDC) supervisory field panel, and Siemens Apogee IX-I/O power and communication modules. No substitutions allowed.

Application Specific Controllers: Field level network devices (VAV boxes, fume hoods, unit ventilators, air conditioning units, heat pumps, etc.) shall be controlled by a TEC (Terminal Equipment Controller) using the appropriate application.

Control Valve and Damper Actuators: Actuators shall be electronic.

Note: Pneumatic Actuators are Not Permitted.

23 09 93 - Sequence of Operation for HVAC Controls:

General:

Control sequences are generally worded for direct acting type of control. Controls shall be direct or reverse acting to match the fail position of the actuator or device.

EMS graphics for building shall include two dynamic link buttons—one will list sequence of operation in layman's terms, and the other, a legend, will list the point ID and corresponding definition.

Setpoints and offsets shall be field adjustable at the controller for electronic controllers via PC communications port and from an operator's work station for DDC systems.

Controls may use Proportional+Integral or Proportional+Integral+Derivative functions to maintain offsets specified, except as noted. Proportional control may be used for space temperature control and level control. Proportional+Integral+Derivative control shall not be used for flow control.

Label function of all control panel alarms switches, indicators, and manual control devices. Label units of analog indicators.

Alarms:

Safety limit alarms shall be manually reset at sensor location. Other alarms may be automatically reset when measured value returns to its normal range or setpoint.

Provide separate discrete inputs for each safety limit control loop to annunciate alarm conditions at workstations.

Analog inputs shall have low/high alarm limits assigned to indicate non-normal operating conditions. Alarm limit values shall be adjustable by operator. Unless noted otherwise, use the following alarm limit values:

Water/liquid system temperature:	+ 3°F from setpoint for 1 minute
Water/liquid system pressure:	+ 10 PSIG from setpoint for 1 minute
Water/liquid system flow:	+ 10% from setpoint for 1 minute
Water/liquid level:	+ 10% from setpoint for 1 minute
Water/liquid pH/conductivity:	+ 10% from setpoint for 1 minute
Air system temperature:	+ 3°F from setpoint for 3 minutes
Air system flow:	+ 10% from setpoint for 1 minute
Space temperature:	+ 1°F from setpoint for 5 minutes
Space pressure:	+ 0.01" W.C. from setpoint for 1 minute

Alarms shall be logged into an alarm event database and shall be annunciated at designated work stations. Type of annunciation shall vary with alarm level type. (Level 1, 2, or 3. Refer to definitions.) Provide software variable to easily change level type for each alarm point.

Control Sequence: Perform as follows.

General Unit Operations:

Lead/Lag Device Operation:

The lead device is defined as the device which starts before any other system devices and stops after all other devices have stopped. The lag device(s) start after the lead device and stop before the lead device. If multiple lag devices exist, the first lag device shall start first, followed by the second, third, etc. Stopping shall occur in reverse order.

The device(s) shall be switched between lead and lag based on runtime. The device with the lowest runtime shall become the lead device when all devices are stopped. The device with the highest runtime shall become the lag device if all devices are operating. If the lead device runs continuously for 1 month, the lag device with the lowest runtime shall be started and the lead device shall be stopped. This lag device shall then become the lead.

On a failure of the lead device as measured by feedback instrumentation (e.g. a pressure switch across a pump, flow switch, current switch, or a motor starter contact), the lag device with the lowest runtime shall become the lead and started. The original lead device shall be disabled and a critical alarm shall be generated.

The operator shall have the ability to override operation through software in the control system.

Variable Frequency Drive (VFD) Powered Equipment:

Variable Frequency Drives (VFD) shall start at the minimum speed setting and shall ramp to the control point. Unless otherwise noted, the VFD shall ramp up and down full scale over 15 seconds. The VFD shall not be allowed to operate at less than 15 Hz. If an electrical disconnect is connected between the VFD and the equipment, the VFD shall not be allowed to operate if the disconnect is open.

Domestic and Industrial Hot Water System:

Domestic and industrial water temperature shall be maintained at 120ºF, field adjustable by immersion thermostat modulating fail closed control valve(s) in water supply.

Provide single control valve as indicated on drawings.

Building Hot Water System:

General:

Hot water at 180°F will be supplied, via underground piping, from the central utility plant.

- Note: Building isolation valves need to be raised, accessible and housed in a covered box, flush with ground level.

The existing outside air temperature (OAT) sensor for the BAS shall be used to measure OAT. When OAT is great than 60ºF, all building hot water valves shall close and hot water pumps shall be locked out.

Building hot water pumps are in series with central plant hot water pumps. The following sequence assumes that the central plant pumps are designed to provide approximately 10 psi pressure differential at the building interface. The building hot water pumps, as described by the following sequence, shall operate only when this pressure difference cannot maintain adequate flow through the hot water coils in the system.

Heating Water Pumps:

Refer to flow diagram.

Hot water pump(s) shall be started and stopped automatically through building automation system.

When more than one pump is provided:

- BAS shall equalize run times on pumps.
- Provide software switch with override to select operating pump. Whenever operating pump fails as proven by LAN communication with VFD, standby pump shall start after 30 second time delay and failed pump shall be de-energized.

Pressure differential sensor/transmitter shall be installed across the hot water supply and return piping at a location representative of the furthest heating coil.

The following start-stop and pump speed control sequence is designed to minimize pump speed and energy consumption while assuring that the flow requirements are met for system coils. Setpoints and timing intervals shall be field-adjustable.

Normal pump VFD modulation: The control system shall monitor the required position of each hot water coil control valve in the system, as determined by the signal to the valve, and select the valve that is second most open. When building hot water pump is operating, the control system shall modulate the pump speed so that, if possible within the limits of the pump capacity, the second most open hot water valve in the system is between 90% and 95% open.

Pump Start: If the building hot water pump is not operating, and if any hot water valve is open more than 95%, the hot water pump shall be started.

Pump Shut Down: If the following 2 conditions exist for longer than 15 minutes, the pump shall be stopped: 1) The pump is at its minimum speed, and 2) The second most open hot water valve is less than 40% open, or as determined by the controls contractor to prevent pump short cycling.

Building Chilled Water System:

General:

Chilled water at 40ºF will be supplied via underground piping from the central utility plant.

- Note: Building isolation valves need to be raised, accessible and housed in a covered box, flush with ground level.

Each building shall have its own OAT sensor. When OAT is below 50ºF, building chilled water valves shall close and chilled water pumps shall be locked out.

Building chilled water pumps are in series with central plant chilled water pumps. The following sequence assumes that the central plant pumps are designed to provide approximately 6 to 8 psi pressure differential at chilled water coil. The building chilled water pumps, as described by the following sequence, shall operate only when this pressure difference cannot maintain adequate flow through the chilled water coils in the system.

Chilled Water Pumps:

Refer to control diagrams.

Hot water pump(s) shall be started and stopped automatically through building automation system.

When more than one pump is provided:

- BAS shall equalize run times on pumps.
- Provide software switch with override to select operating pump. Whenever operating pump fails as proven by LAN communication with VFD, standby pump shall start after 30 second time delay and failed pump shall be de-energized.

Pressure differential sensor/transmitter shall be installed across the chilled water supply and return piping in a location representative of the furthest cooling coil in order to monitor system.

The following start-stop and pump speed control sequence is designed to minimize pump speed and energy consumption while assuring that the flow requirements are met for system coils. Setpoints and timing intervals shall be field-adjustable.

Normal pump VFD modulation: The control system shall monitor the required position of each water hot coil control valve in the system, as determined by the signal to the valve, and select the valve that is second most open. When building chilled water pump is operating, the control system shall modulate the pump speed so that, if possible within the limits of the pump capacity, the second most open hot water valve in the system is between 90% and 95% open.

Pump Start: If the building chilled water pump is not operating, and if any chilled water valve is open more than 95%, the chilled water pump shall be started.

Pump Shut Down: If the following 2 conditions exist for longer than 15 minutes, the pump shall be stopped: 1) The pump is at its minimum speed, and 2) The second most open chilled water valve is less than 40% open, or as determined by the controls contractor to prevent pump short cycling.

Reheat Coils:

Space thermostat shall modulate control valve (FO) to each coil to maintain space setpoint temperature.

Air Terminal Devices:

General:

Air terminal devices to be included in all equipment schedules.

Space temperature and ventilation control shall be DDC with electric actuation.

Control Contractor shall furnish unit controls and actuators to air terminal device manufacturer for factory installation and adjustment.

System shall be capable of remote setpoint change of the following parameters through the DDC panels and remote PC:

- Space temperature.
- Maximum and minimum flow rate for each supply and exhaust air valve.

Space temperature setpoint shall be capable of adjustment between 55°F and 85°F with a throttling range of 10°F between full heating and full cooling.

Where heating and cooling devices are operated in sequence, each device shall have a separate space temperature setpoint. Cooling setpoint shall always be greater than the heating setpoint, with an adjustable deadband between.

Reheat coil control valves in laboratory spaces shall be normally closed. Those in non-laboratory spaces shall be normally open.

Variable Volume Supply With Reheat - (Sequence 1):

Supply air valve shall be pressure independent.

Space temperature sensor shall modulate in sequence supply air valve and reheat coil control valve to maintain space temperature setpoints. On a drop in space temperature below cooling setpoint, supply air valve shall modulate from maximum to minimum air flow quantities as scheduled. On a continued drop in space temperature below heating setpoint, reheat coil control valve shall modulate open to maintain space setpoint temperature. On a rise in space temperature, the reverse shall occur.

Constant Volume Reheat With Supply / Exhaust Air Tracking - (Sequence 2):

Supply and exhaust air valve shall be pressure independent.

Air flow sensors shall measure both supply and exhaust air flow rates. Air flow sensor shall modulate electric actuator on supply air valve to maintain constant volume air flow rate as scheduled. Exhaust air valve shall modulate in response to supply air valve to maintain scheduled cfm differential between measured supply and exhaust air flows.

Where multiple supply and/or exhaust air valves are required to meet air flow requirements, associated air valves shall be modulated in parallel from the same control signal.

Space temperature sensor shall modulate reheat coil control valve to maintain space temperature setpoint. Where reheat coils are not provided, this requirement does not apply.

Variable Volume Supply With Cooling Only (No Reheat) - (Sequence 3):

Supply air valve shall be pressure independent.

Space temperature sensor shall modulate supply air valve to maintain space temperature setpoints. On a drop in space temperature below cooling setpoint, supply air valve shall modulate from maximum to minimum air flow quantities as scheduled. On a continued drop in space temperature below cooling setpoint, control valve shall maintain minimum position. On a rise in space temperature, the reverse shall occur.

Constant Volume Reheat - (Sequence 4):

Supply air valve shall be pressure independent.

Air flow sensor shall measure supply air flow rate. Air flow sensor shall modulate electric actuator on supply air valve to maintain constant volume air flow rate as scheduled.

When space temperature falls below heating setpoint, space temperature sensor shall modulate reheat coil control valve to maintain space temperature setpoint.

Air Handling Unit (AHU #):

Note: Use where AHU provides heating, cooling and ventilation

General:

- Space temperature and ventilation control shall be DDC with electric actuation.
- System is designed as single duct, variable volume system.
- System is designed for minimum and maximum outside air with an economizer control.
- System fans consist of field built-up air handling unit supply fan and return fan.

Unit Operating Mode:

In addition to the normal operating modes, system shall be provided with an emergency smoke control operating mode.

Unit Operation:

Unit operation shall be automatic and activated through building automation system.

During occupied cycle, unit supply and return fans shall operate continuously with outside air and relief air dampers open to their minimum positions and return air damper open fully, unless overridden by economizer or discharge air temperature controls.

Unit discharge air temperature controller to operate as indicated under Unit Discharge Air Temperature Control.

Interlocking:

Return fan shall be interlocked with supply fan so that return fan operates whenever supply fan operates. Return fan shall start before supply fan is allowed to start.

Before each supply and return fan is allowed to start, respective fan discharge and return dampers shall be fully opened and be proven by end switches. Whenever fans stop, outside air and relief dampers shall close fully and return damper shall open fully.

See humidifier sequence for humidifier lockout.

If unit is shut down for any reason, a BAS Level 3 alarm shall alert the system operator that cooling for Elevator Machine Room is not available.

System Air Volume Control:

Supply and return fan volume control will be accomplished using variable frequency drive(s). Arrange controls so that fans will start on low speed. On failure of fan volume control signal, fans shall go to low speed and stop. Speed setpoint signal input to VFD shall be communicated from Siemens field panel via FLN. Provide alarm indication to BAS for VFD fault indication.

Provide static pressure sensing stations in supply duct where indicated on drawings. Pressure controller shall modulate supply fan VFD to maintain static pressure setpoint. Static pressure setpoint shall be reset to maintain the second most open VAV box between 90% and 95% open.

Supply and return fans shall be provided with flow stations to measure cfm of supply air, return air, and outside air.

Return Fan Speed Control: Modulate the return fan speed to track the supply fan volume less an amount equal to the calculated sum of building exhaust minus the differential necessary to maintain building pressurization.

For air handlers serving spaces such as gymnasiums, auditoriums, and theaters, provide a high quality CO_2 sensor in the return air duct and configure controls to reset supply volume to maintain a maximum 800 ppm CO_2 in the building return air.

Unit Discharge Air Temperature Control:

Unit discharge air controller shall modulate control valve on cooling coil to maintain discharge air temperature setpoint.

Cooling coil control valve shall be locked in closed position whenever outside air temperature is below 60°F.

Cooling coil control valve shall be closed whenever unit supply fan is not operating.

Mixed Air / Economizer Control:

Provide dry-bulb type economizer control.

For outdoor air temperature below return air temperature, an air temperature sensor in the mixed air plenum shall modulate outside air damper to maintain mixed air temperature setpoint.

When outside air dry-bulb temperature rises above return air temperature, control shall override mixed air control and open outside air damper to its minimum position as determined by the TAB contractor.

Miscellaneous:

Provide high static pressure limit switch located in unit discharge to shutdown unit supply fan when pressure exceeds high limit. Indicate alarm to BAS.

Provide low static pressure limit switch located in inlet duct of return fan. Shutdown return fan when pressure falls below the low limit of negative pressure.

Provide minimum position controller for outside air dampers.

Air Handling Unit (AHU #):

Note: Use where AHU provides heating and ventilation, only

General:

System is designed as ventilation, heating only, single duct, [variable] OR [constant] volume system.

System fans consist of modular air handling unit with supply fan.

Unit Operating Mode:

System shall operate with two modes; occupied and unoccupied cycle.

In addition to the normal operating modes, system shall be provided with an emergency smoke control operating mode.

Unit Operation:

Unit operation shall be automatic and activated through building automation system.

Building Automation System (BAS) will program time initiated functions.

Interlocking:

Return fan shall be interlocked with supply fan so that return fan operates whenever supply fan operates. Return fan shall start before supply fan is allowed to start.

Before each supply and return fan is allowed to start, outside air damper shall be fully opened and be proven by end switches. Whenever fans stop, outside air dampers shall close.

System Air Volume Control:

During occupied mode, the supply fan continues to run, or is commanded on, which will open the outside air damper and, after fully open, a damper auxiliary switch starts the supply fan.

During unoccupied mode, the supply fan stops. The outside air damper is fully closed.

Unit Discharge Air Temperature Control:

Unit discharge air controller shall modulate control valve on hot water coil to maintain discharge air temperature setpoint.

Heating coil control valve shall be locked in closed position whenever outside air temperature is above 65°F. The BAS shall monitor all of the space temperatures shown on the plans for rooms served by this AHU.

Smoke / Fire Alarm Mode:

Smoke Detectors:

Duct-mounted Smoke detectors will be furnished, installed, and wired to Fire Alarm Control Panel (FACP).

The fire alarm system shall be interfaced to the air handling unit supply fan starter control circuit to shut down unit fan when Fire Alarm System is in alarm condition.

Duct smoke detectors will be provided for all air handling units over 2000 cfm.

Combination Fire / Smoke Dampers (Duct Mains and Occupancy Separation Walls):

This specification applies to fire/smoke dampers located in ducts penetrating occupancy separation walls.

Duct-mounted fire/smoke dampers will be provided under Section 23 33 00 – Air Duct Accessories.

Provide interlock through FACP between fire damper and air handling unit fans serving respective system. Unit fan shall not be allowed to start until all system fire/smoke dampers have been opened.

Upon initiation of fire alarm mode control sequence, all AHU supply and return fans shall stop and their smoke/isolation dampers shall close, and then all combination fire/smoke dampers throughout the building shall close.

Fire alarm mode control sequence shall be terminated upon reset of fire alarm control panel. Fire alarm control panel will provide signal to BAS indicating panel has been reset. Following the fire alarm panel reset, air handling unit supply and return fans shall be automatically restarted through the BAS and other fire alarm control sequences shall revert back to normal operating sequences.

Return Fans:

Return fan shall be electrically interlocked with associated air handling unit.

General Exhaust Fans (EF #):

General exhaust fans shall be started and stopped by the DDC system, based on the building occupancy schedule, or interlocked with the air handler associated with the same zone.

Restroom Exhaust Fans (EF #):

Restroom exhaust fans shall be started and stopped by the DDC system, based on the building occupancy schedule

Exhaust Fans (EF #):

Fume hood exhaust fans operate continuously, and are capable of being started and stopped by the DDC control system.

23 20 00 - HVAC Piping and Pumps
Pumps: B&G, PACO, Armstrong, or approved equal, base-mounted or vertical pumps; no TACO, per Campus Standard.

Pump VFD: ABB model ACH550, or approved equal, with integral disconnect without bypass.
Above ground pipes: copper pipe, ProPress allowed for 2” or smaller.

Underground pipes: welded steel, pre-insulated with jacket. Maximum heating water supply temperature drop: 2°F drop from design based on 40°F ground temperature.

23 21 00 - Hydronic Piping and Pumps
Hydronic Piping: [Underground, Aboveground, Ground-Loop Heat-Pump]

Hydronic Pumps: [In-Line Centrifugal Hydronic; Base-mounted, Centrifugal Hydronic; Vertical-Mounted, Double-Suction Centrifugal Hydronic; Vertical-Turbine Hydronic]

Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

23 25 00 - HVAC Water Treatment
Hydronic System:
Water Treatment for Hydronic System: Coordinate with Project Manager to obtain current vendor’s product specifications for Contractor requirements on treatment for Hydronic System.

Campus Standard: Water Treatment is provided by a vendor. Specifications shall include requirement for Contractor to treat the Hydronic System in accordance with the vendor’s product specifications within a limited number of days after the system is charged. The specifications shall include flushing, passivation, treatment with a cleaner and scale inhibitor, and circulation through a temporary 20 micron, maximum filter set.

Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

23 41 00 - Particulate Air Filtration
Standard: Provide MERV 13 air filters, with extended surface for long life, and plastic reinforced frames to resist degradation due to coastal moisture. Specify filters with low pressure drop.

23 52 00 - Heating Boilers
Boilers: high-efficiency, fully-modulating with minimum 10:1 turndown. No staging boilers.

23 54 00 - Furnaces
Standard: Specify high efficiency (>90%) condensing furnace, with long life stainless steel heat exchanger.

Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

23 60 00 - Central Cooling Equipment
Preferred Manufacturers: Carrier, York, or Trane.
Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

23 73 00 - Indoor Central-Station Air-Handling Units
Campus Preferred Manufacturers: Trane, or Pace.

Heating and Cooling Coils: 4-Pipe verses 2-Pipe Interfaces:

Campus strongly prefers to have separate heating and cooling coils. However, if coils serve both heating and cooling in a 2-pipe configuration, control and isolation valves shall prevent water flow from one closed loop system to the other using positive isolation switchover valves (such as motorized ball valves).

Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

23 74 00 - Packaged Outdoor HVAC Equipment
Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

Thermostats: 7-day Programmable:

- Thermostats shall retain program and time settings during power failure without use of backup batteries.
- If possible, thermostats shall be tied to Siemens Energy Management System (EMS), or be BacNET compatible.
- Package or split air conditioning units shall have a minimum SEER of 17.0.

23 75 00 - Custom-Packaged Outdoor HVAC Equipment
Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

Unit Construction:

General: Consider preferences and proximity to coastal inland location.
Prefer Heavy-Duty enclosures, hardware, hinges, latches and gaskets. Goal is longevity, leak proof, and easy servicing.
Consider Copper and copper coils for longevity.

Products:
Preferred Manufacturers: Trane or Pace.

Unit Base
Fabricated from heavy-duty structural steel. Sized to provide floor height to accommodate cooling coil drain trap height. Weld steel solid at connection points. Size perimeter steel to allow for rigging and handling, and provide removable lifting lugs.

Unit Floor
Fabricated 16 gauge, minimum galvanized steel plate welded to structural members. Not acceptable - drive screw attachment. Seal floor joints and seams with acrylic latex sealant (ASTM C834-76 (1981), or polyurethane sealant, ASTM C-920, Type s, Grade NS, Class 25, USDA approved. Insulated floor and base drain pans to have same thermal and acoustical performance as per unit housing. Support insulation with 20 gauge, minimum galvanized steel liner with joints sealed to provide continuous vapor barrier.

Unit Housing
Fabricated of 2-inch thick, double wall, self-supporting with internal support structure or supported by structural frame work, modular panel type construction. Outer panel face 16 gauge G90 galvanized steel, minimum, and inner panel face 20 gauge G90 galvanized steel, minimum. Sealed inside and outside panel joints and seams with gasket and caulking to prevent air and water vapor leakage. Standing seam water-tight joints and sloped 1/8-inch per foot roof.

Access Doors
24-inch by 72-inch access door with ¼-inch thick wire glass window, 12-inch by 12-inch, minimum, in each unit section. Thermal break construction to comply with Unit Housing. Door hardware includes corrosion resistant metal hinges, or continuous piano hinge, with 2 handles, minimum operable for either side of stainless steel, or aluminum alloy.

Division 25 - Integrated Automation
25 05 00 - Schedules for Integrated Automation
"The purpose of this Guide is to assist CSU campuses in properly metering of utilities and other services at each building in the Campus." This Guide covers what utilities and services must or should be metered, what devices can be used for metering, and how these devices are to be commissioned and calibrated to ensure accurate data.

HVAC power shall be separately metered from lighting and plug loads.

Website: http://www.calstate.edu/cpdc/ae/gsf/documents/CSU_Metering_Guide.pdf

Division 26 - Electrical

26 00 00 - Electrical

Fire Ignition and Arc Flash:

Comply with NFPA 70E – Standard for Electrical Safety in the Workplace and IEEE 1584-2002 – Guide for Performing Arc Flash Calculations. Perform an Arc Flash Hazard Analysis to determine the Flash Hazard Boundary, Incident Energy, Hazard Risk Category, Shock Hazard Voltage, Required Glove Class, Limited Approach Boundary, Restricted Approach Boundary and Prohibited Approach Boundary.

Submit arc flash label with above data to university representative for approval. Label will be consistent with existing labels on campus.

At a minimum, the following electrical equipment will be studied: All Switchgear, Distribution Boards, Panel Boards, Equipment Disconnects, VFD Enclosures, Motor Controllers, Motor Control Centers, MCC Individual Cells (Buckets), Medium Voltage Switchgear Enclosures, Medium Voltage Transformer Enclosures, control panels, lighting control panels and relay enclosures.

Arc flash calculations will be completed using the SKM software to match the rest of the campus study. Submit the arc flash report prepared by the registered professional engineers of record that performed the study. Include the SKM files and the database used to perform the study, on a CT and submit with the completed study. Install required arc flash labels on campus equipment.

26 05 00 - Common Work Results for Electrical

Any electrical equipment operating over 250V phase to ground or above a 225A buss rating will be located in an electrical room with limited access.

All electrical rooms will not be designed with regard to floor plan so that access to other facilities of the building (machine rooms, telecom equipment, etc.) will have to pass through the electrical rooms.

Electrical rooms with be compliant with NEC requirements regarding panic hardware and secondary exiting based on installed equipment ratings.

All general building wiring conductors to be THHN/THWN. #14 and larger gauges to be stranded wire.

All conductors regardless of size will be ordered and installed in their corresponding phase color. Phase tape is not acceptable.

Color Coding:

The following color code will be adhered to:

208/120V Systems:	Phase A:	Black
	Phase B:	Red
	Phase C:	Blue
	Neutral :	White
240/139V Systems	Phase A:	Black
	Phase B:	Purple
	Phase C:	Blue
480/277V Systems	Phase A :	Brown
	Phase B:	Orange
	Phase C:	Yellow
	Neutral:	Gray

240/120V Systems	Phase A:	Black
	Phase B:	Red
	Neutral:	White
Medium Voltage Systems	Phase A:	1 Bands White Tape
	Phase B:	2 Bands White Tape
	Phase C:	3 Bands White Tape

All systems:

Switch Leg/travelers:	Pink or Purple
Ground:	Green

Remove reference to armored cable, armored cable is not permitted.

Equipment Disconnects and Connections:
Equipment disconnects are to be rated as the "Heavy Duty" type. Provide fused disconnects as un-fused disconnects are not acceptable.

Use type LFMC to connect equipment to disconnecting means.

Equipment operating at 480V must use a hard wired disconnect or an interlocked pin and sleeve or interlocked twist lock.

Enclosures:
Enclosures in exterior areas are to be NEMA 3R or 4X.

Conduit entry into 3R enclosures will be via the bottom whenever possible. 3R enclosures entered from the side or the top will have "Myers Hubs" or factory hubs as appropriate.

Conduit entry into 4X enclosures will employ the use of "Myers Hubs" or equivalent. Sealing lock washers are not permitted.

26 05 26 - Grounding and Bonding for Electrical Systems
Grounding will be installed to conform to the National Electrical Code and applicable recommendations in the IEEE Standard.

A copper, wall mounted ground bus bar will be provided in the main electrical to serve as the central grounding point for connection of equipment grounds and system grounds to the grounding electrode system. It will connect to the main switchboard, to the main incoming water line, to the UFER grounding electrode, to building steel, to the ground bus bars located in each electrical room, and to the ground bus bars located in the BDF and IDF rooms.

26 05 33 - Raceway and Boxes for Electrical Systems
Raceway methods and selections that meet campus standards:
- All fittings used in the assembly of raceways are to be of the "steel type". Die cast fittings are not acceptable.
- Die Cast condulets are not acceptable. Provide Form 7 Cast Iron or Form 7 Aluminum condulets where condulets are required. Condulets must be installed in accessible locations.
- The use of "factory fittings" for EMT, IMC and Rigid pipe is not acceptable. All conduits will be field bent and cut to fit using field tools and equipment. PVC factory elbows and fittings are acceptable were PVC has been determined to be an acceptable raceway.

Raceway methods for below grad and in slab work:
Schedule 40PVC or Rigid Metallic Conduit (RMC) with corrosion protection tape.

All transitions for below grade to above grade are to be type RMC. RMC with 10 mill corrosion protection tape to extend 6 inches below grade or finished surface.

Raceway methods above grade to be concealed in walls or ceilings:
RMC, EMT in any length is acceptable. FMC and LFMC is acceptable if length is restricted to 20 feet or less between pulling points.

Raceway methods above grade to be concealed in concrete or block encasements:
Type RMC, Type RNC Schedule 40 or Type LFMC. All penetrations out of encasement shall use type RMC with a minimum of 3" of encasement of the RMC.

Raceway methods above grade outside in exposed locations:
Type RMC. EMT is not acceptable in outdoor locations.

Raceway methods exposed inside buildings but not concealed.
- Efforts to conceal wiring raceways will be made to the greatest extent that is financially feasible as possible.
- Wiremold 500 Series or equivalent for all class rooms, office spaces, hallways and similar locations.
- Wiremold 4000 Series or equivalent for common power and data raceway. See communications section.
- EMT or RMC for all exposed locations **other than** office spaces, hallways, class rooms and similar locations.
- Labs, where houses, workshops and similar locations will employ the use of type RMC to a point of 10' and lower aff in all locations.
- LFMC for equipment disconnects to equipment in all locations.

Raceway methods for use in food service areas:
Food service, food processing and food handling areas shall use Aluminum Rigid Metallic Conduit in all locations where exposed to food and food products.

All fittings to be watertight with "Myers Hubs" and NEMA 4X enclosures.

Wiring Methods Permitted with restrictions:
- Flexible Metal Conduit (Type FMC): For use in concealed locations only. Not permitted in installations requiring over 20 feet between accessible boxes, etc.
- Rigid Nonmetallic Conduit (Type RNC) Schedule 40: For use below grade and for concrete encasement only. Not permitted where exposed to sunlight.
- Rigid Nonmetallic Conduit (Type RNC) Schedule 80: Four use below grade and for concrete encasement only. May be allowed in areas that have a highly corrosive atmosphere. Written consent required before this method of raceway is employed.
- Electrical Metallic Tubing (Type EMT): Not permitted in wet or outside locations. Not permitted in mechanical rooms that house plumbing, heating or pumping equipment. Permitted in exposed surface installations for labs, workshops or similar occupancies where located 10' or greater above finished floor. Permitted and preferred in all concealed locations.

Wiring Methods Not allowed:
- Armored Cable (Type AC)
- Flat Cable Assemblies (Type FC)
- Metal Clad Cable (Type MC)
- Mineral-Insulated, Metal-Sheathed Cable (Type MI)
- Nonmetallic-Sheathed Cable (Type NM, NMC and NMS)
- Service Entrance Cable (Type SE and USE)
- Underground Feeder and Branch Circuit Cable (Type UF)
- High Density Polyethylene Conduit (Type HDPE)
- Nonmetallic Underground Conduit with Conductors (Type NUCC)
- Liquid tight Flexible Nonmetallic Conduit (Type LFNC)
- Electrical Nonmetallic Tubing (ENT)
- Examples of raceway applications for common installations:
- Exterior underground: RNC with RMC where pipe transitions to above grade.
- Exterior of building: RMC in all locations. Where vibration or flexibility is required type LFMC.
- Interior of building concealed in walls: Preferred: EMT Acceptable: FMC in lengths not to exceed 20 feet between pulling locations.
- Interior of buildings exposed (industrial): Type EMT, RMC (10 feet or less AFF) or LFMC.
- Interior of buildings exposed to water: Type RMC
- Interior of buildings exposed (offices and classrooms) Wiremold

Keys for Electrical Equipment
Cabinets and Equipment: Provide 2 keys per panel or piece of equipment. Provide key tags for each set of keys. Coordinate with Section 08 06 05 – Key Schedule

Padlocks: Coordinate need for and location of padlocks with Section 08 06 05 –Schedule

26 05 53 - Identification for Electrical Systems
Submittals:
Provide a sample of all label type for the particular project to the University for approval prior to the execution of any label or identification placement.
Identification of power receptacles, equipment disconnects and other utilization devices and equipment
Provide a thermoplastic engraved legend, white background with black letters that denotes the panel designation and circuit designation per below example:

1LA 2,4,6

Where “1LA” is the panel designation and “2,4,6” is the circuit designation.
Minimum size for power receptacles is .5” high by 1.5” wide with .25” letters.
Affix to the device using 30 year silicone adhesive.

Labels for Panel boards, Distribution Boards and Switchgear
Panel boards are to be named using the following scheme: **#XY**
- Where # denotes the floor number where the equipment is located
- Where X denotes the voltage:
- L = 208/120V 3 Phase or 240/120V 1 Phase
- M = 240V 3 Phase
- H = 480/277V 3 Phase or 480V

- Where Y denotes the panel designation on that floor. A, B, C and so forth

Sub panels will be labeled as follows: **#XY-$**
Where $ is the number 1, 2, 3 etc. of the sub panel feed from the panel board

Distribution boards will be named using the following scheme: **#**DB**XY**
- Where # denotes the floor number where the equipment is located
- Where X denotes the voltage:
- L = 208/120V 3 Phase or 240/120V 1 Phase
- M = 240V 3 Phase
- H = 480/277V 3 Phase or 480V
- Where Y denotes the Distribution Board designation on that floor. A, B, C and so forth

Main Switchgear will be named as follows: **#-**MDB-**$**
- Where # is the building number and $ is the switchgear number if there is more than one.
- See University Representative if further clarification is needed. Use 30 year silicon adhesive to attach legend to panel enclosure.

Medium Voltage Conductors and Equipment
Provide a waterproof tag on each end of termination and in each manhole or pull box indicating the Feeder # from the drawings, the from location, the to location and the approximate feet at the location of the label

Provide thermoplastic engraved legends with white letters on a black background indicating the asset identification number as shown on the drawings. Labels are to be a minimum of 1” high by 3” long with .5” letters. Use 30 year silicon adhesive to attach legend to the enclosure.

Junction and Pull Boxes
Exposed locations: Provide a thermoplastic engraved legend with white letters on a black background indicating the contents, panel numbers and circuit numbers of conductors and equipment in the box. Use 30 year silicon adhesive to attach legend to the box.
Concealed locations: Write legibly from a distance of 15 feet on the box with an indelible ink marker as the contents, panel numbers and circuit numbers of the conductors and equipment in the box.

Circuit Directories
Circuit Directories are to be installed in a permanent enclosure on the inside of the door of the equipment that they are in reference to.
Circuit Directories are to be neatly typed and include the panel designation, date and the source of electricity for the panel.
Circuit directories are to provide enough description of each load or circuit such that an operator can distinguish a particular load in questions from all other loads in the panel without the use of a circuit tracer or similar tools.   Provide a map or diagram with directory if appropriate.

Wire Identification
All wires are to be installed in their corresponding phase colors. See above in section 26 05 00.
Panel connections will have circuit number tape on the ungrounded phase conductors to indicate their specific circuit number.

Panel connections will have circuit number tape on the grounded conductors (neutrals) to indicate their specific circuit numbers.

Other Electrical Identification
All VFD, Lighting Controllers, Relay Cabinets and thermoplastic engraved legend with white letters on black background indicating the equipment designation. Use 30 year silicon adhesive to attach legend. All power outlets, power disconnects and lighting switches will have an engraved plastic legend indicating their feed panel and circuit number. Glue engraved legend in place with silicone.

26 06 00 - Schedules for Electrical
The campus has specified some equipment as a "Sole Source Justification" to match the existing infrastructure in functionality. Note references to specific manufactures in the following sections.

26 09 00 - Instrumentation and control for Electrical Systems
The campus uses a SCADA system for the control and monitoring of the campus high and medium voltage system as well as the acquisition of power meter data back to Facility Service's automated recharge system.

All remote terminal units (RTUs) shall match the existing terminal units that are currently in service. Existing terminal units are the Square D Quantum and Momentum series PLCs.

All electric meters will be connected to the SCADA system for remote monitoring of total energy usage as well as real time data. Meters will match existing meters in the area at which they are installed and will wither be the Square D PM 820 (strike reference to Electroindustries) or the Electroindustires Shark 200. Consult a University Representative for compunctions options which will include either a mod bus connection or an Ethernet connection. Provide necessary hardware and cabling to facilitate meter connection and integration.

26 10 00 - Medium-Voltage Electrical Distribution
Qualifications for Medium Voltage Electrical Work
- In addition to a C10 electrical contractor's license the qualified installer will have the following:
- All employees of the C-10 electrical contractor will hold a valid "State Certified General Electrician" card issues by the California Department of Industrial Relations' Division of Apprenticeship Standards.
- Person(s) performing medium voltage terminations will possess a certification indicating that they have attended formal training on splicing and terminating medium voltage cables in the types of terminations being performed. Successful completion of the course and examination will be provided to the University prior to the start of any work.
- Person(s) performing medium voltage work or installation will have at least 5 years of verifiable experience performing such work. Submit a record of employment or projects for the University to review and approve.

Acceptance Testing for Medium Voltage Work
The installing contractor will retain the services of a 3rd party testing firm to perform acceptance tests on all medium voltage equipment and cables. The 3rd part testing firm's qualifications will be submitted to the University with samples of the test forms that are scheduled for use before testing takes place. The University must approve in writing of the 3rd party testing firms qualifications prior to performing testing.

The university will be present in the field for all tests, inspections and times at which equipment is first placed into service. Notify University Representative 10 working days prior to performing any tests.

All equipment will be placed in its final installation location and fastened securely before acceptance testing is to take place. Acceptance test performed prior to equipment installed in its final installation location will not be considered or accepted.

All cable test will be performed after cable installation is fully complete with the exceptions of termination make up. All manhole cable racking, arc proofing, duct sealing, etc. will be completed before cable is to be acceptance tested.

All cable tests will be performed within 12 hours of their planned energization time. Cable test performed longer than 12 hours before energization will require a retest at the contractor's expense as test over 12 hours old will not be considered. The University will not accept cable that has been tested more than 2 separate times and recommends that the contractor sequence work appropriately so that multiple cable tests are not required as testing stresses cable.

Distribution Standards:
Medium voltage underground system conductors will be installed in Schedule 40 conduit. At any location that the conduit must transition above grade (such as at a pole riser) that transition will be constructed in rigid steel to a point of 15 feet above grade.

Medium voltage underground conduits will be 5" and include 1 spare for each installed feeder.

Medium voltage underground conduits will be encased in red concrete with a minimum of 5 sack mix. The encasement shall be 6 inches from the edge of the encasement to any part of the conduit on the top and 2 sides of the conduit and 3 inches on the bottom of the conduit.

Overhead distribution construction will only be permitted to be used as an acceptable method of distribution construction by the Director of Facility Planning and Capital Projects.

Overhead distribution facilities will be constructed per the current PG&E standards including phase separation compliant with current California Wildlife and Raptor Compliance.

Provide shop drawings detailing the proposed pole framing on each type of framing scenario that is to be used in the project. Submit to University for approval prior to commencing work.

Equipment for Medium Voltage Work
Provide submittals for each piece of the medium voltage installation for the University to review and approve. Submitted items include but are not limited to: Transformers, Vaults, Cables, Conduit, Ground Wire and Fittings, Secondary Wire, Medium Voltage Terminations, Low voltage Terminations, Sectionalizing Switches, Load Interrupters, Cable Accessories, Circuit Breakers and Fuses.

Splices
Splices in the medium voltage network must be shown on the drawings or approved by the University in advance.

No splices will be permitted in manholes or other structures below grade.

Splices will be performed in above grade junction cabinets using separable connectors.

Overhead cable splices will take place at poletop dead-end locations only. Tensioned splices will not be accepted.

Medium Voltage Distribution Cables:
All cable with be 15 KV rated Type MV-105 220mils 133% EPR copper tape shielded cable.

All medium voltage cables will be fire taped in all manholes, switches and transformers from the point of conduit entry to the cable termination.

Cable installed in manholes and vaults will have sufficient slack to go once around the perimeter of the vault + 5 feet. Cable will be racked and fire taped.

Unit Substations:
All unit substations (transformers) will be pad mounted, oil filled units located outdoors adjacent to the building being served.

All transformers will be provided with loop feed bushings with the corresponding switches to allow feeds A, B and transformer primary to be switched independently of each other.

All transformers will be provided with oil immersed fusing on the primary input of the transformer.

All transformers will be provided with 1 spare set of fuses attached to the primary compartment door.

All transformers will be mounted on a transformer utility vault. This vault will have a manhole lid located at the front of the transformer doors and be an open bottom type vault installed on a minimum of 18’ of crushed granite.

Provide a sump pump connected to the building power via an emergency circuit (if available) and the storm drain.

Primary and secondary conduits will be terminated in this vault and route to their appropriate switch or building main service gear.

Under no circumstances will dry type unit substations or transformers be permitted.

Acceptance test all medium voltage transformers per the current NETA ATS by a 3rd parting testing firm that has been pre-approved by the university

Provide a starting DGA analysis of the transformer oil after the transformer has been energized and loaded for a period not less than 1 week. All dissolved gas levels shall be at “condition 1” per the IEEE standards for DGA analysis.

Switchgear
All critical campus loads will be feed from CPP Switchgear to provide the greatest factor of reliability. Consult University Representative for specifics with each individual project.

All switchgear will be 3 phase gang operated 15KV, 60Hz, dead front, single side access pad mounted sectionalizing switchgear. Each switchgear shall consist of a single self-supporting outdoor enclosure with SF6 insulated switches and a remote terminal unit for SCADA control. Accessory components shall be completely factory assembled, tested and then tested after installation. Interrupter switches and fuses shall be enclosed within an inner grounded steel compartment for electrical isolation and protection from contamination. Switched terminals shall be equipped with bushings for 600A continuous ratings. Fused terminals and buss terminals shall be equipped with bushing wells rated at 200A continuous.

All switchgear shall employ the use for programmable, resettable fusing of the 200A continuous terminals. Fuse controller shall provide dry contacts for the status of the load switch's' position.

All pad mounted switchgear shall be installed on a utility vault to facilitate pulling and racking of cables.

Acceptance test all medium voltage switchgear per the current NETA ATS by a 3rd parting testing firm that has been pre-approved by the University.

Medium Voltage Circuit Protection:
Submit qualifications of the Registered Professional Electrical Engineering firm to perform the coordination study. Firms must have at least (5) years of experience performing coordination studies on medium voltage equipment.

A coordination study shall be performed on all new medium voltage equipment being installed to maintain established selectivity using IEEE standards.

Submit coordination study to the University Representative for review and approval.

Upon University approval 3rd party testing firm will implement recommended coordination study settings and test equipment for compliance to the recommended settings.

26 24 00 - Switchboards and Panel boards
All switchboards, panel boards, distribution boards, etc. that have a rating above 225A or a phase to ground voltage above 150V shall use distribution breakers that can be installed and removed without the manipulation of line side hardware (Nuts, Bolts or Screws) on the line side of the circuit breaker. The manipulation of any one circuit breaker must not disturb or interrupt the operation of any other or adjacent circuit breaker in the assembly

All circuit breakers in panel boards with rating less than or equal to 225A and a phase to ground voltage below 150V, ranging from 5A to 100A and ranging from 1 pole to 3 pole shall have trip indication. This indication will be located on the breaker and in the form of a illuminated led or a fluorescent flag. The indicator that will be visible when the breaker is in the tripped state and not visible when the breaker is in the on or off state.

Load centers or other panel board that expose the live bussing when the cover is removed are not acceptable. All panel boards shall employ the use of a dead front in addition to the panel board surface or flush cover.

Provide 20% of the panel board, switchboard or distribution board breaker mounting space as spare space.

All low voltage distribution equipment, panel boards, switchboards, distribution boards, etc. are to be located inside the building or structure. Outdoor distribution equipment is not permitted.

All electrical distribution equipment shall be located in an electrical room or similar occupancy that is not accessible to the general public.

All electrical distribution equipment shall be provided with locking covers or provisions for padlocks.

All circuit breakers that are carry a rating over 70A 1, 2 or 3 pole shall have permanently installed provisions to lock the breakers in the off position.

Flush mounted panel boards shall have (1) .75 inch conduit stubbed into the accessible ceiling space above the panel per 3 available circuit breaker spaces.

All circuit breakers that have trip or protection parameters that are adjustable will have facilities to seal of and protect these parameters form further tampering or adjustment.

Circuit breaker will be fully rated for the fault current that they will encounter in their installation. The use of series rated devices to achieve a system that is of sufficient fault current duty for the installation conditions is not acceptable.

Perform 3rd party acceptance testing for switchboards and panelboards per the current NETA ATS. Test each breaker 400A 3 pole or greater using NETA ATS.

Low voltage transformers are to have an energy rating of CSL3 or higher.
Perform 3rd party acceptance testing for transformers per the current NETA ATS.

Motor starters that require overload protection shall use solid state current sensing type overload protection. Thermal element overload protection is not acceptable.

Low Voltage Coordination Study and Implementation
Submit qualifications of the Registered Professional Electrical Engineering firm to perform the coordination study. Firms must have at least (5) years of experience performing coordination studies on low voltage (600V or less) equipment.

A coordination study shall be performed on all new medium voltage equipment being installed to maintain established selectivity using IEEE standards.

Submit coordination study to the University Representative for review and approval.

Upon University approval 3rd party testing firm will implement recommended coordination study settings a test equipment for compliance to the recommended settings. Contact University representative for sealing of devices after setting.

Emergency Power Systems
For larger building projects provide a central emergency lighting system that is supplied by an emergency generator.

Generators should be APCD Compliant by using Tier 4 Final Prime movers. Natural gas or propane generators may also be considered depending on application.

Centralized emergency battery systems are not permitted.

Small projects where an emergency generator is not feasible, provide emergency lighting fixtures with integral battery backup to provide the required minimum level of illumination.

Lighting

LED Lighting technologies are to be employed wherever possible. See the appropriate University representative to determine the current standards that are in place as this sector is rapidly changing. At a minimum, the following standards are in effect at the time of this document revision:

Exterior lighting is to be LED, linear fluorescent or CFL. No HPS permitted.

Interior lighting is to be LED, linear fluorescent or CFL. HID lights are only permitted in special occupancies that are pre-approved by the University Representative. HPS or HID lighting fixtures are discouraged but may be the option in some applications.

In all light fixtures linear fluorescent fixtures that use lamps that exceed 4'in length are not permitted. The campus standard fluorescent lamp is the F25T84100K lamp.

Lighting Controls

The campus has standardized its lighting controls to the Wattstopper product line.

Design considerations:

All areas of construction are to be compliant with the latest in effect revision of California Title 24 with the following supplementary requirements by functional use:

- Office Spaces are to use line voltage dual technology occupancy sensors. Ceiling sensors are preferred over wall switch sensors.
- Classrooms and lab spaces are to use a local DLM system that takes advantage of the energy saving opportunities that are available in the space. These may include but are not limited to daylight harvesting, occupancy sensors, dimming and the use of bi-level fixtures.
- Equipment rooms, janitor closets, etc. shall be outfitted with a time switch that will permit the lights to be on for a fixed value of time pending no input from an occupant.
- Equipment rooms with rotating machinery or electric rooms shall be outfitted with lighting controls do not automatically switch the lights off without interface from the room occupant.
- Hallways, stairwells, building entrances and exterior lighting shall be routed through an Automation Level lighting control panel. This panel shall have the ability to perform astronomical clock functions, possess an exterior photo cell sensor, and have programmable switches located in strategic locations to the spaces they serve as well as in the main custodial rooms for each building.
- Mechanical time clocks are not permitted.

26 30 00 - Facility Electrical Power Generating and Storing Equipment

Emergency Power:

Provide a 175kW, 480V, 3-Phase, 4 Wire natural gas/ propane generator in a weatherproof, sound attenuated enclosure for emergency loads. The generators are to include (1) 250A distribution breaker that will serve two loads:

A 100A, 480V, 3-Phase, 4 pole transfer switch, which in turn will feed a 100A, 480/277V panel board. The panelboard will power all life safety egress lighting.

A 150A, 480V, 3-Phase, 4 pole transfer switch, which in turn will feed a 150A, 480/277V panel board. This will support sewage lift pumps and the telecom equipment via a 30kVA, 480V:208/120V transformer and 120/208V panelboards.

26 51 00 - Interior Lighting
Light fixtures and systems will be selected for efficiency, durability, maintenance ease, and to accentuate the area architecture. Indoor lighting will be tailored to building's needs and theme.

The illumination levels will conform to the latest edition of Illuminating Engineering Society (IES) Guidelines and user requirements.

The lighting power densities will exceed current requirements California Energy Code, Title 24, Part 6 by a minimum of 20%.

Task light fixtures will be considered in offices to reduce the overall lighting power density.

All spaces will be illuminated with efficient light fixtures (a minimum efficiency of 90%) equipped with efficient lamps and ballasts. The campus standard lamp is 4' linear 25WT8 (4100K) fluorescent.

LED light fixtures may be considered to meet efficiencies required for LEED Certification. All LED light fixtures must be presented to the Owner with full documentation of fixture construction, lamp life, actual installation locations, maintenance, and warranties for approval. Provide a sample if requested by the Owner.

26 52 00 - Emergency Lighting
Egress lighting power will be provided by emergency battery packs integral to the lighting fixtures. Egress lighting will be provided in corridors, lobbies, and along designated paths of egress. Illumination will conform to the current CBC requirement of an average of one foot-candle level at floor level during loss of normal power.

26 53 00 - Exit Signs
LED Exit signs will be provided at all exits and along the path of egress and will be specification grade and vandal resistant. Exit signs mounted at heights below 7'-0" will be protected from physical damage by an architectural system and require tamperproof tools to gain entry to the sign.

26 56 00 - Exterior Lighting
Safety and Security: Pathways to, from and around the site should be appropriate lit and avoid areas of darkness or isolation. Campus approved exterior light fixtures will be installed on the outside building perimeter as needed. All exterior light fixtures will be full cutoff. Design will be based on LED lamping.

Division 27 - Communications

27 00 00 – Communications

Cal Poly Telecommunication Standards;
Website: http://telecommunications.calpoly.edu/telecommstandards/index.html

CSU Design Standards
CSU Telecommunications Infrastructure Planning (TIP) Standard.
TIP recommendations are to be implemented as the Campus minimum standards.
Website: http://www.calstate.edu/cpdc/ae/gsf/TIP_Guidelines/index.shtml
BICSI Telecommunications Distribution Methods Manual (TDMM), 11th Edition.
Website: http://www.bicsi.org/content/index.aspx?file=tdmpubs.htm

Design Engineer Qualifications
BICSI Registered Communications Distribution Designer (RCDD).

Description
This document contains the General and Supplementary Conditions that are a part of the requirements for the work under this Division of the Specifications.

The term “Telecommunications” systems/components is understood to include TELEPHONE, DATA, CATV systems/components and Radio (RF) systems and components when such systems and/or components are part of the Cal Poly Campus Project.

Quality Assurance
Comply with the current applicable codes, ordinances, and regulations of the authority or authorities having jurisdiction, the rules, regulations and requirements of the utility companies serving the project and the campus’ insurance underwriter.

Drawings, specifications, codes and standards are minimum requirements. Where requirements differ, the most stringent apply.

All equipment and installations shall meet or exceed minimum requirements of ADA, ANSI, ASTM, IEEE, TIP, NEC, NEMA, NFPA, OSHA, UL, ITS Telecommunications Standards Document (TSD), the Manufacturer and the State Fire Marshall.

Should any change in drawings or specifications be required to comply with governing regulations, notify and receive written approval from the Cal Poly ITS Telecom group representative prior to submitting your bid.

Execute work in strict accordance with the best practices of the trades in a thorough, substantial, workmanlike manner by competent workmen. Provide a competent, experienced, full-time Superintendent and Project Manager who are authorized to make decisions on behalf of the Contractor.

All telecommunications technicians must have a manufacturer’s certification for the Structured Cable System (SCS) components that they are installing.

Provide all components of a complete system specified within all project documents, specifications and drawings.

All unused component parts from all system installations shall be delivered to the Cal Poly ITS Telecomm group and shall not disposed of or thrown away.

Contractors shall use the strictest manufacturer written recommendations, specifications, and the Cal Poly ITS Campus Telecommunications Standards Document.

Submittals
Review of submittals shall be for general compliance with the design concept and Contract Documents.

Comments or absence of comments shall not relieve the Contractor from compliance with the Contract Documents. The Contractor remains solely responsible for details and accuracy, for confirming and correlating all quantities and dimensions, for selecting fabrication processes, for techniques of construction, for performing the work in a safe manner, and for coordinating the work with that of other trades.

No part of the work shall be started in the shop or in the field until the shop drawings and samples for that portion of the work have been submitted and accepted by the Cal Poly ITS Telecomm group.

A minimum period of five working days, exclusive of transmittal time, shall be required in the Cal Poly ITS Telecomm representative’s office each time a shop drawing, product data and/or samples shall be submitted for review. This time period shall be considered by the Contractor in the scheduling of the work.

Submit materials and equipment by manufacturer, trade name, and model number. Include copies of applicable brochure or catalog material.

Maintenance and operating manuals shall not be acceptable substitutes for shop drawings.

Identify each sheet of printed submittal pages (using arrows, underlining or circling) to show applicable sizes, types, model numbers, ratings, capacities and options actually being proposed. Cross out non-applicable information. Note specified features such as materials or paint finish.

Maintain a complete set of reviewed and stamped shop drawings and product data on site.

Samples
Samples as requested shall be physical examples that represent materials, equipment or workmanship and establish standards by which the work will be judged. Samples shall not be returned to the Contractor.

Test Reports
Pre-Installation Testing Reports: Submit two sets of manufacturers' or field-testing reports for those materials identified in the individual system Specification Sections as requiring that such reports shall be submitted.

Post-Installation Testing Reports: Submit a minimum of two sets of field-testing reports for those materials identified in the individual system Specification Sections as requiring that such reports shall be submitted.

Vender/Contractor/Supplier Information
Submit a complete typed list of all telecommunications infrastructure equipment manufacturers and material suppliers for the equipment proposed to be provided on this project, as well as names of all subcontractors.

Contractor must supply current manufacturer's certification for all employees involved in the installation of all materials contained as part of the Structured Cabling System.

Warranty info
Submit a copy of all relevant warranty information.

Product Documentation:
Documentation for submittals in the form of catalog cuts, manufacturer specifications, and other supporting printed material shall be bound in a single binder, tabbed and separated by specification section, and submitted in its entirety for review and eventual delivery to the Cal Poly ITS Telecomm group.

Shop Drawings
After the Contract is awarded, provide complete shop drawings as requested for each relevant section. Prior to submission, certify that the shop drawings shall be in compliance with the Contract Documents. Modify any work, which proceeds prior to receiving accepted shop drawings as required to comply with the Contract Documents and the shop drawings.

Shop drawings for Telecommunications Room (TR) layout, specifically including details for wall-fields and rack mounting layouts shall be approved by a representative of the Cal Poly ITS Telecomm group prior to installation.

For each room or area of the building containing telecommunications infrastructure equipment, submit the following:

Floor plans, at not less than 1/8" scale, showing routing of telecommunications conduits, cable trays and other pathways.

Riser diagrams showing types, quantities and schematic routing of all telecommunications backbone pathways, cabling and the TBB.

Enlarged plan views and elevation layout drawings for the telecommunications Entrance Facility (EF) room, Telecommunications Rooms (TRs) and all other designated telecommunications Equipment Rooms (ERs) indicating the equipment in the exact location in which it is intended to be installed. These plans shall be of a scale not less than ¼ inch = 1'-0". They shall be prepared in the following manner:

Indicate the physical boundaries of the space including door swings and ceiling heights and ceiling types (as applicable).
Illustrate all telecommunications hardware proposed to be contained therein. Include top and bottom elevations of all telecommunications hardware. The Drawings shall be prepared utilizing the dimensions contained in the individual equipment submittals. Indicate code and manufacturer's required clearances.

Illustrate all other equipment therein such as conduits, detectors, luminaires, ducts, registers, pull boxes, wire-ways, structural elements, etc.

Illustrate concrete pads, curbs, etc.

Indicate dimensions to confirm compliance with code-required clearances.

The work described in shop drawing submissions shall be carefully checked by all trades for clearances (including those required for maintenance and servicing), field conditions, maintenance of architectural conditions and coordination with other trades on the job.

Each submitted shop drawing shall include a certification that related job conditions have been checked by the Contractor and each Subcontractor and that conflicts do not exist.

The Contractor shall not be relieved of the responsibility for dimensions or errors that may be contained on submissions, or for deviations from the requirements of the Contract Documents. The noting of some errors but overlooking others does not grant the Contractor permission to proceed in error. Regardless of any information contained in the shop drawings, product data and samples, the Contract Documents govern the work and shall be neither waived nor superseded in any way by the review of shop drawings, product data and samples.

Inadequate or incomplete shop drawings, product data and/or samples shall not be reviewed and shall be returned to the Contractor for resubmittal.

Indicate the following on the lower right hand corner of each shop drawing and on the front cover of each product data brochure cover: The submittal identification number; title of the sheet or brochure; name and location of the project; names of the Architect, Engineer, Contractor, Subcontractor, manufacturer, supplier, and vendor; the date of submittal; and the date of each correction, version and revision. Number all pages and drawings in product data brochures consecutively from beginning to end. Unless the above information is included, the submittal shall be returned for re-submission. Resubmittals of product data or brochures shall include a cover letter summarizing the corrections made in response to the review comments.

The telecommunications room layout submittals and the related telecommunications equipment submittals shall be submitted concurrently. Failure to submit concurrently shall result in the immediate return of the submittal marked REVISE AND RESUBMIT.

Quality Control

Campus Communications Inspector

The work shall require Special Inspection by the Campus Communications Inspector, and shall be scheduled in advance through the Trustees Inspector.

Campus has had concerns in the past with unclear specifications and below standard installations. Telecommunications work is highly specialized and requires additional care and adherence to standards to provide trouble-free service to the Campus. Just as with other project work, the Trustees will withhold payment for work not completed to specifications.

The Contractor shall be responsible to schedule inspections and insure that work is not covered before it is approved. Covered work will be required to be uncovered for inspection.

Work not installed to CSU design requirements, Campus Standards and specifications will be removed, and replaced with work complying, and then re-inspected. Determination as to the acceptance of the work shall be by the Campus Communications Inspector. The specifications need to clearly include these requirements.

Moisture Intrusion

Piping

The telecom rooms and surrounding walls shall not contain floor drains, plumbing cleanouts, waste lines, wet standpipes, storm drain lines, roof drain lines, etc. Incoming Outside Plant (OSP) conduits shall be weather stopped. Coordinate with MEP Engineer.

Fire Sprinklers

Use dry pipe if fire sprinklers are required to prevent damage to telecom equipment from leaking pipes.

Coordination of work

The Contract Documents establish scope, materials and quality but are not detailed installation instructions. Drawings are diagrammatic.

Coordinate work with related trades and furnish, in writing, any information necessary to permit the work of related trades to be installed satisfactorily and with the least possible conflict or delay.

The telecommunications infrastructure drawings show the general arrangement of equipment and appurtenances. Follow the appropriate drawings as closely as the actual construction and the work of other trades, will permit. Provide offsets, fittings, and accessories, which may be required but not shown on the Drawings. Investigate the site, and review drawings of other trades to determine conditions affecting the work, and provide such work and accessories as may be required to accommodate such conditions.

The locations of cable termination fields, faceplates, patch panels, equipment racks and other equipment indicated on the Drawings are approximately correct, but they are understood to be subject to such revision as may be found necessary or desirable at the time the work is installed in consequence of increase or reduction of the number of faceplates, or in order to meet field conditions, or to coordinate with modular requirements of ceilings, or to simplify the work, or for other legitimate causes. The final designs shall be accepted by the Cal Poly ITS Telecomm group prior to installation.

Exercise particular caution with reference to the location of outlets/faceplates, racks, blocks, patch panels, control panels, switches, etc., and have precise and definite locations accepted by the Cal Poly ITS Telecomm group representative before proceeding with the installation.

The Drawings show only the general run of raceways and approximate locations of faceplates. Any significant changes in location of faceplates, cabinets, etc., necessary in order to meet field conditions shall be brought to the immediate attention of the Cal Poly ITS Telecomm group representative for review before such alterations are made. Modifications shall be made at no additional cost to the University.

Verify with the Cal Poly ITS Telecomm group representative the exact location and mounting height of faceplates and equipment not dimensionally located on the Drawings. For power distribution to equipment located in equipment racks see the Cal Poly Labeling, Design and Syntax Standard in Appendix B.

Faceplate/cable labels in the form of alpha/numeric characters are used where shown to indicate the faceplate and cable designation numbers in cable termination fields (terminal blocks and/or patch panels). Show the actual faceplate/cable numbers on the as-built Record Drawings, on the associated typed termination field labels and in the printed and computer readable cabling schedules. Where faceplate/cable-numbering information is not indicated, request clarification from the Cal Poly ITS Telecomm group representative.

Wherever work interconnects with work of other trades, coordinate with other trades to insure that they have the information necessary so that they may properly install the necessary connections and equipment. Identify items (remote ballast, pull boxes, etc.) requiring access in order that the Ceiling Trade will know where to install access doors and panels.

Furnish and set sleeves for passage of telecommunications risers through structural masonry and concrete walls and floors and elsewhere as required for the proper protection of each telecommunications riser passing through building surfaces.

Provide appropriate firestop materials around all pipes, conduits, ducts, sleeves, etc. which pass through rated walls, partitions and floors.
Provide detailed information on openings and holes required in precast members for telecommunications work.

Provide required supports and hangers for conduit and equipment, designed so as not to exceed allowable loadings of structures.

Examine and compare the Contract Drawings and Specifications with the Drawings and Specifications of other trades, and report any discrepancies between them to the Cal Poly ITS Telecomm group representative and obtain written instructions for changes necessary in the work. Install and coordinate the work in cooperation with other related trades. Before installation, make proper provisions to avoid interferences.

Before commencing work, examine adjoining work on which this work is in any way affected and report conditions, which prevent performance of the work. Become thoroughly familiar with actual existing conditions to which connections must be made or which must be changed or altered.

Adjust location of conduits, panels, equipment, etc., to accommodate the work to prevent interferences, both anticipated and encountered. Determine the exact route and location of each conduit prior to fabrication.

Right-of-Way: Lines which pitch shall have the right-of-way over those which do not pitch.
For example: condensate, steam, and plumbing drains normally have right-of-way. Lines whose elevations cannot be changed have right-of-way over lines whose elevations can be changed.

Provide offsets, transitions and changes in direction of conduit as required to maintain proper headroom and pitch on sloping lines.

In cases of doubt as to the work intended, or in the event of need for explanation, request supplementary instructions from the Cal Poly ITS Telecomm group.

Coordinate with Cal Poly ITS Telecommunications group representative for access into existing campus telecommunication spaces.

Cutting and Patching
Where cutting, channeling, chasing or drilling of floors, walls, partitions, ceilings or other surfaces shall be necessary for the proper installation, support or anchorage of conduit or other equipment, layout the work carefully in advance. Repair any damage to the building, piping, equipment and/or defaced finished plaster, woodwork, metalwork, etc. using skilled tradespeople of the trades required at no additional cost to the campus.

Do not cut, channel, chase or drill unfinished masonry, tile, etc., unless permission from the Cal Poly ITS Telecomm group representative is obtained. If permission is granted, perform this work in a manner acceptable to the Cal Poly ITS Telecomm group representative.

Where conduit or equipment is mounted on a painted finished surface, or a surface to be painted, paint to match the surface. Cold galvanize bare metal whenever support channels are cut.

Provide slots, chases, openings and recesses through floors, walls, ceilings, and roofs as required.

Where these openings are not provided, provide cutting and patching to accommodate penetrations at no additional cost to the campus.

Work by Others- Scheduling
The Telecommunications System requires work to be performed by others to meet the full project completion deadline. The Contractor shall be responsible for working with Campus Telecommunication Services to schedule and coordinate the installation of equipment to insure an operating system by the project completion deadline. Telecom work needs to be performed concurrently with Contractor work on the project, and sufficient time allowed for installation and testing. The Consultants need to work with the Trustees Representative and Campus Telecommunication Services to understand the sequence for turn-over of completed spaces and durations of Campus telecom work. The specifications need to clearly include these requirements.

27 01 30 - Operation and Maintenance of Voice Communications
Telecommunication Spaces:
Refers to specific rooms or spaces dedicated to telecommunications services only. Telecommunication rooms shall open off a corridor or to the building exterior, and shall not be used to access other rooms. The three main categories of spaces are: (1) Service Entrance (room), (2) Equipment Room, and (3) Telecommunications Rooms. These spaces have a distinct function, but are all very inter-dependent, and shall be provided in university projects. A brief description and requirements for each of these “spaces” is listed below.

- Service Entrance (Room):
    - Description: Room where outside cables are terminated and interconnected with backbone cables.
    - Provides facilities and supporting hardware for large splice containers, cable termination mountings and possibly copper cable electrical protectors.

- If this is a stand-alone space, it is not expected to house any network or telecommunications equipment.
- Design Requirements:
  - Locate on a lower level and within fifty (50) feet of an outside wall.
  - Provide space with direct access to the inter-building (entrance) conduit.
  - Situate to provide a direct pathway to the equipment room or backbone (riser) distribution spaces.
  - Do not locate in, or directly adjacent to the building's electrical service entrance, transformer room or mechanical room.
- Size: Stand-alone Service Entrance Room: Provide 5 feet by 7 feet, minimum for a 10,000 square feet or smaller building. Allow 25%, minimum for future expansion for infrastructure, pathways, wall penetrations, patch panels, blocks, etc. See TIP for additional detailed space information.
- Doors and Door Hardware:
  - Doors shall open outward.
  - Doors shall be secured with a 112-series key lock.
  - Coordinate key lock with Architect and work in Section 08 71 00 – Door Hardware.
- Environment:
  - Stand-alone Service Entrance Room: No special air handling required. Provide environment same as for a standard office of equivalent size. Air conditioning needs to be independently controlled.
  - Coordinate with Mechanical Engineer and work of Division 23 – Heating, Ventilating and Air Conditioning.
- Grounding: Route building ground cable to the Service Entrance (Room) to connect telecommunications master grounding bar.
- Electrical:
  - Provide 2 dedicated 20 amp circuits.
  - Coordinate with Electrical Engineer and work of Division 26 - Electrical.
- Installation Requirements: Comply with BICSI TDMM, 11th Edition.

- Equipment Room (Building's Main Telecommunication Room):
  - Description: Centrally located; houses telecommunications network equipment to service users throughout the building. Equipment housed in room: Telephone equipment, local area network switches, video distribution equipment, and cable interconnect and cross-connect hardware.
  - Design Requirements:
    - Locate near Service Entrance Room.
    - Provide access from outside the building to allow for installation of large equipment cabinets (36" wide by 96" tall).
    - Multi-story Buildings: Stack and locate Equipment Room and Telecommunication Rooms near the center of the building; within 290 feet, maximum cable pathway distance to furthermost user outlet. The average cable pathway distance shall be 150 feet or less.
    - Plumbing Pipes & Fixtures: Do not place water or drainage pipes directly over, or near Equipment Room. (Coordinate with Mechanical Engineer and work of Division 22 – Plumbing.)
    - Electrical & Equipment Interference: Do not place Equipment Room adjacent to potential sources of electrical interference, such as electrical power supply transformers, motors, generators or elevator equipment. (Coordinate with Electrical Engineer and work of Division 26 – Electrical.)
  - Size:

   - When Combined With Service Entrance Room: Provide an additional 200 square feet, minimum.
   - Stand-alone Room: Provide 10 feet by 15 feet, minimum.
   - Allow 25%, minimum for future expansion for infrastructure, pathways, wall penetrations, patch panels, blocks, etc.
   - See TIP for additional detailed space information and other design factors that could increase room size requirements.
 - Finishes: Keep space opened from floor to ceiling. Do not provide a false ceiling.
 - Structural Loading: Provide floor loading of 100 pounds per square foot (distributed loading), minimum. (Coordinate with Structural Engineer.)
 - Doors and Door Hardware:
   - Doors shall open outward.
   - Doors shall be secured with a 112-series key lock. (Coordinate key lock with Section 08 21 00 – Door Hardware.)
 - Environment:
   - Provide positive airflow and cooling, even when the main building systems are shut down. Air conditioning needs to be independently controlled.
   - Consider separate air handlers or stand-alone cooling systems, thermostatically controlled and designed to operate 24 hours per day, 365 days a year.
   - If served by an auxiliary power supply, connect air handling systems to backup power generation system.
   - Coordinate with Mechanical Engineer and work of Division 23 – Heating, Ventilating and Air Conditioning.
   - Provide air changes to exhaust fumes from UPS batteries (hydrogen gas) to meet indoor air quality standards.
 - Grounding: Route building ground cable to the Service Entrance (Room) to connect telecommunications master grounding bar.
 - Electrical:
   - Provide 225 amp, 208V Panel, possible new transformer.
   - Provide 30-minute battery backup system (uninterruptible) power with capacity to support 3 times the planned capacity.
   - Coordinate with Electrical Engineer and work in Division 26 - Electrical.
 - Installation Requirements: Comply with BICSI TDMM, 11th Edition.
- Telecommunications Room:
 - Description: Supports cable and equipment between the Equipment Room (building main telecommunications room) and User (Station) Locations. Space to terminate the cables from station outlets and backbone (riser) systems and house the local area network equipment, phone equipment, cable cross-connects and video distribution equipment.
 - Location:
   - Multi-story Buildings: Stack and locate Equipment Room and Telecommunication Rooms near the center of the building; within 290 feet, maximum cable pathway distance to furthermost user outlet. The average cable pathway distance shall be 150 feet or less.
   - Dedicated Room: Room shall not be shared with electrical, janitorial, fire alarms, security systems or storage.
   - Overhead Obstructions: Eliminate.
   - Damage or Interference from Other Sources:
   - Plumbing Pipes & Fixtures: Do not place water or drainage pipes directly over, or near Telecommunications Room. (Coordinate with Mechanical Engineer and work of Division 22 – Plumbing.)

    - Electrical & Equipment Interference: Do not place Telecommunications Room adjacent to potential sources of electrical interference, such as electrical power supply transformers, motors, generators or elevator equipment. (Coordinate with Division 26 – Electrical.)
    - Dust, Airborne Contaminants and Physical Hazards: Do not place Telecommunications Room where potential for dust, contaminants or physical hazard may occur.
  - Size:
    - Room: Provide 8 feet by 10 feet, minimum.
    - Allow 25%, minimum for future expansion for infrastructure, pathways, wall penetrations, patch panels, blocks, etc.
    - See TIP for additional detailed space information and other design factors that could increase room size requirements.
  - Finishes:
    - Keep space opened from floor to ceiling. Do not provide a false ceiling.
    - Provide sealed concrete or tiled floor; carpeting not allowed.
    - (Coordinate with Division 9 – Finishes.)
  - Doors and Door Hardware:
    - Keep space opened from floor to ceiling. Do not provide a false ceiling.
    - Provide sealed concrete or tiled floor; carpeting not allowed.
    - (Coordinate with Division 9 – Finishes.)
  - Environment:
    - Provide positive airflow and cooling, even when the main building systems are shut down. Provide environment same as for a standard office of equivalent size. Air conditioning needs to be independently controlled.
    - Consider separate air handlers or stand-alone cooling systems, thermostatically controlled and designed to operate 24 hours per day, 365 days a year.
    - Coordinate with Mechanical Engineer and work of Division 23 – Heating, Ventilating and Air Conditioning.
  - Lighting:
    - Provide 50 foot-candles at 36 inches above the floor. Consider location of cable trays and racks in designing the lighting to meet the requirements. Telecom installers spend several hours working in the spaces to connect patch panels and cabling; labeling patch panels, and racks; and testing cables.
    - Coordinate with Electrical Engineer and Section 26 51 00 – Interior Lighting.
  - Electrical:
    - Provide 100 amp, 208V Panel, possible new transformer.
    - Provide 30-minute battery backup system (uninterruptible) power with capacity to support 3 times the planned capacity.
    - Coordinate with Division 26 - Electrical.
  - Installation Requirements: Comply with BICSI TDMM, 11th Edition.
- Telecommunications Pathway “Spaces” for Interbuilding Distribution System:
  - See Division 33 – Utilities. Allow 25%, minimum for future expansion for infrastructure, pathways, wall penetrations, etc.
  - Installation Requirements: Comply with BICSI TDMM, 11th Edition.
- Telecommunications Pathway “Spaces” for Intrabuilding Backbone:
  - Description: Horizontal and vertical pathways in building distribution system.
  - Used for placement of telecommunications media between Service Entrance Room, Equipment Room and Individual Telecommunication Room(s).
  - Design Requirements:

    - Vertical Backbone (riser) Conduits: Provide three conduits, minimum. Refer to TIP Standards for other design factors that could increase the number of conduits.
    - Pull Boxes: Provide in conduit runs 100 feet or greater, and where there are more than 2 – 90 degree bends. Cable Slack: Provide one complete wrap inside pull box.
  - Size:
    - Backbone Conduits and Sleeves: 4-inch, minimum.
    - Allow 25%, minimum for future expansion for infrastructure, pathways, wall penetrations, patch panels, blocks, etc.
  - Cable Fill: 40 percent, maximum.
  - Prohibited: Using Telecommunications Pathways for other purposes, such as Fire Alarm, Security Alarm, Building System Controls, Security Camera Power, and Video Signals.
  - Installation Requirements: Comply with BICSI TDMM, 11th Edition.
  - Horizontal Pathways:
    - Description: Facilities supporting installation and maintenance of cables between Telecommunications Room and Station Outlet Locations.
  - Design Requirements:
    - Cable Tray:
    - Solid bottom, aluminum, NEMA Class Designation 12B (75 pounds per linear foot).
    - 18 inches wide; and 3 inches deep, minimum.
    - Seismic Bracing Standards: Install to prevent horizontal, lateral and vertical movement for Seismic Zone 4.
    - Refer to TIP Standards for additional information.
    - Grounding and Bonding: Provide continuous bonding of pathways. Refer to Section 27 05 26 – Grounding and Bonding, and TIP Standards.
    - Size: Allow 25%, minimum for future expansion for pathways, wall penetrations, patch panels, blocks, etc.
    - Cable Fill: 40 percent, maximum.
    - Wall Outlet:
    - Outlet Box: 4 11/16 inch square.
    - Conduit: 1 ¼-inch conduit installed with a total of 180 degrees of bend, maximum.
    - Protective Bushings: Provide at ends of conduit and penetrations.
    - Prohibited: Daisy-chained and back-to-back mounting using a common feeder conduit.
    - Floor Boxes:
    - Outlet Box: Sized to meet Cat 6 standards.
    - Note: Depending on the configuration, floor boxes are usually too small for the required services, and services end up being dropped from the ceiling or provided by another means. Engineer needs to confirm the amount of services allow for the Cat 6 to be installed without compromising the cable signals.
    - Conduit: 1 ¼-inch conduit installed with a total of 180 degrees of bend, maximum.
    - Protective Bushings: Provide at ends of conduit and penetrations.
    - Prohibited: Daisy-chained and back-to-back mounting using a common feeder conduit.
    - Prohibited: Using Telecommunications Pathways for other purposes, such as Fire Alarm, Security Alarm, Building System Controls, Security Camera Power, and Video Signals.
    - Installation Requirements: Comply with BICSI TDMM, 11th Edition.
- Technology Instructional Spaces:
  - Description: Spaces with a significant amount of network, computing and display hardware to secure and connect to equipment within the room and to the Telecommunications Room.
  - Note: This is not telecommunications space, but it has critical, functional requirements that may not be listed in other program documents. A Technology Instructional Space is a room with a

heavy concentration of computer workstations where students and the instructor utilize this equipment on a regular basis. This can range from a few high-end computing platforms used for specific applications, to a general use computing lab or a lecture hall with facilities for student access to power and network services.

- Design Requirements:
- Location: Phone Outlet: Locate near an exit door.
- Allow 25%, minimum for future expansion for infrastructure, pathways, wall penetrations, etc.
- Refer to TIP Standards for other design requirements.
- Environment:
  - Provide air handling capacity to accommodate heat generated by network, computer and display hardware.
  - Coordinate with Mechanical Engineer and work of Division 23 – Heating, Ventilating and Air Conditioning.
- Electrical:
  - Provide 1-20 amp dedicated circuit per 4 workstations.
  - Provide 1-20 amp dedicated circuit per 2 printers.
- Installation Requirements: Comply with BICSI TDMM, 11th Edition.

- Office, Classroom and Laboratory Spaces:
  - Description:
    - Offices: Provide two 3-port telecommunications outlets, on opposite walls near electrical outlets.
    - Larger Offices and Open Suites: Provide one 3-port telecommunications outlets per 75 square feet, minimum, but no less than one outlet for every two electrical outlet.
    - Classrooms and Labs: Provide one phone outlet near an exit door. Provide additional telecommunications outlets in every classroom and lab. See TIP for additional classroom and laboratory design factors.
  - Size: Allow 25%, minimum for future expansion for infrastructure, pathways, wall penetrations, etc.
  - Installation Requirements: Comply with BICSI TDMM, 11th Edition.

27 05 26 - Grounding and Bonding

Grounding:

Ground all equipment racks, ladder racks, pathways and penetrations to the Telecommunications Main Grounding Bus Bar (TMGB). Proper grounding of telecommunications related infrastructure requires a very specific design prepared in coordination with (but separate from) the overall electrical grounding system within a building. Refer to TIP Standards – Telecommunications Grounding System, Page 3-11.

Telecommunications Bonding Backbone (TBB): Standard: TIA/EIA 607.

Performance Requirements: Grounding System Resistance: 5 ohms, maximum per NEC.

Quality Assurance:

Comply with Grounding, Bonding and Electrical Protection Chapter of the TIP Standards (Latest version), BICSI TDM Manual, TIA/EIA 607, and NFPA 70.

Wire:

Material: Stranded copper.
Grounding Conductor: Copper conductor insulated (white and gray).

Bonding Conductor: Copper conductor bare and insulated (green).

Mechanical Connections:
Product Description: Exothermic materials, accessories, and tools for preparing and making permanent field connections between grounding system components.

Installation:
Install in accordance with TIP Standards, BICSI TDM Manual, TIA/EIA 607, and NFPA 70.
Cables and bus bars: Identify and label per Section 27 05 53 – Identification for Communications System.
Permanently attach equipment and grounding conductors prior to energizing equipment.

Field Quality Control:
Test: In accordance with TIP Standards, BICSI TDM Manual, TIA/EIA 607, and NFPA 70.

27 05 28 - Pathways for Communications System
Pathways: Each pathway must be continuous, accessible, viable and usable after completion of the construction. Provide entry and exit points for each section of pathway. Pay special attention to fire-rated and plenum ceilings where hold down clips are used on acoustical tiles making access difficult without damaging the ceiling tiles.

Telephone Termination Backboards:

- Provide fire retardant treated; ¾ inch thick, 8-foot high plywood on all walls in the telecom rooms, starting 4-inches above floor.

Paint: Intumescent Systems:

- Benjamin Moore & Company:
- Primer: Moorcraft Super Spec Alkyd Enamel Undercoater & Primer Sealer (C245).
- 1st and 2nd Coats: 200 Latex Fire Retardant Coating (M59).

Contego International, Inc. and Zinsser Company, Inc.

- Primer: Zinsser Bulls Eye 1-2-3, Kilz Premium.
- 1st and 2nd Coats: Contego Intumescent Latex (Thin Film).
- Color: White.

27 05 28.36 – Cable Trays for Communications Systems:
Metal Ladder-Type Cable Tray:
Manufacturers: Chatsworth Products, Inc., Model Telco-Style Cable Runway (Phone: 800-834-4969; Website: www.chatsworth.com), or Cooper B-Line, Inc., Model Two-Side Rail System (Phone 618-654-2184; Website: www.g-line.com)

Project Description:
NEMA VE 1, Class 12C ladder type tray.
Material: Steel.
Inside Width: 12 inches, minimum.
Inside Depth: 1½ inches.
Straight Section Rung Spacing: 12 inches on center.

Furnish manufacturer's accessories, including wall support brackets, connectors, hold-down clamps, grounding clamps, bonding jumpers, and color – gray.

27 05 53 - Identification for Communications
Labeling: Nomenclature and location shall follow requirements exactly.

Manufacturers:
- Anthony-Lee Associates, Inc. (Phone: 800-275-8911; Website: www.anthony-lee.com)
- Impact, Inc. (Phone: 877-549-7600)
- Sharpmark USA (Phone: 407-321-7394;
- Other acceptable labeling systems:

Systems provided by manufacturer with equipment, such as protectors, termination blocks, punch down block, racks, and modular patch panels, specified in Section 27 13 43 – Communications Services Cabling

Nameplates: General: Engraved laminated plastic nameplate fastened with adhesive. Size: ½ inch high letters. Color: Black lettering on yellow background. Legend: Cable trays, existing and new.
Message:
WARNING! DO NOT USE CABLE TRAY AS WALKWAY, LADDER, OR SUPPORT. USE ONLY AS MECHANICAL SUPPORT FOR CABLES AND TUBING!

Labels: General: Size as required for message. Color: Black lettering on white background. Mechanically produced labels. Text Size: ½ inch high letters, minimum, except as approved by Trustees. Legend: Refer to Schedules and drawings.

Refer to Schedules and drawings.

Cable Markers:
- Wrap-Around Type: Vinyl or polyester. Self-laminating, wrap-round style with clear portion wraps around cable and laminates legend. Use before or after termination. Size: Length to completely wrap around cable and laminate legend. Width to allow 1/8 inch space between edges and text.
- Split sleeve Type: Heat shrink polyolefin. Slip over open end of cable: full-circle. Use before termination. Size: Length as recommended by Manufacturer. Width to allow 1/8 inch space between edges and text.
- General: Color: Black lettering on white background. Mechanically produced labels. Test size: ¼ inch high letters, minimum, except as approved by Trustees.
- Legend: Backbone Cable between Campus Telephone Utility connection and Building Service Entrance. Riser Cable within Building between Main Data Frame Room#_____ and Telecom Room#_____. Horizontal Cable to each outlet from Main Data Frame Room #_____ and Telecom Room #____.
- Message: Refer to Schedules and drawings.

Stencils: General: Clean cut symbols and letters. Size: 1 inch high letters. Colors: Black lettering on white background.

Stencil and Background Paint by Manufacturers:
- Benjamin Moore & Company (info@benjaminmoore.com)

- Frazee Paint and Wall Covering (mbrower@grazee.com)
- The Sherwin-Williams Company (www.sherwin-williams.com)

Painting Systems:
- Benjamin Moore: Primer – Eco Spec Interior Latex Flat 219; 1st and 2nd Coat, and Stencil lettering – Eco-Spec Latex Semi-Glass 224.
- Frazee Paint: Primer – 066 Envirokote Primer; 1st and 2nd Coat, and Stencil lettering – 032 Envirokote SG.
- Sherwin-Williams: Primer – S-W PrepRite Masonry Primer; 1st and 2nd Coat, and Stencil lettering – S-W ProClassic Waterborne Acrylic Semi-Gloss, B31 Series.

Legend: Vault Walls at Ducts for Backbone Cable between Campus Telephone utility connection and Building Service Entrance.

Installation:
Rack Labeling Installation:
Order of labeling: Labeling and layouts shall be placed in ascending alpha-numerical order.

1
2
3
4

or

1A
1B
1C
1D

Refer to Schedules for examples.
Provide label for each Rack. Place label at top, front of Rack, except if label is obscured from view, place on ladder raceway, centered directly above Rack.
Examples: **RACK 1, RACK 2, RACK 3,** etc.

Patch Panel Labeling Installation:
- Center on extreme left and right edges on each Panel where space is available. Do not cover individual Port Labeling areas. Start at top of Rack and label sequentially downward, except do not count the following types of items as "Panels:"
- Horizontal Wire Management at top, bottom and between each Panel and Active Device.
- Active Devices – Hubs Switches, Routers, UPS's, etc.
- Label: Rack # Panel #
- Examples: R1P1, R1P2, R2P2

Patch Panel Port Labeling Installation:
- Label each Port.
- Label (Port to Faceplate in Rooms): Room # and Faceplate / Jack #.
- Label (Individual Ports): Rack # Panel # Port ##.
- Examples: R1P1t01, R1P2t02, R2P3Pt18

27 10 00 - Structured Cabling
Cabling – Voice and Data
There is no differentiation between voice and data on structured cabling. Category 6 cable may be used for either or both. For example, a 4-port communication faceplate may have 3 data and 1 voice.

Related Sections:
Firestopping - Existing and new firestopping at fire rated openings for conduit and cable trays. Campus Preference for wall and floor penetrations of cabling for telecommunications, fire alarm and security is Specified Technologies Inc., EZ Path Fire Rated Pathways Model Series.

27 11 13 - Communications Entrance Protection

Protectors (Copper Cable):
Manufacturer:
Not Recommended: Circa.

189 ET1 Building Entrance Terminals:
Steel; five-pin tin/lead alloy plated contact over spring tempered phosphor base metal with an operating temperature of -13 degrees F to 149 degrees F. UL 497 Listed for Primary Protection. 100 pair configurations. 5-pin gas-tube unit protectors. Input Stubs – 710 Style Connector and Output Stubs 66 Style Termination Block.

27 11 16 - Communications Cabinets, Racks, Frames and Enclosures

Rack Mounted Fiber Optic Cabinet
Description: Optical fiber enclosure to house fiber pigtails fused to optical cable terminated with SC and ST connectors.

Manufacturer: Panduit – Substitutions are not permitted.

Racks Standards:
- Provide 6 inches vertical wire management on both side, front and back of each equipment rack.
- Provide non-coated racks and star washers. Note: No painted or coated equipment racks and accessories.
- Racks shall be installed plumb, true and square, otherwise equipment doesn't fit into the racks.

Description: Mounting Brackets for mounting cable tray ladder rack to top of rack. Mill Finish Aluminum and Velcro cable ties.

Free-Standing Racks:
Front and Rear channels complying with ANSI/TIA/EIA 568-B.1. Top Cable Trough with waterfall and built-in Path/Horizontal Cable Distribution Separator. EIA Hole Pattern front and rear with 19-inch assembly, vertical patch cord cable management rings, pre-drilled base for floor attachment, and 7-foot (45 rack units).

27 11 19 - Communications Termination Blocks and Patch Panels

Submittals:
Shop Drawings: Patch panel and rack layouts. Telecommunications Representative shall review and approve shop drawings.

Patch Panels:
- Preferred Manufacturer: AMP.
- Provide 48 ports.
- Note: 24 ports not allowed.

- Provide two rack units (RU) horizontal wire management above, below and between all patch panels. Refer to the "Campus Standard Positions of Items in Telecomm Rack."

Termination Blocks

- Standards: 66 Block / Rack Patch Panel for voice grade Telecom cabling to equipment racks. Note: 110 blocks are no longer used, and are not permitted.
- Description: Punch Down Block: 66M1-50 Cross Connect Blocks, 50-pair.

Modular Patch Panels

Description: Category 6 Horizontal Patch Panels with rack-mounted assembly of terminals and accessory patch cords. 48 ports, 3.5 inches high by 19 inches wide. Comply with ANIS/TIA/EIA 568-B.2-1.

27 13 43 - Communications Services Cabling

System Description:

- Comply with TIP Standards. Exception: Terminate Multimode fibers with ST connectors, and singlemode fibers with SC connectors.
- Backbone Cable: Use OSP UTP (Outside Plant, Twisted Pair) and optical fiber backbone cables between Campus Telephone Utility connection and Building Service Entrance. Utilize fusion splices and pigtails, only.
- Coordinate work Firestopping.

Products:

Backbone Cable:

OSP UTP Cable: TIA/EIA 568, 150-ohm, twisted-pair cable, pairs sized as per TIP Standards, 24 AWG copper conductor.

Optical Fiber Cable: UV-Resistant, Flame Retardant Outer Jacket, Gel-free design fully waterblocked using craft-friendly, water-swellable yarns and Tape, 3.0 mm buffer tube, fillers, ripcords, dielectric cable construction and central member, color-coded fibers and buffer tubes, UL Listed (OFNR and FT-4). Prefer Hybrid consisting of Multimode and Single Mode Fibers configured as follows: Multimode Fiber to be 62.5/125 µm (850/1300 nm) fiber type, 3.5/1.0 dB/km maximum attenuation, 200/500 MHz•km minimum OED Bandwidth, 220/ - MHz•km minimum effective Modal Bandwidth, 300/550 m serial gigabit Ethernet distance, and 33/ - m serial 10 gigabit Ethernet distance. Single Mode Fiber to be 1310/1383/1550 nm fiber type, 0.4/0.4/0.3 dB/km maximum attenuation, 5000/ - / - m serial gigabit Ethernet distance, and 10000/40000m serial 10 gigabit Ethernet distance. Contractor's Option is to provide Optical Fiber Cable for Backbone Cables as either "Hybrid" or separate Multimode Fiber and Single Mode Fiber Cables.

Manufacturer:

Optical Fiber Cable: Corning Cable Systems – FREEDM Loose Tube Gel-Free Cables. Substitutions must be submitted for Campus approval, and manufacturer must be an ISO 9001 registered company.

Execution:

Prohibited: Sharing pathways and fire-rated penetrations with other than telecommunications cabling.

Installation:
Install per TIA/EIA 568-B.1, and manufacturer's installation instructions.

Install following BICSI installation guidelines.

Cable Slack:

Telecom Rooms: 1 times perimeter of room; 30 feet, minimum.

Vaults: 1.5 times perimeter of vault; coil around vault on wall.

27 14 00 - Communications Riser Cabling

System Description:

- Comply with TIP Standards. Exception: Terminate Multimode fibers with ST connectors, and singlemode fibers with SC connectors.
- Cable: Riser Cable: Use UTP (twisted pair) and optical fiber backbone cables within building between Main Data Frame Room and Telecom Room.

Riser Cable:
Unshielded Cable: TIA/EIA 568, 150-ohm shielded, twisted-pair cable, 100 pairs, 24 AWG copper conductor.

Optical Fiber Cable: UV-Resistant, flame retardant outer jacket, gel-free design fully waterblocked using craft-friendly, water-swellable yarns and Tape, 3.0 mm buffer tube, fillers, ripcord, dielectric cable construction and central member, color-coded fibers and buffer tubes, UL Listed (OFNR and FT-4). Prefer Hybrid consisting of Multimode and Single Mode Fibers configured as follows: Multimode Fiber to be 62.5/125 µm (850/1300 nm) fiber type, 3.5/1.0 dB/km maximum attenuation, 200/500 MHz•km minimum OED Bandwidth, 220/ - MHz•km minimum effective Modal Bandwidth, 300/550 m serial gigabit Ethernet distance, and 33/ - m serial 10 gigabit Ethernet distance. Single Mode Fiber to be 1310/1383/1550 nm fiber type, 0.4/0.4/0.3 dB/km maximum attenuation, 5000/ - / - m serial gigabit Ethernet distance, and 10000/40000m serial 10 gigabit Ethernet distance. Contractor's Option is to provide Optical Fiber Cable for Riser Cables as either "Hybrid" or separate Multimode Fiber and Single Mode Fiber Cables.

Manufacturer:
Optical Fiber Cable: Corning Cable Systems – FREEDM Loose Tube Gel-Free Cables. Substitutions must be submitted for Campus approval, and manufacturer must be an ISO 9001 registered company. Cable Strain Relief: For vertical cable risers.

Execution:
Prohibited: Sharing pathways and fire-rated penetrations with other than telecommunications cabling.

Installation:
Install cable strain relief on vertical cable risers.
Install per TIA/EIA 568-B.1, and manufacturer's installation instructions.
Install following BICSI installation guidelines.
Cable Slack: Telecom Rooms: 15 feet, minimum; coil in Cable Tray.

27 15 00 - Communications Horizontal Cabling

System Description:

Comply with TIP Standards. Exception: Terminate Multimode fibers with ST connectors, and singlemode fibers with SC connectors.

Horizontal Cable: Use Cat 6 cables to each work area outlet from Main Data Frame Room and Telecom Room(s).

Products:

Horizontal Cable:

- Physical Characteristics: 100-ohm Category 6 unshielded twisted pair (UTP) cable; 023 AWG (0.58 mm), bare copper wire insulated with FEP (plenum) copper conductor, jacketed with flame-retardant PVC, 4 twisted pairs, FEP insulated (<u>Not Allowed</u>: 2x2 and 3x1 construction); plenum rated and NFPA 262, CMP flame rating; complying with ANSI/ICEA S-80-576 and ANSI/TIA/EIA 568-B.2-1; suitable for environment. 0.022-inch (0.56 mm) maximum conductor diameter, 0.22-inch (5.6 mm) cable diameter, 28 lb./kft. (41.3 kg/km) nominal cable weight, 25 lb. (110 N) maximum installation tension, 400 N minimum ultimate breaking strength per ASTM D 4695, 1.0-inch (25.4 mm) minimum bend radius at minus 20 degrees Celsius without jacket or insulation cracking.
- Cable color is blue and pair color coding is as follows:

Pairs	Colors		Code
Pair 1	White/Blue;	Blue	W-BL; BL
Pair 2	White/Orange;	Orange	W-O; O
Pair 3	White/Green;	Green	W-G; G
Pair 4	White/Brown	Brown	W-BR; BR

- Transmission Characteristics: DC Resistance per ASTM D4566, 9.38 ohms per 100 m maximum at 20 degrees Celsius, 3 percent (when measured at or corrected to 20 degrees Celsius) unbalance between any two conductors of any pair. 4.4 Nf maximum mutual capacitance of any pair at 1 kHz for 100 m of cable, and 330 pF per 100 m maximum capacitance unbalance to ground at 1 kHz of any pair.

Cable Management:

- Spaces (including Plenum):
    - Cable Ties: Reusable Hook and Loop fasteners, straps or tape, flexible, fire retardant, plenum rated, meeting UL 94-V2, FAR 2.5 853 A/B, and NEC Section 300-22 paragraph (C) and (D).
    - “J”-Hooks: May be used to support small bundles of 3 faceplates or 12 cables, maximum.
    - “Arlington” Loops: Plenum rated; may be used to support small bundles.
    - Prohibited: Nylon Cable ties. Connection cable management ties, hooks and loops to suspended ceiling elements.
- Work Area Outlets:

    Faceplates shall be UL listed. Match faceplate color used or other utilities in building. Exception: For Faceplates, match surface raceway color, and use stainless steel wall-mounted telephone faceplates. For Voice and Data Jacks, if match is not possible, Campus preference is Almond, then White.
- Preferred Manufacturer:
    - Faceplates: AMP, 4-Port (#557502), and 2-Port (compatible with and to be used for Wiremold 4600 Raceway).
    - Voice and Data Jacks: AMP, Model 1375055.

- Website: http://www.ampnetconnect.com/product_groups.asp?grp_id=1697

- Execution:
  - Prohibited: Sharing pathways and fire-rated penetrations with other than telecommunications cabling.
  - Installation:
    - Install per TIA/EIA 568-B.1, and manufacturer's installation instructions.
    - Install following BICSI installation guidelines.
    - Install cabling clear of obstructions:
      - Water, waste and fire sprinkler piping.
      - Suspended ceiling components.
      - HVAC units and ducting.
      - Building system control cabling.
      - Wiring for door bells and door access, and transformers.
      - Antenna feed lines.
      - Protruding nails, screws, and sharp or rough objects.
    - Install cabling with <u>minimum</u> separation:
      - Crossing electrical conduits at 90 degree angle: 2 inches.
      - Parallel with electrical conduits: 6 inches.
      - From electronic devices: 12 inches.
      - From fluorescent lights: 6 inches.
      - From fluorescent light ballasts: 12 inches.
      - From hot surfaces above 104 degrees F: 12 inches.
      - From ceiling tiles: 6 inches.
    - Cable Slack.
      - Provide slack loop at building structural seismic joints.
      - Telecom Rooms: 15 feet, minimum; dressed in Cable Tray.
      - J-boxes at Work Area Outlets: 1 foot, minimum; coil in J-box.
      - Leave pathways clear and accessible for maintenance, moves, additions and changes.

Field Quality Control:
Testing Equipment: Bi-directional fiber tests shall be done using optical power meter and light sources, and OTDR equipment.

Testing is to demonstrate product capability, and compliance with specification installation and performance requirements. Test instrumentation for fiber optic plant (OTDR = optical time domain reflectometer) is easy to use, but difficult to interpret. For that reason, Campus requires the person doing the testing to have been trained and approved by the testing equipment manufacturer in using the equipment. The testing shall be observed by the Campus Communications Inspector, and results submitted for review and acceptance. If the test data in the opinion of the Campus Communications Inspector indicates improper testing, then the tests shall be redone by a different experienced and qualified tester to the satisfaction of the Campus Communications Inspector. If the test data indicates too great a drop, the length of cable and associated termination shall be replaced and retested. Testing shall be conducted on each fiber optic cable link. The specifications need to reflect these Campus Standards and Requirements.

Inspect and text optical fiber cables in accordance with NETA ATS, except Section 4. Perform inspections and tests listed in NETA ATS, Section 7.25.

Inspect and test copper cables and terminations in accordance with TIA/EIA 568.

27 16 19 - Communications Patch Cords, Station Cords, and Cross Connect Wire
Fiber Optic Pigtails: Color coded. Size: 6 feet, minimum.

27 32 23 - Elevator Telephones
Coordinate requirements with Section 14 24 00 – Hydraulic Elevators.

27 32 26 - Ring-Down Emergency Telephones
Pedestal Unit:

- Code Blue CB 1-s: Vandal resistant, ADA compliant, 12.75 inches diameter by 91.5 inches high.
- Communication system: Code Blue CB3100, single touch communications device button with automatically dial preprogrammed number that shall simultaneously activate blue strobe light.
- Website: http://www.codeblue.com/Central/cb01s.html

Wall Unit:

- Code Blue CB 2-s: Vandal resistant, ADA compliant, stainless steel, wall mounted, 42 inches high by 12 inches wide by 8.75 inches deep.
- Communication system: Code Blue CB3100, single touch communications device button with automatically dial preprogrammed number that shall simultaneously activate blue strobe light.
- Website: http://www.codeblue.com/Central/cb02s.html

Division 28 - Electronic Safety and Security

28 01 00 - Operation and Maintenance of Electronic Safety and Security
Electronic Monitoring and Control:

Fire Alarm Testing:

Standard: Provide bypasses for testing signals, dampers, HVAC shutdowns, etc.

28 05 28 - Pathways for Electronic Safety and Security
Fire Alarm Cables:
Standard: Fire alarm sensor and network cables shall be in conduit.

Keys for Cabinets and Padlocks:

- Cabinets and Equipment: Provide 2 keys per panel. Coordinate with Section 08 06 05 – Key Schedule.
- Padlocks: Coordinate with Section 08 06 05 – Key Schedule.
- Closeout Submittal: Provide panel keys separated and labeled. Provide location, room number, quantity, manufacturer name and model numbers of keys, and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

28 31 00 - Fire Detection and Alarm
Fire Alarm System: General:
Fire Alarm System: Manufactured by an ISO 9001 certified company and meet requirements of BS EN9001: ANSI/ASQC Q9001-1994.

In-place Campus Alarm Central Station System – Provide panel components and programming of panel to work within Campus system. The system requires telephone line communication, and communication formats in Ademco Contact ID, or Radionics Modem IIe.

Scope:
Basic Performance:
- Alarm, trouble and supervisory signals from all intelligent reporting devices: Encoded on NFPA Style 4 (Class B) Signaling Line Circuits (SLC).
- Initiation Device Circuits (IDC): Wired Class A (NFPA Style D) as part of an addressable device connected by the SLC Circuit.
- Notification Appliance Circuits (NAC): Wired Class A (NFPA Style Z) as part of an addressable device connected by the SLC Circuit.
- On Style 6 or 7 (Class A) configurations on a single ground fault or open circuit on the system Signaling Line Circuit: Not cause system malfunction, loss of operating power, or the ability to report an alarm.
- Alarm signals arriving at the FACP: Not lost following a primary power failure (or outage) until the alarm signal is processed and recorded.
- NAC speaker circuits: Arrange such that there is a minimum of one speaker circuit per floor of the building or smoke zone, whichever is greater.
- Audio amplifiers and tone generating equipment: Electrically supervised for normal and abnormal conditions.
- NAC speaker circuits and control equipment: Arrange such that loss of any 1 speaker circuit will not cause the loss of any other speaker circuit in the system.
- 2-way telephone communication circuits: Supervised for open and short circuit conditions.

Basic System Functional Operation: When a fire alarm condition is detected and reported by one of the system initiating devices, the following functions shall immediately occur.
- The system alarm LED on the system display shall flash.
- A local piezo electric signal in the control panel shall sound.
- A backlit LCD display shall indicate all information associated with the fire alarm condition, including the type of alarm point and its location within the protected premises.
- Printing and history storage equipment shall long the information associated with each new fire alarm control panel condition, along with time and date of occurrence.
- All system output programs assigned via control-by-event interlock programming to be activated by the particular point in the alarm shall be executed, and the associated system outputs (notification appliances and/or relays) shall be activated.

Submittals:
- Product data: Submit for each item, including details of construction relative to materials, dimensions of individual components, profiles, and finishes.
- Shop Drawings: Submit with manufacturer’s name(s), model numbers, ratings, power requirements; equipment layout, device arrangement, wiring point-to-point diagrams, and conduit layouts; annunciator layout, configurations and terminations; sections of typical trim members; and attachments to other work.
- Operating and Maintenance Manuals: Include manufacturer’s name(s), technical data sheets; wiring diagrams indicating internal wiring for each device and interconnections between the items of equipment; clear, concise description of operation to properly operate the equipment and systems. Submit with shop drawings.

- Certifications for Installation Supervisor and Maintenance Personnel: From major equipment manufacturer certifying personnel are authorized representatives of the major equipment manufacturer. Include names and addresses in the certification. Submit with shop drawings.
- Comply with Section 01 33 00 – Submittal Procedures.

Post Contract Maintenance:
Provide complete maintenance and repair service for fire alarm systems from a factory trained, authorized representative of the major equipment manufacturer for a period of five (5) years after expiration of the guaranty.

Products:
Main Fire Alarm Control Panel or Network Node
Purpose: Communicate with and control intelligent addressable smoke and thermal (heat) detectors, addressable modules, printer, annunciators, and other system controlled devices. Use 2-way telephone communication circuits, supervised for open and short circuit conditions.

Standard:
- Manufacturer: Notifier by Honeywell International Inc.
- Models: ONYX Series with voice notification capabilities
- Website: http://www.notifier.com/products/controlpanels.htm
- Note: Fire•Lite Alarms by Honeywell will be considered by Campus Facility Services on a case-by-case basis. No other substitutions allowed.

Operator Control:
- Acknowledge Switch: Upon activation in response to new alarms and/or troubles: Silence local panel piezo electric signal and change alarm and trouble LEDs from flashing mode to steady-ON mode. If multiple alarm or trouble conditions exist, depression of switch shall advance LCD display to next alarm or trouble condition.
- Depression of switch: Silences remote annunciator piezo sounders.

Alarm Silence Switch: Upon activation: Programmed alarm notification appliances and relays shall return to normal condition after an alarm condition. Notification circuits and relay selection silenceable by switch shall be field programmable with confines of applicable standards. PACP software shall include silence inhibit and auto-silence timers.

Alarm Activate (Drill) Switch: Shall activate notification appliance circuits. Drill function shall latch until panel is silenced or reset.

System Reset Switch: Upon activation: Electronically-latched initiating devices, appliances and software zones, plus associated output devices and circuits, shall return to normal condition.

Lamp Test: Upon activation: Local system LEDs shall activate, each segment of the liquid crystal display shall light, and panel software revision shall display.

Signal Bypass Switch for Alarm Testing:
  - Horns and strobes shall remain silent.
  - Dampers shall remain open.

System Capacity and General Operation:

Control Panel and Network Nodes:

- Provide or be capable of expansion to 636 intelligent / addressable devices.
- Include Form-C alarm, trouble, supervisory, and security relays rated at 2.0 amps @ 30 VDC, minimum.
- Include Four Class B (NFPA Style Y) or Class A (NFPA Style Z) programmable Notification Appliance Circuits.
- Support 8 additional output modules with 8 circuits each, minimum (64 circuits) for signal, speaker, telephone, or relay. Circuits shall be either Class A (NFPA Style Z), or Class B (NFPA Style Y). {Note to Designer: Coordinate specifications and drawings.}
- Operator Interface Control and Annunciation Panel: Full featured with backlit Liquid Crystal Display (LCD), individual color coded system status LEDs and alphanumeric keypad with easy touch rubber keys for field programming and control of fire alarm system.
- Special Tools, PROM Programmers and PC Based Programmers: Not required to program, configure and expand system in the field.
- Memory ICs: Not required to be replaced to facilitate programming changes.
- Programming of System:
- Allow programming of an input to activate an output or group of outputs.
- Support 20 logic equations, minimum, including “and,” “or,” and “not,” or time delay equations for advanced programming.
- Logic equations shall require a PC with software utility design for programming.
- Not Allowed: Systems with limited programming (such as general alarm), complicated programming (such as a diode matrix), and requiring a laptop personal computer.
- Features:
- Drift compensation to extend detector accuracy of life, and include a smoothing feature allowing transient noise signals to be filtered out.
- Detector sensitivity test, meeting requirements of NFPA 72, Chapter 7.
- Maintenance alert, with two levels (maintenance alert / maintenance urgent), to warn of excessive smoke detector dirt and dust accumulation.
- Nine sensitivity levels for alarm to be selected by detector. Alarm level range between 0.5 to 2.35 percent per foot for photoelectric detectors, and 0.5 to 2.5 percent per foot for ionization detectors. Support sensitive advanced detection laser detectors with an alarm level range of 0.03 percent per foot to 1.0 percent per foot. Include nine levels, minimum of Pre-alarm, selected by detector, to indicate impending alarms to maintenance personnel.
- Display or print system reports.
- Alarm verification, with counters and trouble indication to alert maintenance personnel with a detector enters verification 20 times.
- PAS pre-signal, meeting NFPA 72 3-8.3 requirements.
- Rapid manual station reporting (under 3 seconds).
- Meet HFPA 72 Chapter 1 requirements for activation of notification circuits within 10 seconds of initiating device activation.
- Periodic detector test, conducted automatically by software.
- Self-optimizing pre-alarm for advanced fire warning, which allows each detector to learn its particular environment and set its pre-alarm level to just above normal peaks.
- Cross zoning with the capability of counting two detectors in alarm, two software zones in alarm, or one smoke detector and one thermal detector.
- Walk test with a check for two detectors set to same address.

- Control-by-time for non-fire operations, with holiday schedules.
- Day/night automatic adjustment of detector sensitivity.
- Device blink control for sleeping areas.
- Coding:
- FACP shall be capable of coding main panel node notification circuits in March Time (120 PPM), Temporal (NFPA 72 A-2-2.2.2), and California Code.
- Panel notification circuits (NAC 1, 2, 3 and 4) shall support Two-Stage operation, Canadian Dual Stage (3 minutes) and Canadian Dual Stage (5 minutes). Two-stage operation shall allow 20 Pluses Per Minute (PPM) on alarm and 120 PPM after 5 minutes, or when a second device activates. (Canadian Dual Stage is the same as Two-Stage, except will only switch to second stage by activation of Drill Switch 3- or 5-minute timer.)
- Provide coding option to synchronize specific strobe lights designed to accept a specific “sync pulse.”
- Network Communication:
- Network Architecture: Local Area Network (LAN), a firmware package that utilizes a peer-to-peer, inherently regenerative communication format and protocol.

Protocol shall be based on ARCNET or equivalent. The network shall use a deterministic token-passing method.

Not acceptable: Collision detection and recovery type protocols; Master, polling computer, central file computer, display controller or other central element (weak link) in the network; cascading of CPUs or master-slave relationships at network level to facilitate network communications.

Nodes may be an intelligent Fire Alarm Control Panel (FACP), Network Control Station PC (NCS) or Network Control Annunciator (NCA).

Network shall be capable of expansion to 103 nodes, minimum.

Failure of a node shall not cause failure of communication degradation of other nodes or change the network communication protocol among surviving nodes located within distance limitations. Each node/panel shall communicate on the network at a baud rate of 312 KBPS (kilo bits per second), minimum.

- Network node addresses shall be capable of storing Event equations. The event equations shall be used to activate outputs on one network node from inputs on other network nodes.
- Network shall be capable of communicating via wire or fiber optic medium. A wire network shall include a fail-safe means of isolating nodes in event of complete power loss to a node.
- Network Repeaters: A network repeater shall be available to increase twisted pair distance capability in 3000-foot increments, minimum. Optionally, a repeater shall be available for fiber optics to increase the wire distance in 10 dB increments, minimum. A mix (hybrid) fiber/wire network repeater shall also be supported.

Not acceptable: Systems with distance limitations, and no means to regenerate signals.

Fiber Optic Network Communication: Network shall support the following:

- Size: 62.5 micrometers / 125 micrometers.
- Type: Multimode, Dual Fiber, and Plenum Rated.
- Distance: 10 dB, maximum total attenuation between network nodes.
- Connector Type: ST.

Central Microprocessor:

- State-of-the-art, high speed, 16-bit RISC device capable of communicating with, monitoring and controlling external interfaces.
- Include an EPROM for system program storage, Flash memory for building-specific program storage, and a “watch doc” timer circuit to detect and report microprocessor failure.

Capable of t-tapping Class B (NFPA Style 4) Signaling Line Circuits (SLCs).

Systems that do not allow t-taps, or have restrictions (for example, in the amount of t-taps, length of t-taps, etc.) are not acceptable.

System Components

- Programmable Electronic Sounders
- Speakers
- Strobe lights meeting requirements of 2001 CBC and UL 1971, fully synchronized.
- Manual Fire Alarm Stations.
- Conventional Photoelectric Area Smoke Detectors.
- Conventional Ionization Type Area Smoke Detectors
- Duct Smoke Detectors.
- Projected Beam Detectors
- Automatic Conventional Heat Detectors
- Water Flow Indicator
- Sprinkler and Standpipe Valve Supervisory Switches
- Alphanumeric LCD Type Annunciator
- Portable Emergency Telephone Handset Jack
- Fixed Emergency Telephone Handset
- Universal Digital Alarm Communicator Transmitter (UDACT)
- Field Wiring Terminal Blocks.
- Printer
- Video Display Terminal (VDT)

Equipment and Material:

- New and manufacturer’s current model.
- Materials, appliances, equipment and devices: Tested and listed by a nationally recognized approval agency for use as part of a protective signaling system, and meeting the National Fire Alarm Code.
- Fasteners and supports: Adequately sized to support the required load.

Conduit: Sized to provide 40 percent, maximum fill of interior cross sectional area.

Wire:

- Initiating Device Circuits and Signaling Line Circuits: New 18 AWG, minimum.
- Notification Appliance Circuits: 14 AWG, minimum.
- Meet local, state and national codes (e.g., NEC Article 760).
- Meet fire alarm system manufacturer recommendations.
- Listed and approved by a recognized testing agency for use with a protective signaling system.
- Wire and Cable not installed in conduit:

- Fire resistance rating suitable for the installation as per NFPA 70 (e.g., FPLR).
- Not be exposed (visible).

Wire for multiplex communication circuit (SLC):
- Twisted and unshielded, and support 12,500 feet, minimum.
- System designed to permit use of IDC and NAC wiring in the same conduit with the SLC communication circuit.

Field Wiring: Electrically supervised for open circuit and ground fault.

Terminal Boxes, Junction Boxes and Cabinets: UL listed for purpose and use.

Capable of t-tapping Class B (NFPA Style 4) Signaling Line Circuits (SLCs).

Execution:
- Initiating Circuits: Arranged to serve like categories, such as manual, smoke, and water flow.
- Mixed Category Circuits: Not allowed.
- Exception: Signaling line circuits connected to intelligent reporting devices.

Fire Control Panel:
- Connect to separate dedicated branch circuit; 20 amperes, maximum.
- Label circuit at main power distribution panel as "FIRE ALARM."
- Primary wiring: 12 AWG.
- Ground panel.

Division 31 - Earthwork

31 10 00 - Site Clearing

All unsuitable materials, as defined in a Soils Engineering Report, shall be removed from the site and disposed of in a suitable location.

Erosion control measures and BMPs shall be implemented in accordance to the SWPPP and dust control shall be employed at all times.

31 22 00 - Grading

Ponding and areas of low flow gradients should be avoided unless part of a designed storm water mitigation measure such as bio-retention swales, or rain gardens. Positive surface drainage or drainage devices such as swales or drain inlets shall be provided wherever required to convey water away from building foundations.

Storm drainage shall be directed away from all buildings. If in the rare case this cannot be accomplished, redundant drainage systems are required. There shall be no potential for ponding water that could flood into a building. Safe overland flow paths must be provided in case storm drains clog or their capacity is exceeded.

Site retaining or planter walls shall have waterproofing and adequate drainage to prevent additional loading. Provide a water proofing system at the backside, and foundation drains at the base. Drains may be tied into the storm drain or an appropriate location. Retaining walls more than 30" in height from top to finished grade are required to have a 42" high guardrail or other approved protective measure.

Provide flat areas for all utility slabs, vaults or boxes.

Finished grade elevations shown on the grading plan shall be to the finished surface elevation at 0.1' increments for soils surfaces and at 0.01' increments for hard finish surfaces. Contractor shall subtract structural thickness of pavements of top soil to achieve rough grading elevations. Proposed grading contours shall be indicated at 1-foot contour intervals.

31 23 33 - Trenching and Backfilling
Standard: Coordinate backfilling with installation of underground duct markers for utilities. Backfill with sand to 12 inches above utility duct. Install duct markers 6 inches below finished surface.

Schedule: Storm and Sanitary Piping: Cover pipe and bedding with ____ inches

Utilities	Fill Type	Bedding and Cover Thickness, minimum	Compaction, minimum	Underground Duct Markers
Water Utilities Including Irrigation	Sand	1" below 3" above		
Sanitary Sewerage Utilities				
Storm Drainage Utilities				
Natural-Gas Distribution Utilities				
Hydronic Utilities			95%	
Electrical Utilities				
Communication			95%	

Division 32 - Exterior Improvements

32 00 00 - Exterior Improvements
Division of State Architect (DSA) reviews design conformance with CBSC Chapter 11 for accessibility requirements. There are differences between this code and the Americans with Disabilities Act (ADA).

The design professional is required to provide compliance to both the CBSC and the ADA. The University is accountable for compliance to both the ADA and the CBSC accessibility requirements. Drawings describing accessibility requirements provide the DSA reviewer with information that shows compliance to the provisions of CBSC Chapter 11B, as well as other related requirements applied to the project from federal and local agencies.

When required, at least one accessible route shall be provided from public transportation stops, accessible parking and accessible passenger loading zones, and public streets or sidewalks to the accessible building entrance they serve. When applicable, at least one accessible route shall connect accessible buildings, facilities, elements, and spaces that are on the same site. The accessible route shall coincide, to the maximum extent feasible, with the approved main campus circulation routes.

32 01 90 - Operation and Maintenance of Planting
Planting Guarantee:
During the Guarantee to Repair Period specified in the General Conditions the Contractor shall be liable for damages to all trees covered by the provisions of this Section. Compensation to the University shall be as outlined below.

Contractor will not be held responsible for damages due to vandalism or freak acts of nature during the guarantee period. Immediately report such conditions to the University's Representative.

Protection and Preservation Procedures

On any site survey map, all trees should be identified whose root systems are likely to be impacted by construction equipment, staging areas, proposed walks and roads, utility corridors, and any cut or fill activities. A tree protection zone (TPZ) shall be established to encompass the critical root zone (CRZ) and maintained for all trees to be preserved in a construction site. TPZ should be determined by the tree size, health, how the species responds to construction damage, and should be adjusted according to specific tree and site factors; calculated as 1.50'x the diameter at breast height (DBH) of the tree in inches. For example, a healthy California black walnut with a 30" diameter. = 1.50' X 30" = 45' radius TPZ.

At no time should the TPZ be less than a radius of 6'. The TPZ must be clearly identified as an exclusion zone, where construction and equipment use is prohibited, and must be fenced before demolition to avoid damage.

A barrier shall be constructed for each tree or grouping of trees to protect the trunk and root systems. Chain link 6' fencing shall be the approved barrier material. No root raking shall be allowed within any tree protection zone at any time during clearing, grading, or construction of a project. No equipment or vehicle shall be parked or construction material stored, or substances poured or disposed of or placed within any tree protection zone at any time during clearing or construction of a project.

Vigilance is required to protect trees on construction sites. Monitoring of tree health during and after construction on a regular, frequent basis shall apply to all trees intended for preservation. Monitoring shall be executed by a registered arborist or appointee and shall include watching for signs of tree stress, such as dieback, leaf loss, or general decline in tree health or appearance.

Pruning Materials

- Pruning materials shall be in accordance with current horticultural practices.
- Pruning sterilant shall be Physan 20 Fertilome Type A, or diluted bleach.

Fencing

Fencing Materials: 11-gauge galvanized 6' high chain-link fence with galvanized steel posts at 10' o.c. minimum.

A continuous 6' high temporary chain-link fence will be erected around trees with a caliper of 4" or larger at the dripline, in order to prevent soil compaction, limb damage, or the accidental introduction of toxic materials into the root zone. Fence can be erected around groups of adjacent trees where possible. Otherwise, fence to be erected around individual tree.

The fence will be removed only at the end of construction, as approved by the University's Representative.

Plant Material Protection:

Provide protection for all plant materials designated to be retained. Contractor is responsible for replacing damaged plant life with approved equivalent.

New and existing plant materials shall not be allowed to deteriorate and shall be maintained in a healthy and vigorous condition during the course of construction and maintenance period.

During the course of construction the Contractor shall take all necessary precautions, as outlined herein, to protect existing plant materials to be preserved from injury and death. Protection shall be given to the roots, trunk, and foliage.

The Contractor shall conduct operations continually to completion, unless weather conditions are prohibitive.

Provide ample water supply of potable quality and sufficient quantity for all operations required under this Section.

Trees subject to the provisions of this Section, which have been injured shall be repaired immediately by a certified Arborist. Repairs shall include removal of rough edges, sprung bark and severely injured branches as directed by the Arborist.

Necessary measures shall be taken to maintain healthy living conditions for existing plant materials to be preserved. Such measures shall include monthly washing of leaves for the removal of dust, regular irrigation, root feeding, etc.

Tree protection fencing shall be installed for the protection of existing trees to be preserved. No construction, demolition, or work of any nature will be allowed within the fenced area without prior written approval by the University's Representative.

- Approval by the University's Representative for work within the fenced area shall not release the Contractor from any of the provisions specified herein for the protection of existing trees to be preserved.
- During the course of construction of approved work within the fence area, no roots shall be cut without prior written approval by the University's Representative.

During construction, the existing site surface drainage patterns shall not be altered within the area of the drip line of existing plant materials.

Contractor shall not alter the existing water table within the area of the drip line of existing plant materials. Do not permit the following within the drip line of any existing tree or shrub to be preserved:

- Storage or parking of automobiles or other vehicles.
- Stockpiling of building materials, refuse or excavated materials.
- Skinning or bruising of bark.
- Use of trees as support posts, power poles, or signposts; anchorage for ropes, guy wires, or power lines; or other similar functions.
- Dumping of poisonous materials on or around plant materials and roots. Such materials include but are not limited to paint, petroleum products, dirty water, or other deleterious materials.
- Cutting roots by utility trenching, foundation digging; placement of curbs and trenches, and other miscellaneous excavation without prior written approval by the University's Representative.
- Damage to the trunk, limbs, or foliage caused by maneuvering vehicles or stacking material or equipment to close to the plant.
- Compaction of the root area by movement of trucks or grading machines; storage of equipment, gravel, earth fill, or construction supplies; etc.

- Excessive water or heat from equipment, utility line construction, or burning of trash under or near shrubs or trees.
- Damage to root system from flooding, erosion, and excessive wetting and drying resulting from watering and other operations.

Excavation around trees

- Excavation within the drip lines of trees shall be done only where absolutely necessary, under the direction of a Certified Arborist and with prior approval from the University's Representative.
- Where trenching for utilities is required within drip lines, tunneling under and around roots shall be by hand digging. Main lateral roots and taproots shall not be cut. Smaller roots that interfere with installation of new work may be cut with prior approval from certified Arborist.
- Where excavation of new construction is required within drip line of trees, hand excavation shall be employed to minimize damage to root system. Roots shall be relocated in backfill areas wherever possible. If large, main lateral roots are encountered, they shall be exposed beyond excavation limits as required to bend, and relocate without breaking. If encountered immediately adjacent to location of new construction and relocation is not practical, roots shall be cut approximately 6 inches back from new construction under the direction of a certified Arborist.
- Exposed roots shall not be allowed to dry out before permanent backfill is placed. Temporary earth cover shall be provided, or roots shall be packed with wet peat moss or four layers of wet, untreated burlap and temporarily supported and protected from damage until permanently relocated and covered with backfill. The cover over the roots shall be wetted to the point of runoff daily.
- Branching structure shall be thinned in accordance with National Arborists Association "Pruning Standards and Principles" to balance loss of root system caused by damage or cutting of root system. Thinning shall not exceed 30 percent of existing branching structure.

Tree Trimming

A Certified Arborist, shall be engaged to direct removal of branches from trees if necessary to protect the health of the tree or if required to clear for construction.

In company with the University's Representative, University and a certified Arborist, ascertain the limbs and roots which are to be trimmed. Clearly mark them to designate the approved point of cutting.

Dead and damaged trees that are determined by the Certified Arborist to be incapable of restoration to normal growth pattern shall be removed.

Cut evenly, using proper tools and skilled workmen, to achieve neat severance with the least possible damage to the tree.

In the case of root cuts, apply wet burlap or other protection, approved as noted herein, to prevent drying out, and maintain in a wet condition as long as necessary for temporary protection.

Landscape Maintenance

Plant material will be maintained throughout the duration of the construction period in a healthful manner. Plant material identified which requires special pruning, insect control, fertilization or other remedial health action will be treated during this period. Methods and rates of pesticide and fertilizer application will be reviewed by the University's Representative prior to approval.

Watering

Plant materials will be watered on a regular basis, at a rate consistent with their particular requirements. Verification of the proposed watering schedule shall be reviewed by the University's Representative prior to commencement of the maintenance.

- The maintenance of the plant materials shall comply with standard horticultural practice for the correct watering, fertilizing, pruning and spraying of the specimen boxed trees.
- The maintenance and quality of the plant materials shall be subject to monthly checks. The dates of these checks shall be outlined in the University's Representative's field notification relating to the establishment of the plant maintenance period. Additional checks shall be scheduled as determined by the University's Representative.
- Contractor shall be responsible for performing periodic inspections of existing plant materials to be protected and relocated throughout the construction period, and submit written proposals to the University's Representative for additional maintenance work as may be required to ensure the health and general well-being of the plant material. Contractor shall retain, at the direction of the University's Representative additional specialists as may be required to perform this work.
- Contractor shall keep plant material free from weeds and debris at all times.

Tree Damage Compensation

Damage to existing tree crowns or roots over I" in diameter shall be immediately reported to the University's Representative.

A Certified Arborist shall direct all repairs to trees. Repairs shall be made promptly after damage occurs to prevent progressive deterioration of damaged trees. Repairs shall be at the Contractor's expense.

Development activities shall be planned to preserve and protect trees on Cal Poly's campus. Any tree that must be removed to accommodate development, or because of damage during storm events, disease, or water/sewer repairs must be shown on the site plan and a method of compensation shall apply as follows:

- For trees and shrubs with diameters up to 4 inches, compensation shall be the actual cost on replacement with an item similar in species, size and shape, including:
    - Actual cost of the item boxed out of ground
    - Transportation and delivery of boxed item to Project site.
    - Planting and staking
    - Maintenance, including watering, fertilizing, pruning, pest control and other care for a period of 90 days.
    - Values for removals of trunks up to:
        - Twelve inches - $7,200
        - Thirteen inches - $8,200
        - Fourteen inches- $9,200
        - Fifteen inches - $10,000
        - Sixteen inches - $11,500
        - Seventeen inches - $12,000
        - Eighteen inches and over, add for each caliper inch - $1,200
- A penalty will be assessed for a limb and root injury of $200 per inch of limb/root diameter for and limb/root greater than 2 inches in diameter, measured where the limb should be pruned in order to make a proper thinning cut or root to the point a clean cut can be made.
- A penalty will be assessed of $20 per square inch of tree trunk area injured. This penalty shall be assessed when it is determined that an entity is responsible for the damage to the tree trunk, but the tree is still healthy enough to remain at the site. An example of this kind of damage would be the

collision of a tractor with the trunk of a mature tree where the bark is peeled back, and the injured area will require repair and healing.

The Landscape Services operating account shall receive and manage the tree replacement program. Payment shall be made to the tree planting and replacement account if there is a problem with site space or readiness.

General: The Contractor guarantees the protection of all plant material included as part of this work, in a healthful manner during the duration of the construction period. Destruction of, or significant damage to, any or all of the plant materials to be protected, as determined by the University's Representative, will result in compensation by the Contractor of 3-36" box trees, installed on the site, for each existing tree damaged.

Assessment:
All damaged trees on Cal Poly's campus shall be assessed by a Certified Arborist or outside consultant using the tree evaluation form or other similar methods preapproved by the Landscape Advisory Committee. Results from the evaluation determine whether the tree should be removed, pruned, or receive treatment such as fertilization and insect/disease control. All actions post-evaluation are updated on the tree inventory list.

Enforcement:
Whenever it is determined that violation of this procedure has occurred, the Facilities representative or designee shall immediately issue written and oral notice to the person or company or department in violation, identifying the nature and location of the violation and specifying that remedial action is necessary to bring the violation into compliance.

Penalties:
The person, company or department in violation shall immediately, conditions permitting, commence remedial action and shall have seven (7) working days after the receipt of the notice, or such longer times as may be specified in the notice, to complete the remedial actions required to bring the activity into compliance with this policy.

32 12 00 - Flexible Paving
Consider use of permeable paving.

32 13 00 - Rigid Paving
Concrete Finish: Stamp location of irrigation sleeves into concrete with a “L.”

All hardscape shall be specified to meet (or exceed) SRI of .28 and comply with aging requirements to comply with current LEED standards. Permeable paving is preferred wherever possible.

Concrete is to include fly ash when feasible.

32 16 00 - Curbs and Gutters
Regulatory Requirements
Access for Persons with Disabilities: Comply with California Building Code and Americans with Disabilities Act Accessibility Guidelines (ADAAG) for site development, walks, and sidewalks to ensure access for persons with disabilities. Form, shape, and finish curb cuts in accordance with requirements of most restrictive code.

32 33 13 – Bicycle Racks
Bicycle Storage: Covered, lockable, high density parking storage is to be provided. Racks and base assembly shall be Hot rolled ASTM A36 solid steel bars welded with GMAW (MIG). industrial high gloss powder coated.

Racks should be secured to concrete with two tamper-resistant stainless steel anchors.

32 33 23 – Site Trash and Litter Receptacles
Trash enclosures: A trash enclosure shall be provided that allows for three appropriately sized bins (landfill, recycle and compost). The trash enclosure should be designed to blend contextually with the surrounding buildings, while screening trash and odors as much as possible with a concrete apron.

32 80 00 – Irrigation
Design Guidelines:
The Irrigation system shall utilize the latest technology in water conservation. Smart controllers with rain sensors, matched precipitation rate spray heads in turf and drip irrigation in planting areas will minimize overspray and run-off. Two bubblers will be placed at each tree location for deep root watering. The irrigation system will be zoned according to site condition, species watering needs, and sun exposure. Quick couplers shall be provided every 75 feet throughout the irrigation system for maintenance.

CSU Requirements:
Conserve water resources, including installing controls to optimize irrigation water, and promoting the use of reclaimed water. Prepare a year-round watering program according to seasonal evapotranspiration data for the region. Decorative fountains should be minimized.

Campus Standard Irrigation System:
Calsense Resource Management Systems (Website: www.calsense.com). Provide a programmable, automatically controlled underground sprinkler irrigation system for landscape planting. Provide components from manufacturers matching standard irrigation equipment used by University, including heads, valves, piping circuits, controls, and accessories, or approved equal. Provide looped layout with isolation valves.

Coordinate with Project Manager to locate inside utility or electrical room with network access and accessible to landscape maintenance staff.

Installed Calsense equipment to be reviewed by manufacturer representative and letter of installation certification shall be provided to campus representative, by contractor, prior to completion.

32 84 00 - Planting Irrigation
Irrigation Controller:
Campus Standard: Calsense CS3000 Ethernet and Radio Controlled. Evapotranspiration (ET)-based controller. Controller shall have non-volatile memory to retain program in memory during temporary power failures. Provide diagram of numbered valves and respective irrigated areas for inside panel of each controller.

Wall Mount Enclosure:
The wall-mounted gray box shall be a completely assembled unit, pre-mounted with the designated controller. The box shall be constructed of weather- and vandal-resistant stainless steel. The wall mount

unit shall come complete with transient and lightning protection board and factory-labeled terminals. The transient protection board shall be pre-mounted in the wall mount unit and shall support field replaceable modules which include terminal strips for the connection of irrigation field wires, 2-Wire cable, and weather monitoring devices such as an ET gage, Tipping Rain Bucket, and Wind gage. The wall mount unit shall feature a security-tight locking mechanism, louvered vents, with splash guards, and bee/wasp screens.

All wall mount units shall come with a 10-year limited warranty and shall be fully UL-approved. Double-Wide, Top Entry Enclosure Assemblies:

Calsense Controller Assembly: SSE-D-R, vandal and weather resistant, made entirely of 304 grade stainless steel with the top being 12 gauge and the body being 14 gauge. The pre-assembled vandal resistant enclosure factory pre-assembled and supplied by controller manufacturer shall come complete with 24 VAC lightning and surge protection and all terminals shall be factory labeled. The pre-assembled enclosure shall come provided with an On/Off switch to isolate the controller along with a GFI receptacle. Specific radio antenna(s) shall be pre-mounted and connected on enclosure. The enclosure shall include 2-7/8”, 1-1/2” thick, 6-pin cylinder, die-cast steel padlock with unique shackles design. The assembly shall carry a full U/L listing. The 38 inch height enclosure with flip top should allow for side by side placement of two controllers. All necessary wiring between the two controllers in order to share central communications and/or flow and weather data shall be pre-wired by the manufacturer for easy installation. The main housing shall be louvered upper and lower body to allow cross flow ventilation. A stainless steel backboard shall be provided for the purpose of mounting electronic and various other types of equipment. The stainless steel backboard shall be mounted on four stainless steel bolts that will allow for easy removal of the backboard. The factory pre-assembled enclosures shall carry a ten year limited warranty.

Top Entry Single Enclosure:
Calsense Controller Assembly: SSE-R, vandal and weather resistant, made entirely of 304 grade stainless steel with the top being 12 gauge and the body being 14 gauge. The pre-assembled vandal resistant enclosure factory pre-assembled and supplied by controller manufacturer shall come complete with 24 VAC lightning and surge protection and all terminals shall be factory labeled. The pre-assembled enclosure shall come provided with an On/Off switch to isolate the controller along with a GFI receptacle. Specific radio antenna(s) shall be pre-mounted and connected on enclosure. The enclosure shall include 2-7/8”, 1-1/2” thick, 6-pin cylinder, die-cast steel padlock with unique shackles design. The assembly shall carry a full U/L listing. The 38 inch height enclosure with flip top should allow for one controller. The main housing shall be louvered upper and lower body to allow cross flow ventilation. A stainless steel backboard shall be provided for the purpose of mounting electronic and various other types of equipment. The stainless steel backboard shall be mounted on four stainless steel bolts that will allow for easy removal of the backboard. The factory pre-assembled enclosures shall carry a ten year limited warranty.

Controller Grounding:
Grounding shall consist of one 5/8-inch x 8-foot copper rod installed per irrigation controller and where multiple controllers *are not* connected to the same ground rod.

The top of each rod shall be installed inside a 10-inch round valve box, with the rod installed as close as practical to the controller. If a pedestal enclosure is used, the ground rod may be installed through the pedestal base. Under no circumstances shall the rods be shortened. A #6 AWG solid copper wire shall be used to connect from the ground lug of the transient protection board to the copper rod. Brass clamps specifically designed to secure the copper wire to the grounding rod shall be used. There shall be no kinks or sharp bends in the wire. Each wire may be wrapped around the rod and brazed in place as an

alternative to clamping. Braze the wire to the rod for at least one circumference of the rod.
Recommended: Control System

Low Voltage System: Expressly for control of automatic control valves for underground sprinkler systems.

Transformer: To convert building service voltage to control voltage of 24 volts.

Circuit Control: The controller shall automatically calculate cycle and soak scheduling to water each station for a fixed cycle time and allow the water to soak in between cycles, maximizing infiltration and minimizing runoff.

Timing Device: The controller shall have the ability to accommodate multiple types of irrigation schedules including irrigating even days, odd days, prescribed days of the week, and interval scheduling ranging from every other day up to every four weeks.

Wiring: Solid copper with UL approval for direct burial in ground. Provide one spare control wire along entire wire routing for each controller for each unused station at controller. Loop 36 inches excess wire into each single valve box and into one valve box in each group of valves.

CS 3000, pedestal mounted controller call out: CS3-2W-S/CS3-EN/CS-2W-2ST/CS-2W-POC/FM XX

Provide CAT-5 or 6 cable in conduit from the nearest router to the controller maximum run 328' including bends and twists.

Two wire plan notes:
Specify number # of decoders needed. Two station decoders. POC decoder included in call out above.
Decoder part number: CS-2W-2ST

Two wire cable: Specify Paige cable P-7354-D in conduit. Maximum run 7,000 feet. 1.25 "Conduit recommended.

Grounding requirements:
- Every 300-400 feet with one 5/8-inch x 8-foot copper grounding rod per irrigation controller and decoder.
- #6 AWG solid copper wire from the copper rod to the field common (white wires in the black harness) of the controller / decoder.

Flow Sensor: The flow sensor used shall be supplied by the same manufacturer as the irrigation controller. The flow sensor shall be wired back to the irrigation controller using two #14 AWG wires, one red, and one black in 1" PVC conduit to connect to the irrigation controller. The maximum wire run between flow meter and controller shall be 2000 ft. The flow meter shall send low voltage digital pulses back to the controller and therefore all electrical connections must be waterproof and be resistant to any moisture entry. It is intended that all wire runs between the controller and flow meter shall be direct pulls and have no splices. If wire splices are unavoidable, they must be installed in a valve box with Spears DS-100 connectors with Spears sealant or 3M Scotch lock No. 3570 connector sealing pack used.

Piping and Fittings
Campus Preferences:

Mainline Pipe and Fittings - PVC Plastic Pipe (3 inches and larger): Rigid unplasticized polyvinyl chloride (PVC) 1120, Type 1, Grade 1, NSF-approved pipe, complying with ASTM D 2241. For mainline to control valve connections, use Schedule 80 PVC threaded both ends.

Mainline Pipe and Fittings - PVC Plastic Pipe (smaller than 3 inches): Rigid unplasticized polyvinyl chloride (PVC) 1120, Type 1, Grade 1, NSF-approved pipe, color – white, complying with ASTM D 1785. For mainline to control valve connections, use Schedule 80 PVC threaded both ends.

Lateral Pipe and Fittings (Downstream of Control Valves): Rigid unplasticized polyvinyl chloride (PVC) 1120, Type 1, Grade 1, NSF-approved, color-white, complying with ASTM D 1785. For pipe and fittings, Schedule 40 solvent weld pipe, and Schedule 40, Type 1, PVC Solvent weld fitting conforming to ASTM D 2466 and ASTM D 1784.
Galvanized Steel Pipe and Fittings: Pipe standard weight, seamless or welded, galvanized conforming to ASTM A 53. Fittings galvanized malleable-iron, threaded fittings conforming to ANSI B 16.3.

Copper Pipe and Fittings: Pipe Type L seamless copper water tube, drawn temper, conforming to ASTM B 88. Fittings wrought copper or cast brass, recessed solder joint type fittings conforming to ANSI B 16.22.

Sleeving: Rigid unplasticized polyvinyl chloride (PBC) 1120, Type 1, Grade 1, NSF-approved pipe, extruded from material conforming to ASTM D 1784, color-white. Schedule 40 solvent weld pipe.

Campus Preference

- Sprinkler Remote Control Valves (RCV) - Angle Valves: Superior Controls Company, Inc.; Model 950A-DWPRS – Electric Diaphragm Angle Valve. Solid brass construction; 200 PSI rating; Website: http://www.bucknersuperior.com/Professionals/Products/AngleValves/950A.aspx
- Sprinkler Remote Control Valves (RCV) -Globe Valves: Superior Controls Company, Inc.; Model 950-DWPRS – Pressure Regulating Valve. Solid brass construction; 200 PSI rating. Website: http://www.bucknersuperior.com/Home/NEWSuperior950.aspx
- Sprinkler Manual Control Valves (MCV): Nibco Inc., 1516 Middlebury Street, Elkhart, IN 46516-4740, Phone: 574-295-3000, Fax: 574-295-3307. Model T-211-YK – Class 125 Bronze Globe Valves. Furnish two valve keys, 3-foot long with tee handles and key end to fit valves. Website: http://www.nibco.com
- Isolation Valves (3-inches and smaller): Nibco Inc., 1516 Middlebury Street, Elkhart, IN 46516-4740, Phone: 574-295-3000, Fax: 574-295-3307. Model T-113-K – 250 PSI CSP Bronze Gate Valve. Screw-in bonnet, non-rising stem, solid wedge, push-on ends with joint restraints, 250 PSI / 17.2 Bar non-shock cold working pressure, conforming to MSS SP-80, size - 2-inches. Website: http://www.nibco.com/assets/PR113KIR.pdf
- Isolation Valves (Larger than 3-inches): Comply with Section 33 11 00 – Water Utility Distribution Piping.
- Quick Coupling Valves, Valve Keys and Key Lug: Rain Bird International, Inc., POB 37, Glendora, CA 91740-0037, Phone: 626-963-9311, Fax: 626-852-7343.
    - Quick Coupling Valves: Rain Bird Model 33DLRC.
      ¾-inch (20/27). Heavy-duty, brass construction, two-piece body design, stainless steel internal valve spring, locking rubber cover. Website: http://www.rainbird.com/landscape/products/valves/quickCouplingValves.htm
    - Locking Cover Keys: Rain Bird Model #2049I. Furnish two cover keys per project. Website: http://www.rainbird.com/landscape/products/valves/lockingcoverkey.htm

- Valve Keys: Rain Bird Model 33DK. ¾-inch (20/27). Key threads into top of quick-coupling valve to provide water access. Furnish two valve keys per quick coupling valve.
  Website: http://www.rainbird.com/landscape/products/valves/valvekeys.htm
- Flex Riser - KBI model FR or approved equal (6" length).
- Triple-Swing Assemblies - Rainbird SA series, KBI TSA-TT series, or approved equal. 12" length for 4" and 6" pop-up sprinklers, and 18" length for 12" pop-up sprinklers. Match sprinkler inlet size.
- Sprinkler Heads - Pop-up height: 4" minimum height for turf sprinklers; 12" minimum height for shrub sprinklers.

Campus Alternate
Two-Wire Path and Decoders:
The 2-Wire option shall provide support for up to one-hundred and twenty-eight (128), 2-Wire stations connected to a single controller and shall provide support for up to 6 points of connection (POC's). The 2-Wire cable shall either be Paige P7354D or Regency's Hunter® Decoder cable with a maximum length of 7,000 ft. A ground rod, 5/8 inch x 8-ft solid copper shall be required every 300-feet along the 2-Wire path as well as a single ground rod at the end of the cable run.

The station decoder shall be a 2-station decoder and shall be able to operate up to 2-solenoids using unique colored wires for each. A single controller shall be able to operate up to 70, 2-station decoders and it shall be intended that all wire runs between valves and 2-Wire decoders shall be direct pulls and have no splices except at the decoder location. All electrical connections must be waterproof and moisture-resistant and shall be done with 3M™ Scotchcast™ 3570G Connector Sealing Packs. The 2-Wire decoders shall use #14 AWG direct burial wire to connect to remote control valves and the maximum wire run between the decoder and the valve shall be 100-feet.

The POC decoder shall operate a single master valve and flow meter (model FM). A single controller shall be able to operate up to six POC decoders with a maximum of 12-POC's in a chain, controllers using *FLOWSENSE*™ technology. The maximum wire run between the POC decoder and flow meter shall be 20-feet while the maximum wire run between the decoder and the master valve shall be 100-feet.

32 91 00 - Planting Preparation

General Standards

- Chemical soil analysis will dictate soil amendment materials to adjust pH and nutrients for optimal plant growth.
- Soil organic matter content will not be less than 2% or greater than 5%
- Soil shall be free of stones larger than ¾" in diameter
- Mulch - Keep 2" minimum from bark of plant.
- Deep Root Planters are not to be used.

Herbicide Use

- Post-emergence (existing weeds): "Roundup" or approve equal.
- Pre-emergence (non-turf areas, prior to seed germination): "Ronstar" or approved equal.
- Post-emergence (turf areas): Speed Zone Southern http://gordonsprofessional.com/products-1.php?PRODUCT_CODE=6561126 or approved equal.

32 93 00 - Plants

Plant Selection

Cal Poly holds a “learn by doing” philosophy, the campus represents a living laboratory; therefore increasing the diversity of tree species is extremely important. The landscape planting should be native, non-invasive, drought tolerant species that increase or enhance ecological biodiversity. The plant palette shall be low water use and low maintenance, specific to the regional climate zone and distinct site ecotones. Species are encouraged to be drought tolerant utilizing California native and adaptive native plant material throughout the site. Low hypoallergenic material is encouraged. Invasive plant material causing economic, environmental, and human health harm is not permitted on site. All plant material is susceptible to the surrounding wildlife and cannot be harmful nor an extensive food source. Species selection, however, is subject to dictation by site conditions and pre-approval by the Landscape Advisory Committee.

- All plant suppliers are required to specialize in growing and cultivating plants specified for no less than 5 years and to certify the quality of the plants.
- Standard trees must meet requirements of a single, straight trunk with well-balanced crown and intact leader in containers or boxes.
- A multi-stem tree must have a minimum of 3 stems rising from the crown of the root ball, branched or pruned naturally according to species and with relationship of caliper, height, and branching.
- Trees size shall be minimum 36” box with shrub and groundcover ranging between 1 to 15- gallon container sizes.

Delivery, Storage, and Handling

Plants shall be protected from the sun, wind, and physical injury and damage during transit and during storage while awaiting planting work. Trees are not to be pruned prior to delivery except by approval from the University.

- Removal of rejected plants from site shall take place immediately and replaced with plants of acceptable quality.
- Plant deliveries are scheduled to avoid lengthy periods of storage and pre-planting maintenance. Sod delivery will be scheduled for tree installation within 24 hours.
- Plants will be handled by root ball or container, not lifted by the trunk, and container grow stock will not be removed from containers before time of planting.
- Water root systems of plants will be stored on-site as often as necessary to prevent container soil from drying out.
- Leaders shall not be pruned prior to or after delivery, either by nursery or by landscape contractor.

Site Preparation

Proper plants for each location shall be chosen to reduce or eliminate the necessity for pruning. A planting hole should be dug no deeper than the root ball when measured from the bottom of the root ball to the trunk flare and twice the size of the root ball diameter. Planting shall proceed only after irrigation systems have been tested and approved.

32 93 43 - Trees

Tree Planting

Trees are to be placed so the crown of the ball is 2” above surrounding grade and backfill with native soil in 6” lifts. Backfill soil should be watered to settle the soil after each lift. Fertilizer tablets should be distributed evenly around root ball at midway. 3” high water basins shall be built around tree pit and thoroughly watered. The same process should be performed for smaller plants but appropriate basin to the relative size of the plant.

Staking/Grates/Guards

- Tree stakes shall be driven 3'-0" into the ground alongside the root ball at 18" apart. Tie trees to stakes at 12" from top of stake and 36" from top of stake utilizing cinch-ties with ties interlocked around trees as approved by the University. Cinch-ties should be stapled to stakes with staples over cinch-tie, not through it.
- Tree grates shall be installed according to manufacturer's recommendations
- Tree guards will be placed only as required to protect the tree from direct contact with vehicles.

Approved Trees for Campus Planting:

Scientific Name	**Common Name**
Acacia pendula	Weeping myall
Alectryon excelus	New Zealand ash or titoki
Angophora costata	Rose gum
Bischofia javanica	Toog tree
Caesalpinia ferrea	Leopard tree
Callistemon salignus	Willow bottlebrush
Castanospermum australe	Moreton Bay chestnut
Cryptocarya rubra	Red laurel
Dalbergia sissoo	Sissoo
Elaeocarpus sylvestris	Japanese blueberry tree
Eucalyptus spathulata	Swamp mallet
Ficus sur	Cape fig
Harpephyllum caffrum	Kaffir plum
Melaleuca styphelioides	Prickly-leaved paperbark
Peltophorum dubium	Yellow poinciana
Quercus hypoleucoides	Silverleaf oak
Robinsonella cordata	Blue hibiscus tree
Sophora secundiflora	Texas mountain laurel
Syzygium smithii	Lilly-pilly
Taxodium huegelii	Montezuma bald cypress
Tupidanthus calyptratus	Mallet flower
Agathis spp.	Kauri pines
Brahea edulis	Guadalupe palm
Brahea armata	Blue Hesper palm
Caryota spp.	Fishtail palm
Cedrela fissilis	Brazilian cedarwood

Scientific Name	Common Name
Ceiba insignis	White floss-silk tree
Cercidiphyllum japonicum	Katsura tree
Chiranthodendron pentadactylon	Monkey hand tree
Dipteronia sinensis	Chinese Dipteronia
Dombeya spp.	Snowball tree
Ficus auriculata	Roxburgh fig
Ficus sycomorus	Sycamore fig
Glyptostrobus pensilis	Water pine
Griselinia spp.	Puka
Harpullia pendula	Tulipwood
Harpullia arborea	Tulipwood
Howea forsteriana	Kentia palm
Howea belmoreana	Curly palm
Lagunaria patersonii	Primrose tree
Liquidambar orientalis	Turkey sweet gum
Liriodendron chinense	Chinese tulip tree
Markhamia lutea	African trumpet tree
Melia azedarach	Chinaberry
Parkinsonia aculeata	Mexican palo verde
Parkinsonia x 'Desert Museum'	Desert museum palo verde
Persea indica	Ornamental avocado
Pinus densiflora	Japanese red pine
Pinus roxburghii	Chir pine
Pinus torreyana	Torrey pine
Pittosporum angustifolium	Willow Pittosporum
Platanus orientalis	Eastern sycamore
*Platanus occidentalis**	Western sycamore
*Platanus Mexicana**	Mexican sycamore
Pseudotsuga macrocarpa	Bigcone spruce
Pseudotsuga menziesii	Douglas-fir
Quercus virginiana	Southern live oak
Quercus engelmannii	Engelmann oak
Sciadopitys verticillata	Japanese umbrella pine
Spathodea campanulata	African tulip tree
Taxus spp.	Yews

Scientific Name	Common Name
Tilia spp.	Lidens
Torreya californica	California nutmeg tree
Tristaniopsis laurina	Water gum
Ulmus pumila	Siberian elm
Sophora japonica	Japanese pagoda tree
Allocasuarina torulosa	Forest oak
Corynocarpus laevigata	New Zealand laurel
Meryta sinclairii	Puka
Olea africana	African olive
Quillaja saponaria	Soapbark tree
Taxodium mucronatum	Montezuma cypress
Trachycarpus wagnerianus	Dwarf Chusan palm
* Riparian Locations Only	

32 96 43 Tree Transplanting

Palm Tree Transplanting and Planting

Size of palms to plant

- Large palms are easier to planted or transplanted.
- Containerized palms, with little or no root disturbance, size is dependent on the capacity of the equipment to move and handle large, heavy specimens safely.

For palms transplanted from one site to another or field-grown plants dug in a nursery, it is Important to select specimens with some visible trunk or stem because they are more tolerant of root disturbance and will reestablish more quickly and successfully.

Do not transplant the following species with visible above-ground trunks:

- *Sabal* (palmetto palm)
- *Bismarckia* (Bismark palm)
- *Latania* (Latan palm)

When to plant/transplant

- Root growth is essential for rapid and successful establishment. Root growth is highest during warmer months.
- Transplant palms in the beginning of the warm season to ensure several months of high temperatures for adequate root growth.
- Container palms can be planted safety year round.
- Size of root ball
    - Most palm species with root balls extending out 12 inches from the trunk and 12 to 24 inches deep are adequate for transplanting.
    - A deeper root ball will help to stabilize taller specimens.
    - The larger the root ball the more successful and quick is establishment.

- Palms with excessively large root balls are more difficult and expensive to move due to their size and weight.
- The hole left after a palm with a large root ball has been dug usually must be filled, requiring more labor and expense, and perhaps even additional soil.

Leaf removal/Tie up

- Consult Cal Poly Landscape Representative to determine if this is necessary. The practice has little value when transplanting in cooler, more humid coastal areas.
- To protect the palm and for ease of handling, tie up leaves during digging, transporting, and replanting. In appropriate situations, untie leaves once the palm is replanted.

Transport

- Palm specimens must be well supported and protected during moving and handling to prevent injury to the apical bud or meristem and trunk.
- Protect palm bark; wounds and injuries are permanent and potential sites for disease and insect entry.
- Some species, like Archontophoenix cunninghamiana (king palm), are sensitive to handling. They and other slender trunked palms with heavy crowns should have a wooden splints attached along the trunk and extending into the leaves to prevent the weight of the crown from damaging the apical bud.
- Tie stems together of the multi-trunked species for additional protection.
- Use nylon and/ or burlap slings and ties to support and grasp palms to prevent injury to the trunk when moving and handling with heavy equipment.
- Stack or shingle palms securely on the vehicle, either standing them up at an angle from the wind or laying them down with the root balls forward and the crowns at the rear.
- Cover root balls and crowns with shade cloth or other protective material during transport to prevent wind and sun damage and excessive drying.

Planting

- Backfill with the same unamended soil excavated from the hole.
- To support the stability of larger palms, consider backfilling with washed builder's sand to pack more easily and uniformly.
- There is no benefit to amending the backfill with organic matter. Use organic matter as mulch several inches deep and several feet out from the palm's base. Tamp out air pockets.
- If stabilization is required for large palms:
  - Use 2 x 4 or 4 x 4 wooden bracing attached against one-foot lengths of 2 x 4 vertically strapped or banded around the trunk.
  - Protect the trunk with nylon, burlap, or other suitable material where the one foot lengths of 2 x 4 are secured.
  - Do not nail into the trunk; nailing will cause permanent wounds, and disease and insect entry sites.
  - Palms may also be secured with guy wires or cable instead of wooden bracing.
  - Do not stabilize palms by planting them deeper in the hole; although some palms survive deep planting, most do not.
- Construct an irrigation berm four to six inches around the root ball and hole.
- Provide a two- to three- inch layer of mulch around the base of the palm to encourage new root growth, conserve moisture, and suppress weeds.

- Irrigate deeply and thoroughly.

Post-planting care
- Schedule irrigations based on need, not a clock or calendar.
- Irrigate sensibly, keeping the root ball and backfill evenly moist but not saturated.
- Keep turf grass and weeds away from the trunk base.
- Maintain a regular and complete fertilizer program once the palm is fully established.

Division 33 - Utilities

33 10 00 - Water Utilities
All underground plumbing utility piping shall have a minimum of 12” sand encasement.

All underground plumbing utility piping shall have a minimum depth of 24” unless otherwise approved by the administrative authority after all existing utilities have been potholed by the contractor.

All underground storm drainage and sanitary sewage piping shall be video-recorded prior to connection to existing systems. These videos shall be presented to campus building inspectors and representatives of the Plumbing Shop.

33 11 13 - Public Water Utility Distribution Piping
All underground domestic water piping 3” or less shall be schedule 80 PVC or HDPE SDR 11.

All underground domestic water piping 4” or larger shall be PVC C900.
All valves for underground domestic water 2” or larger shall be epoxy coated resilient wedge gate valves with 2” operating head.

Preferred Manufactures: Mueller Co. or Clow Valve.

Valve well sleeve shall be 8” diameter unless otherwise authorized by the University.

Valve sleeve shall come up at minimum half the depth of the G5 box.

33 12 13.13 - Water Supply Backflow Preventer Assemblies
Preferred manufacturers are Zurn and Wilkins.

All backflow preventers shall meet AWWA standards.

All backflow preventers installed by the contractor will be rested by a certified tester and documentation will be turned over to a campus representative before lines are put into service.

33 12 19 - Water Utility Distribution Fire Hydrants
Fire hydrant spacing shall comply with CFC and City of San Luis Obispo Fire Marshall’s requirements.

All site building will be fully protected by fire sprinklers so a 75% reduction in fire flow can be applied with the authorization of the City San Luis Obispo’s Fire Marshall.

Reduced pressure principle back flow preventers (RP devices) are required at any connection that would potentially contaminate the water supply. The RP device shall be a double detector check valve assembly at the connections to the fire risers and stand pipes.

All water pipes shall be installed with a minimum of 36" of cover.

Water meters shall be installed at every water service connection per the Plumbing Narrative.

Valves shall be arranged so that the system can be operated in a manner where no more than 5 fire protection apparatuses (hydrants, risers, stand pies, etc.) will be removed from service if repairs to the system are necessary.

Vertical clearance of all water pipes shall comply with State Department of Health separation requirements and water shall always be above the sanitary sewer.

Horizontal clearance shall comply with State Department of Health separation requirements with a minimum 10' separation of sewer and water measured from outside of pipe.

Fire hydrants shall be placed at a maximum of 300' from all portions of the buildings per City of San Luis Obispo Fire Marshall's requirements.

Fire department connections shall be placed a maximum of 40' from a fire hydrant per City of San Luis Obispo Fire Marshall's requirements.

Provide a minimum residual pressure at the roof elevation of 20 psi at fire flow (1,500 gpm) + average day demand (ADD).

Pipes less than 4" diameter shall be schedule 80 PVC or SDR11 HDPE.

Pipes from 4" to 12" in diameter shall be C900 DR18 PVC.

All fittings shall be ductile iron.

All gate valves shall be non-rising stem with resilient wedge seats.

2" operating nut is preferred on all underground gate valves.

All fire hydrants shall comply with City of San Luis Obispo Public Works Engineering Standards. requirements and have solid bronze bodies, shall be wet barrel, with one 2-1/2", and two 4-1/2" connections.

Backflow devices for fire connection shall be double detector check valves, with lockable OS&Y stems, leak detection meters, and shall employ reduced pressure principle check valves.
Tee is the preferred valve method for hydrants.

33 31 00 - Sanitary Utility Sewerage Piping
All sanitary sewer lines 4" or larger shall be SDR-35 PVC, lines 3" and smaller shall be DWV PVC or cast iron.

All sanitary sewer manholes shall meet City of San Luis Obispo Public Works Engineering Standards. Transition from existing clay and cast iron pipe to PVC shall be made with Ferno Rubber coupling.

Provide documents or certified test results indicating the pipe furnished meets all specified requirements. Satisfactory documents include pipe manufacturer certificate indicating that the pipe has been:

1. Sampled
2. Tested
3. Inspected

In compliance with the ASTM specifications.

Sanitary Sewer Design CriteriaAll sanitary sewer laterals and mains shall be installed with a minimum of 48” of cover.

Curved sewer mains shall be installed with a minimum curvature in compliance with manufacturer’s recommendations. Joint deflections are not permissible.

All sanitary sewer mains shall have a minimum 8” diameter.

Sewer mains shall be designed with a minimum 2 fps and a maximum 10 fps half-full flow velocity and the following minimum slopes:

- 8” 0.50%
- 10” 0.35%

Force mains shall be designed with a minimum velocity of 4 fps and a maximum velocity of 8 fps.

Sewer laterals shall be 4” minimum diameter with a minimum slope of 2%.

Sewer mains shall have manholes at 300’ maximum spacing, at angle points, at the beginning or ending of curves, clean outs may be used at the beginning of runs less than 150’ long, otherwise a manhole is required.

Service laterals shall connect to the main with a wye pipe fitting and a clean out at the building connections and angle points. Laterals longer than 100’ long shall require an additional clean out at approximately mid-point or at 100’ maximum spacing.

Sewer pipes shall be installed a minimum of 12” below water lines in compliance with State Department of Health separation requirements.

Horizontal clearance between sewer and potable water mains shall be a minimum of 10’ from outside of pipe in compliance with State Department of Health separation requirements.

Lift Station Design Criteria

Pumps shall be an alternating duplex system with Individual pumps sized for PDD (250 gpm); Total dynamic head = ±120’. A triplex pumping system may be installed at the Design Builder’s discretion and with University approval.

Accessories shall include an alarm system tied into a central monitoring station and odor control.

A stand by generator shall be included in the project to provide emergency power for the lift station. The generator must be sized to power both pumps in a duplex system or at least two pumps in a triplex system.

Sanitary Sewer Materials
New sewer main pipeline material must be HDPE. When repairing existing pipeline, comply with all sewer repair sections.

All gravity mains and laterals shall be SDR 35 PVC.

All force mains shall be Schedule 80 PVC or C900 DR 14 PVC.

All Manholes shall be pre-cast concrete with eccentric cones, and traffic rated covers and frames.

Lift station wet wells shall be precast or cast in place concrete lined with a minimum 125 mils of type B polyurethane coating.

Lift station pumps shall be submersible pumps with oil filled, explosion proof motors with moisture sensors. Pumps shall be mounted to a factory provided rail system.

High Density Polyethylene
Use:
1. Virgin grade
2. High molecular weight
3. Standard Dimension Ration (SDR) 17
4. Iron Pipe (IPS)

High Density Polyethylene (HDPE) pipe made in diameter and tolerances in compliance with the latest version of ASTM D3035.

Furnish complete with all fabricated fittings, and other appurtenances as necessary, for a complete and functional system.

The pipe must be free of:
1. Visible cracks
2. Holes
3. Foreign inclusions, or
4. Other defects

Any pipe not meeting these criteria will be rejected.

The pipe must be clearly marked with the following:
1. Name and trademark of manufacturer
2. Nominal pipe size
3. Dimension ration
4. The letters PE followed by the polyethylene grade per the latest version ASTM D1248
5. Hydrostatic design basis in psi
6. Manufacturing standard reference
7. A production code from which the date and place of manufacture can be determined.

The material must be listed by the Plastic Pipe Institute (PPI) with a designation PE 3408 and have a minimum cell classification of:

1. 345434C
2. D, or
3. E

As described in latest version of ASTM D3350.

Pipe material must meet the requirements for:

1. Type III
2. Class B or C
3. Category 5
4. Grade P34

Material as described in latest version of ASTM D1248

Provide pipe with interior wall color of:

1. White
2. Gray or
3. Light green

Provide pipe with exterior wall color of:

1. Black
2. Gray or
3. Light green

Provide submittals on furnished pipe from manufacturer certifying pipe is in compliance with:

1. Specifications
2. Codes
3. Standards

Any pipe segment that has cut in the pipe wall exceeding 10 percent of the wall thickness must be cut out and removed from the site.

Store pipe so that it is not deformed.

Polyvinyl Chloride (PVC) Pipe
Furnish pipe in 20-foot lengths with integral wall belled ends and elastomeric joint and solid wall. Pipe and fittings must be free of imperfections and clearly marked with name of manufacturer.

Minimum pipe stiffness (F/y) at 5 percent deflection is 46 psi for all sizes when calculated in compliance with ASTM Designation D 2412.

Pipe must have minimum Standard Dimension Ration (SDR) of 35 and pipe stiffness of 46 psi.

Pipe color must be green.

PVC Pipe 4 to 15 Inch Diameter
PVC Pipe must conform to the requirement of latest version of ASTM specification D 3034.

PVC Pipe 18 to 27 Inch Diameter
PVC Pipe must conform to the requirement of latest version of ASTM Standard Specifications F 679.

PVC Pipe 30 to 48 Inch Diameter
PVC Pipe must conform to the requirement of latest version of ASTM Standard Specifications F 794.

Ductile Iron Pipe
Ductile iron pipe must be:

1. Centrifugally cast.
2. Ductile iron pipe.
3. Gasketed push on joints appropriate for use in a wastewater environment such as Polychloroprene, Ethylene Propylene Diene Monomer, or an approved equal.
4. A pressure class 150 for pipe with 3 feet or less of cover.
5. A pressure class of 350 for pipes with 3 feet or less of cover or exposed above grade.
6. Coated on exterior.
7. Lined with fusion bonded epoxy, polyurethane or approved equal.

Ductile iron pipe must be encase in polyethylene casing material. Casing material must be:

1. Tube type
2. Conform to the latest ANSI/AWWA C105 Standard.

Polyethylene casing must extend over:

1. Tees
2. Bends
3. Couplers at the end of a Section of ductile iron where it connects to a different type of pipe
4. Close casing at the end (dead end) of a pipe

Exposure to air and sunlight must be kept to minimum for either type “A” or type “C” encasement material.

Sewer Lateral Pipe
New and repaired sewer lateral pipe may be:

1. PVC SDR 35
2. PVC Schedule 40
3. HDPE SDR 17
4. ABS Schedule 40

Pipe joints must be gled or fused.

Joints and Fittings

HDPE
HDPE Pipe and fittings must be in compliance with the latest version of:

1. ASTM F714
2. ASTM D3261

Joints and Fittings for HDPE must be of the same manufacturer as the pipe and the same SDR rating.

PVC
PVC pipe must have a rubber ring bell and spigot joints providing a water tight seal and allowing for contraction and expansion. The bell must consist of an integral wall Section stiffened with two PCV retainer rings that securely lock the solid cross Section rubber ring into position.

All fittings and accessories must be as manufactured and furnished by the pipe supplier, or approved equal, and have bell and/or spigot configurations identical to that of the pipe. All fittings must be of the same material as the pipe and same SDR rating, unless specified otherwise.

Ductile Iron
Use restrained fittings for exposed ductile iron pipe, such as bridge crossings. Restrained fittings must be Flex-Ring by American Ductile Iron, TY FLEX by U.S. Pipe, or approved equal which use a factory weld as part of the restraining system.

Repair Joint
Use strong back RC couplings or equal meeting the following requirements:

1. Flexible sewer couplings and transition couplings
2. Comprised of an elastomeric sealing component
3. Type 316 series stainless steel tension components (end clamps and shear rings).
4. Shear rings must have am minimum thickness of 0.012 inches
5. End clamps must have “bolts” as their means of tightening (not worm gears).

Couplings must be appropriately sized for the pipe materials being joined, without the need for bushings.

HDPE Pipe with fused ends must be repaired with HDPE pipe with gused joints. Strong back couplings must not be used.

Sewer Lateral Joints (New and Replacement)
Sewer lateral pipe must be joined using glued joints and fittings or fused.

Concrete
Use class 2 concrete.

Use minor concrete for:

1. Manholes
2. Pipe junctions
3. Jacketing

Use fifteen percent approved pozzolan replacement for manhole construction.

Precast concrete manhole Sections must comply with the most current version of ASTM specification C-478-61T or AASHTO-M170.

All manholes must be watertight, and the floor sloped for a smooth monolithic trowel finish. The interior finish of the manholes must be smooth.

Mortar
To make mortar, use one part of Type II Portland cement and two parts sharped grained particles that are:

1. Clean
2. Hard
3. All passing a # 4 sieve

Mix mortar in a machine or water tight box. Accurately measure and thoroughly mix mortar to a uniform consistency. Use mortar immediacy after mixing. Do not remix mortar that begins to harden prior to placement.

Pipe Installation

Sanitary sewer lines must be water tight. Install pipe to ensure the system is water tight throughlout the component parts, particularly at the pipe joint.

Do not:
1. Cut
2. Gouge
3. Score or
4. Damage pipes

When:
1. Unloading
2. Handling
3. Storing
4. Installing

Pipe Laying

Lay the pipe in perfect conformity to the design line and grade obtained for each pipe by measuring down from a tightly stretched line running parallel with the grade.

Lay all pipes continuously uphill.

Install pipe and fittings for underground gravity sewers in compliance with the latest version of ASTM Standard D-2321. Lay bell and spigot pipe, with the bell of the pipe upgrade.

Pipe Bursting and Reaming

Install sewer pipe by pneumatic pipe bursting or pipe reaming. Install pipe in compliance with the pipe manufacturer's recommendations. For pipe busting installation, use pneumatically operated equipment with a pipe bursting head attached to HDPE pipe.

Locate, expose, disconnect and isolate existing sewer laterals from sewer main before pipe installation work begins. When pipe reaming, you must prevent drilling fluid from entering sewer laterals.

Submit to the Engineer for review and approval a sewer installation plan which includes insertion and reception pit location.

For pipe bursting work, use a constant tension pneumatic tool used in conjunction with a constant tension hydraulic winch. Size the winch based on the diameter and the depth of the pipe to be replaced. The constant tension winch must be sufficient sized to pull one continuous length of pipe between approved winching points.

The void created by the device must be sufficient in size to accommodate the pipe which is installed immediately after the void is formed. The void must not be so large that pipe displacement or pavement settling occurs. Allow new sewer pipe to relax for twelve hours prior to final connection to manholes.

If you cannot complete pipe bursting or reaming without damage to existing closely places lines or pavement, you may request authorization from the Engineer to place new pipe with traditional open-cut trenching. If you encounter an obstruction that prevents the bursting or reaming tool from continuing, you must:

1. Stop the operation
2. Notify the Engineer
3. Excavate the obstruction
4. Remove the obstruction.

Any pavement heaving, or utility damage caused by pipe bursting or reaming work must be repaired at no additional cost to the city or utility company.

If you use any material or method that is not approved by the Engineer, you must remove the work and replace as directed by the Engineer.

If an obstruction is found during testing, remove the obstruction. Remove and replace Section of pipe is damaged.

HDPE Pipe Joint

Join HDPE pipe by:

1. Heat fusion welding
2. Electrofusion fitting or
3. Equal as approved by the Engineer.

All connections to the sewer pipe must be water tight, flush with the edges of the sewer pipe with clean uniform cuts.

Perform heat fusion welding in compliance with the pipe manufactuer’s recommendations and ASTM D2657. Fusion equipment used must be capable of meeting all conditions recommended by the pipe manufacturer including, but not limited to:

1. Fusion temperature
2. Alignment
3. Fusion pressure

Fusion equipment must only be operated by technicians who have been certified by the pipe manufacturer or supplier. Document and furnish to the Engineer technicians certification in a submittal.

Use a fire-retardant bag or suitable enclosure for the heater plate to facilitate control of heating process and to protect the heater plate surfaces from dirt and other debris when not in use.

Clean heater plate surfaces regularly to prevent accumulation of fusion welding residues or other substances that may result in faulty pipe joining. The heater plate must be equipped with suitable means to measure the temperature of plate surfaces and to assure uniform heating such as thermometers or pyrometers.

Joint strength must be equal to that of the adjacent pipe. Clean the pipe ends with a cotton or non-synthetic cloth to remove:

1. Dirt
2. Water
3. Grease

4. Other foreign materials.

Cut pipe ends square and carefully aligned just prior to heating.

After achieving the proper melt pattern, bring the pipe ends together in a firm, rapid motion applying sufficient pressure to form a pipe bead (1/8 to 3/16 inch in height) around and inside the entire circumference of the pipe. Remove pipe bead before welding the next joint of pipe.

Use tools designed for and approved by the manufacturer and supplier for joining pipe.

Sand Traps

Furnish and install sand traps or other debris catching measure approved by the Engineer during the work. Debris catching devices must always be installed during construction. You assume all costs associated with any damage resulting from construction materials entering the wastewater system or treatment facility.

Bypass Pumping

Submit a bypass pumping plan for approval by Engineer at the pre-construction meeting. At a minimum the plan must include:

1. Pump size and type
2. Backup pump size and type
3. Contingency plan for pump failure to ensure continuous bypass operations.

The bypass system must be free from leaks. The bypass pumping plan must address access to:

1. Driveways
2. Cross streets
3. Pedestrian crossings.

Manholes

Construct manholes per Engineering Standards.

Sewer Laterals

Sewer laterals must be tied over as shown. Notify the Engineer immediately upon discovering any lateral not shown, or any lateral that appears to be dry and out of service. The Engineer will then determine if lateral is out of service and not needed. If the lateral is out of service and not needed, the Engineer will direct you to abandon the lateral. In these cases, the pay item for lateral tie over will not be reduced and will be paid to compensate you for all time, equipment, labor, materials for sewer lateral abandonment.

Existing Sewer

Existing Manholes

Existing manholes must be:

1. Adjusted to grade
2. Remodeled or
3. Abandoned

As shown and in compliance with Engineering Standards and Section 15.

Existing manholes may have large cast in place concrete bases. No additional payment will be made for the removal of existing bases as needed to complete the work.

Oversize manholes may require a manufactured concrete reduction ring prior to setting the new manhole ring and cover.

Abandonment of Sewerlines

Sewer facilities taken out of service must be abandoned in compliance with engineering Standard 6050.

Provide the Engineer 48-hour notice prior to abandoning sewer laterals. Cut off the sewer lateral at the main and plug pipes with class 3 concrete a minimum of 12-inches into the pipe, away from the sewer main pipe. Provide a minimum five-foot by five-foot excavation with shoring at the sewer main, adequate for University to remove the existing wye and replace it with new section of pipe. Provide:

1. Backfill
2. Compaction
3. Surface restoration

For the excavation.

Repair

Repair cut sewer facilities using new pipe of material the same diameter. If the existing sewer pipe material complies with materials, use that same pipe material.

Center a continuous Section of new pipe at the repair location. Repair must be water tight and placed at the same grade. Prior to backfilling excavation, place level on repaired portion of sewer, in the prewsence of the Engineer, to confirm line and grade. Backfill, compact and restore surface improvements.

Repair must be documented with:

1. Location
2. Repairs made
3. Photos
4. Guarantee letter
5. Interior video inspection of pipeline, when directed by the Engineer.

Provide hardcopy of all documents to owner. Provide electronically, all documents to the Engineer.

Testing

Air Test

After the pipeline is in place and the joints made, you must air test the sewer in the presence of the Engineer. Air test procedure is as follows:

1. A maximum of 400 feet of sewer pipe will be tested at one time.
2. Plug and brace securely all outlets.
3. Introduce air into test Section until internal pressure is 4.0 psi. If sewer pipe is placed in ground water, calculate ground water pressure and add that additional pressure to internal pressure used for test.
4. Maintain an internal test pressure by adding air as need for a minimum time of 2 minutes.
5. Measure the time required for pressure to drop from 3.5 psi to 2.5 psi. Do not introduce new air into test Section during measurement.

Minimum permissible pressure discharge time as follows in seconds
(time to drop pressure from 3.5 psi to 2.5 psi)

Sewer Main		4-inch Sewer Lateral				
Diameter	Length	Sewer Lateral Length				
Inches	Feet	0 feet	100 feet	200 feet	300 feet	400 feet
6 & 8	0	0 seconds	20 seconds	40 seconds	50 seconds	70 seconds
	50	40 seconds	50 seconds	70 seconds	90 seconds	80 seconds
	100	70 seconds	90 seconds	100 seconds	100 seconds	90 seconds
	150	110 seconds	120 seconds	110 seconds	100 seconds	100 seconds
	200	140 seconds	120 seconds	110 seconds	110 seconds	100 seconds
	300	140 seconds	130 seconds	120 seconds	110 seconds	110 seconds
	400	140 seconds	130 seconds	120 seconds	120 seconds	110 seconds
10	50	50 seconds	70 seconds	90 seconds	100 seconds	90 seconds
	100	110 seconds	130 seconds	120 seconds	110 seconds	110 seconds
	200	170 seconds	150 seconds	140 seconds	130 seconds	120 seconds
	300	170 seconds	160 seconds	150 seconds	140 seconds	130 seconds
	400	170 seconds	160 seconds	150 seconds	150 seconds	140 seconds
12	50	80 seconds	100 seconds	110 seconds	110 seconds	110 seconds
	100	160 seconds	170 seconds	150 seconds	140 seconds	130 seconds
	200	200 seconds	180 seconds	170 seconds	160 seconds	150 seconds
	300	200 seconds	190 seconds	180 seconds	170 seconds	160 seconds
	400	200 seconds	190 seconds	180 seconds	180 seconds	170 seconds
15	50	120 seconds	140 seconds	160 seconds	140 seconds	130 seconds
	100	250 seconds	220 seconds	190 seconds	170 seconds	160 seconds
	200	260 seconds	230 seconds	220 seconds	200 seconds	190 seconds
	300	260 seconds	240 seconds	230 seconds	220 seconds	210 seconds
	400	260 seconds	240 seconds	230 seconds	220 seconds	220 seconds

Sewer Main		6-inch Sewer Lateral				
Diameter	Length	Sewer Lateral Length				
Inches	Feet	0 feet	100 feet	200 feet	300 feet	400 feet
6 & 8	0	0 seconds	40 seconds	80 seconds	100 seconds	100 seconds
	50	40 seconds	70 seconds	110 seconds	110 seconds	110 seconds
	100	70 seconds	110 seconds	120 seconds	110 seconds	110 seconds
	150	110 seconds	120 seconds	120 seconds	120 seconds	110 seconds
	200	140 seconds	130 seconds	120 seconds	120 seconds	120 seconds
	300	140 seconds	130 seconds	120 seconds	120 seconds	120 seconds
	400	140 seconds	130 seconds	130 seconds	120 seconds	120 seconds
10	50	50 seconds	90 seconds	120 seconds	120 seconds	110 seconds
	100	110 seconds	140 seconds	130 seconds	130 seconds	120 seconds
	200	170 seconds	150 seconds	140 seconds	140 seconds	130 seconds
	300	170 seconds	160 seconds	150 seconds	140 seconds	140 seconds
	400	170 seconds	160 seconds	150 seconds	150 seconds	140 seconds
12	50	80 seconds	120 seconds	140 seconds	130 seconds	120 seconds
	100	160 seconds	170 seconds	150 seconds	140 seconds	140 seconds
	200	200 seconds	180 seconds	170 seconds	160 seconds	150 seconds
	300	200 seconds	190 seconds	180 seconds	170 seconds	160 seconds
	400	200 seconds	190 seconds	180 seconds	180 seconds	170 seconds
15	50	120 seconds	160 seconds	160 seconds	150 seconds	140 seconds
	100	20 seconds	210 seconds	190 seconds	170 seconds	160 seconds
	200	260 seconds	230 seconds	210 seconds	200 seconds	190 seconds
	300	260 seconds	240 seconds	220 seconds	210 seconds	200 seconds
	400	260 seconds	240 seconds	230 seconds	220 seconds	210 seconds

PVC Joints
Joint tightness is measured by assembling two Sections of pipe in compliance with the manufacturer's recommendations.

Subject the joint to an internal hydrostatic pressure of 25 psi for one hour. Consider any leakage a failure of the test requirements.

Testing of Force Mains
Test force mains according to the following procedure:

Fill each section of pipe with water and expel all air. Allow pipe to set for a minimum of 24 hours. Refill pipe and pressure pipe to:

1. 150 psi, or
2. Service pressure plus an additional 50 psi

Whichever is greater. Maintain pressure for two hours. Replace any portion of line that fails and retest. Maximum allowable leakage is 4.17 gallons per hour per mile per nominal inch of diameter.

Manhole Vacuum Testing
Vacuum test all newly constructed manholes prior to placing any backfill around manhole and again after manhole is raised to finish grade. Provide the Engineer 24-hour notice prior to each test.

You must prepare the manhole as follows:

1. Plug all inlets to the manhole
2. Place a test head in the top of the manhole
3. Inflate a seal

Place a vacuum of 10 inches of mercury on the manhole and measure the time for the vacuum to drop to 9 inches of mercury. The manhole meets requirements if the measured time for the vacuum drop meets or exceeds the value from the following table:

Manhole Depth	Manhole Diameter	
	4 feet	5 feet
4 feet	10 seconds	13 seconds
6 feet	15 seconds	20 seconds
8 feet	20 seconds	26 seconds
10 feet	25 seconds	33 seconds
12 feet	30 seconds	39 seconds
14 feet	35 seconds	46 seconds
16 feet	40 seconds	52 seconds
18 feet	45 seconds	59 seconds
20 feet	50 seconds	65 seconds

If the manhole fails the vacuum test, provide the necessary repairs to make the manhole pass the vacuum test.

33 40 00 - Storm Drainage Utilities
Storm water management systems must meet RWQCB post-construction requirements as listed in the Post-Construction Storm water management Requirements for Development Project in the Central Coast Region (Resolution No. R3-2013-0032).

Project storm drain systems shall be designed for the 25-year design storm with 1' of freeboard at inlets, and overland relief shall be provided for the 100-year design storm.

Pipes less than 12" in diameter shall be SDR 35 PVC at a minimum of 0.5% slope; all pipe 12" or greater shall be dual wall type S Corrugated Polyethylene Pipe (CPP) with water tight joints at a minimum of 0.5% slope. All pipes shall have a minimum cover of 24". All pipes within the street shall be a minimum of 18" diameter.

Manholes or catch basins are required at angle points, at beginning or ending of curves, or on runs longer than 300'.

Provide minimum cover on all pipes in accordance with manufacturer's recommendations and not less than 12".

Provide a minimum of 12" of clearance to all other underground utilities.

Provide a minimum of 12" of cover from bio-retention swale gravel beds to top of pipes.

All manholes, catch basins, and junction boxes shall be concrete with cast iron lids or grates. All grates and lids shall be ADA compliant, bicycle friendly, and shall be traffic rated where there is a potential for vehicular traffic, including on sidewalks. Galvanized grates and lids are not allowed to prevent heavy metal contamination of storm water.

The bio-retention swale overflow inlets and drain pipe shall be designed to have adequate capacity to convey the 100-year storm with a minimum of 12" of free board in the swale.

Roof drains down spouts shall not directly connect to the storm drain system and the runoff should be directed to vegetated areas where possible. The intent of the storm drain system is to allow for bio-filtration of the storm water runoff and infiltration. Drainage should be conveyed in surface swale or by sheet flow where possible and pipes should be avoided unless necessary.

Storage can be provided in high porosity aggregate or perforated pipes or storm chambers surrounded by high porosity aggregate. Storage can also be provided in the aggregate under porous pavers. The system should be designed to infiltrate completely in 48 hours or less. Excess runoff to these systems shall be connected to the storm drain collection system.

Contractors are responsible for preparing a Storm Water Control Plan (SWCP) in accordance with Post-Construction Storm water management Requirements for Development Project in the Central Coast Region (Resolution No. R3-2013-0032), to be approved by the RWQCB.

Deflection
Following the:

1. Placement
2. Backfill

3. Compaction

Prior to permanent pavement, clean and measure pipe for obstruction such as:

1. Deflections
2. Joint offsets
3. Lateral pipe intrusions.

Allowable internal diameter is determined using appropriate size mandrel. Prior to use, the mandrel must be certified by the Engineer or by another entity approved by the Engineer. Use of an:

1. Uncertified mandrel or
2. An altered mandrel

Will invalidate test. If the mandrel fails to pass, the pipe will be deemed to be over deflected.

The mandrel must:

1. Be rigid
2. Be nonadjustable
3. Have an odd-numbering-legs (9 legs minimum)
4. Have an effective length not less than its nominal diameter
5. Be fabricated of steel or aluminum
6. Be fitted with pulling rings at each end
7. Be stamped or engraved indicating the:
8. Pip material specification
9. Nominal size
10. Mandrel outside diameter.

Using the manufacturer's specified internal diameter of pipe, maximum vertical deflection must not exceed:

1. 95 percent – for nominal diameter pipe less than or equal to 12 inches
2. 96 percent – for nominal diameter pipe less than or equal to 30 inches
3. 97 percent – for nominal diameter pipe greater than 40 inches

For pipes equal to or smaller than 24 inches in internal diameter, pull the mandrel through the pipe by hand. For pipes greater than 24 inches in internal diameter, deflections may be determined by mandrel or by a method submitted to and approved by the Engineer. If a mandrel is selected it must conform to the requirements in this section.

Any over deflected pipe must be uncovered to remove the compact soil loading. Once uncovered if the pipe can pass the mandrel it may remain. If not, remove and replace the damaged pipe. In all cases, the Engineer will determine whether the pipe may remain or must be replaced. Any pipe subjected to any method or process other than uncovering, even if successful to remove over deflection, must be removed and replaced with a new Section of pipe.

All costs incurred by you attributable to:

1. Mandrel testing
2. Deflection testing
3. Repairs
4. Any delays

Are borne by you at no cost to the University.

Television Inspection (Video Inspection)

The University will video inspect (CCTV) all public sewer pipe systems prior to acceptance. Provide the Engineer three weeks' notice prior to placement of final paving or surface restoration. Allow one working day per 2,000 linear feet of sewer main to be video inspected. Installations which do not conform to the requirement must be reconstructed and re-video inspected at your expense.

If you are required to submit video inspection to the Engineer for review, furnish video on flash drive. CCTV reports shall be National Association of Sewer Service Companies (NASSCO) Pipeline Assessment and Certification Program (PACP) certified with no modifications. Every section of sewer (manhole to manhole) shall be identified by audio and alphanumeric on the video display and shall include:

1. Project name
2. Municipality
3. Street name
4. University designated GIS manhole numbers
5. Sewer diameter and length
6. Date of inspection

Video inspection recordings and reports must be completed for gravity conveyance systems per the following requirements:

1. The pipeline video inspection must be submitted on USB drive.
2. The inspection recording must be of adequate resolution to display pipe, potential pipe defects, and pipe joints.
3. Audio and written notes must be recorded in the video.
4. A 1-inch cylindrical gauge may be required in the video inspection to determine if the pipe segment has a grade deficiency.
5. General convention of the recording will travel downstream and must automatically track the pipeline length (in feet) from the start of the inspection to the end of the inspection.
6. Inspection report must include the project location, a scaled plan of the pipe segment(s), and a summary of inspection findings.

Cleaning
After the final air test has been satisfactorily completed, sewer shall be cleaned using High-Velocity (Hydro-cleaning) equipment only. Cleaning shall be with clean water with a minimum 2,000 psi @ 50 gpm standard cleaning nozzle. Cleaning shall be performed starting at the furthest upstream segment (manhole to manhole) proceeding downstream. Each segment shall be cleaned from its downstream manhole. All debris shall be vacuumed and removed from each set-up location.

All foreign material must be removed from:

1. Pipes
2. Manholes
3. Cleanouts

Prior to being placed into service. Remove all material from sand traps or debris catchers in manholes prior to removing the sand trap or debris catcher.

33 41 00 - Storm Utility Drainage Piping
All storm sewer lines 4” or larger shall be SDR-35 PVC, lines 3” and smaller shall be DWV PVC or cast iron.

All manholes shall meet [City of San Luis Obispo Public Works Engineering Standards].

Transition from existing clay and cast iron pipe to PVC shall be made with Ferno Rubber coupling.

33 51 00 - Natural-Gas Distribution

Cal Poly owns and maintains the underground natural gas distribution systems throughout the campus. Gas distribution pressure is 5 PSI. The Southern California Gas Companies also has a 60 PSI distribution system.

Provide a gas regulator and meter at each new building.

For new buildings to be connected to the campus natural gas system, the anticipated additional gas demand should be identified early during preliminary planning. This anticipated gas demand should be submitted to the Principal Engineer for Cal Poly Physical Planning and Construction. Improvements to the campus system may be required to accommodate the additional demand. The Principal Engineer shall identify a suitable point of connection to the campus system and what system improvements may be necessary to accommodate the new building.

33 51 13 - Natural-Gas Piping

All underground gas piping shall be polyethylene SDR-11 yellow and installed by a contractor/employee with current certifications in the type of fusion they are using.

All underground isolation valves for gas lines shall be polyethylene with 2” operating nut.

Acceptable Piping Materials

- Above Grade 2" and smaller: Schedule 40 black steel pipe with treaded 150 pound black malleable iron fittings. Paint all exterior and exposed piping to prevent rust.
- Above Grade 2-1/2" and larger: Schedule 40 black steel pipe with welded steel fittings. Paint all exterior and exposed piping to prevent rust.

Building Gas Pressure:

Regulate gas before entering buildings from 5 PSI to 7 inches water column. Size the building gas distribution system based on a pressure of 7 inches water column. For some special high gas use applications such as large boilers, a 2 PSI mechanical room gas pressure may be allowed when approved by the University Representative and permitted by code.

Regulators:

Gas Regulators should be specified to include an under pressure shut off feature to automatically shut off the gas in the event of a downstream pipe rupture.

Earthquake Valves:

Provide seismic gas shut off valves on gas services to all buildings which are intended to be used for student housing or as a residence. Seismic gas shut off valves are not required on non-residential buildings.

Gas Meters

- Gas meters may be either diaphragm bellows or turbine type as appropriate for the application. Gas meters shall record in cubic feet.

- Except where covered under c., below, provide valving and connections to allow for the use of a temporary meter bypass (3/4 inch hose) when removing the meter for servicing. This bypass should be configured such that the meter may be removed while still supplying limited gas service to keep pilot lights lit.
- For large academic buildings (greater than 30,000 gross square feet), commercial kitchens, and gas service which supplies an emergency generator, provide valuing and permanent piping for a full sized bypass of the meter. The bypass shall be configured to allow for meter removal without disruption of gas service.

Gas piping shall not be routed under building floor slabs.

Prior to installing new gas appliances in existing buildings, verify the gas pressure with the University Representative. In some cases, the gas pressure has been increased to medium pressure to compensate for under sized gas piping. Provide an additional gas regulator near the appliance in the event that the existing building gas pressure exceeds the listing on the regulator to be furnished with the gas appliance. (Note: Gas appliances are often furnished with regulators which are only listed for a maximum inlet gas pressures of either 9 or 14 inches water column.) Provide regulator with a flow limiting orifice or with vent piped to outdoors as required by code. Review regulator location and safety issues with the University's Representative on a case by case basis. As a general rule, medium pressure gas should be reduced to low pressure prior to piping through spaces which are normally occupied. Medium pressure gas shall not be permitted in occupied spaces used for student housing or a residence.

Valving:
Gas piping systems shall be provided with manual isolation valves at the following locations:
- At building points of entry.
- At each floor where branch piping connects to a riser.
- At each gas appliance.
- At branch connections to underground system mains.

Gas meters to be installed with all boilers, generators, engine, etc., that may need a permit to operate, and that use natural gas, to allow for gas reporting requirements.